

How Much Is the Option to Refinance Worth, and Who Captures the Realized Gains?

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Abstract

The right to refinance is an option to replace a mortgage when rates fall. Lenders price this option, but its value is opaque to borrowers at the time of origination. When the option is exercised, it produces a benefit to homeowners similar to that of a cash-settled derivative instrument. But the gain is not apparent in measured income or as a change in household wealth. This paper estimates both over the period from 1990 through 2026. Given financial conditions at origination and assuming borrowers refinance optimally, the option lowered expected mortgage costs by 29 basis points on average and by 103 basis points during 2022–2024. When borrowers exercised the option, retiring an above-market mortgage at par produced exercise payoffs of about \$5.4 trillion over 1990–2024, in constant 2026 dollars. That payoff is a transfer from mortgage investors to borrowers, not a change in aggregate wealth. Because many of those mortgages would later have ended through another refinance or a move, the survival-weighted use value of the interest savings is smaller, about \$1.9 trillion. Most of the difference between the option's value at origination and its realized payoff resulted from rate declines that markets did not anticipate. The gains were unevenly distributed. For example, Black borrowers held 7.4% of mortgage debt but received 4.4% of the gains.

JEL Codes: G21, G12, D14, D31, E52, R31

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1 Introduction

The right to refinance a fixed-rate mortgage is one of the most valuable financial options most households will ever hold. When rates fall borrowers can swap into cheaper loans. If rates rise borrowers keep their existing rate. The mortgage rate quoted at origination thus overstates what borrowers should expect to pay over the life of the loan, by an amount that depends on financial conditions at the time and on the borrower's future refinancing behavior. Investors who buy mortgages price this option, but its value is opaque and hard to measure for individual borrowers, and no public time series of its value exists.

The financial benefits households gain from refinancing are also opaque. Exercising the refinance option is economically the same as settling a financial derivative in cash, but it produces no measured income flow. It is recorded neither as income (e.g. in household surveys), because no cash changes hands, nor as a change in measured wealth, because the refinanced balance is unchanged. Because the option is broadly held and valuable, and exercised at different times and at different rates across homeowners, this source of income is likely to be economically important and unequally distributed.

This paper provides a coherent measure of how much the option was worth at origination for every month from 1990 to 2026, assuming optimal refinancing; how much lenders charged households for the option; how much benefit households ultimately realized from exercising it; and how that benefit was distributed across households.

To measure the ex-ante option value, I compute the steady mortgage rate that would leave a borrower as well off, over the expected life of the loan, as the quoted rate paired with the right to refinance. I call this steady rate the option-adjusted cost rate, and the gap between it and the quoted rate is what the option is worth to the borrower. Measuring this option value requires a rule for when borrowers refinance, a model of future interest rates anchored to financial conditions prevailing at origination, and a simulation over future rate paths to estimate the distribution of option values. I compute this value for every month from January 1990 through mid-2026. Borrowers are assumed to follow the optimal refinancing rule of Agarwal et al. (2013). In the main specification, I assume interest rates follow a standard term-structure model (Hull–White) calibrated to the observed yield

curve each month. Investors in mortgage-backed securities price the same option. The spread on those securities net of the option-adjusted spread is what investors charge for bearing prepayment risk. That charge is the market price of the option, the second of the paper's measures. I compare it month by month with the model value.

To measure the benefit borrowers realized from exercising the option, I model the mortgage as a debt the household can retire early at par. When rates fall the loan is worth more than its nominal balance, but the borrower pays it off at the balance by refinancing. Net of costs, that difference is the exercise payoff, the amount a separately traded call on the loan would have paid in cash. The payoff values the retired payments as if the old loan would otherwise have run to maturity. Many of those mortgages would instead have ended through a later refinance or a move, so I also report a survival-weighted use value that counts the interest savings only over the life the old loan would probably have had. (Households may also benefit during refinancing by extracting equity or lowering payments through re-amortizing to a new 30-year term, but these are not changes in wealth.) I measure and distribute this gain using near-universe data on mortgage applications and originations from the Home Mortgage Disclosure Act, loan-level histories from Freddie Mac, and the Survey of Consumer Finances.

On average over the thirty-six years, the option is worth about 29 basis points off the quoted rate. Its value varies substantially over time with the rate environment, rising above a full percentage point during the 2022–2024 rate spike. The market has generally charged more. The market price exceeded the model value in most years, and over the full sample the model value averages about a quarter of the spread on mortgage-backed securities over Treasuries. Part of the difference is the premium investors require for bearing prepayment risk. Many borrowers fail to refinance when they should, which works in the other direction and should narrow the gap.

Over 1990–2024, interest rates declined more than markets expected, and the payoff to households exceeded both the price paid and the value implied at origination. Measured as the value of retiring an above-market liability at par at prevailing market rates, exercised payoffs totaled roughly \$5.4 trillion over 1990–2024, in constant 2026 dollars. If instead each month's saving is weighted by the probability the loan survives, the household use value is about \$1.9 trillion. Most of the payoff

was luck: rates fell farther than markets or forecasters expected, so the gain was a surprise.

Higher-income households collected most of the dollars largely because they hold most of the mortgage debt. Black borrowers held 7.4% of mortgage debt but received only 4.4% of the gains. White borrowers' share of the gains matched their share of debt. One reason is exercise: refinancing take-up, estimated from millions of loan-month records, rises with credit score and with loan balance, so small-balance borrowers refinance the least. Approval is a second channel: across 185 million refinance applications, Black applicants were denied at roughly twice the white rate, before any adjustment for credit characteristics.

1.1 Related literature

The paper relates to three literatures. The first studies the mortgage borrower's refinancing decision, a canonical problem of household finance (Campbell, 2006). Agarwal et al. (2013) solve the borrower's refinancing problem in closed form. A large empirical literature shows that the resulting option is exercised imperfectly: roughly 20% of unconstrained borrowers fail to refinance when it is optimal (Keys et al., 2016), deeply in-the-money options go unexercised (Deng and Quigley, 2003), inaction is widespread and tied to inattention and complexity (Andersen et al., 2020; Berger et al., 2025), and documentation requirements and out-of-pocket costs block refinancing precisely when it is most valuable (DeFusco and Mondragon, 2020). Gerardi et al. (2023) show that Black and Hispanic homeowners are substantially less likely to refinance, which they argue amplifies the redistributive consequences of monetary policy, and Berger et al. (2024) and Fisher et al. (2024) study how refinancing frictions redistribute wealth across borrowers with heterogeneous rates. This literature measures how the option is used and misused. However, it typically does not assess how much the option is worth at origination. It also often focuses on how refinancing changes the monthly payment, which conflates the change in loan duration (replacing, say, the remaining 15 years on the old mortgage with a new 30-year term), cash extracted, and the change in the interest rate; only the last is the financial gain from the option itself.

A second literature values the prepayment option from the investor's side. Contingent-claims models of mortgage-backed securities (MBS) (Dunn and McConnell, 1981; Schwartz and Torous,

1989; Stanton, 1995; Deng et al., 2000) price the borrower's right to prepay as the investor's risk. Gabaix et al. (2007) and Boyarchenko et al. (2019) show that bearing that risk commands a premium in MBS spreads. Related work on product choice and monetary transmission (the fixed-versus-adjustable decision, Campbell and Cocco, 2003; Koijen et al., 2009, and the refinancing channel of monetary policy, Berger et al., 2021; Eichenbaum et al., 2022) establishes why the option matters for household choices and for policy. However, these models price the investor's expected costs, which are not necessarily a direct counterpart to a borrower's expected cost of financing a home. Kalotay et al. (2004) and Kalotay and Fu (2009) apply the same tools from the borrower's side, valuing the refinancing option in a single loan to compare mortgage offers on an option-adjusted APR. That work prices one loan on one date. No study appears to provide a time series of the borrower-side option value at origination, or to set it against the market's charge and the payoffs borrowers later realized.

A third literature measures what borrowers gain from refinancing (or leave on the table). This literature frequently focuses on the change in the monthly payment or the discounted stream of payment differences (Hurst and Stafford, 2004; Keys et al., 2016; Andersen et al., 2020), or on the cash-flow relief refinancing delivers to constrained households (Beraja et al., 2019), evidence that matters for understanding the macroeconomic and consumption responses to interest rate changes. But the payment change from refinancing reflects a combination of lower rates, extending the loan's duration (e.g., back to 30 years), and equity extraction. Only the first component is a realized gain, and the distinction is quantitatively large: in 2020, only about \$88 of the average \$233 decline in monthly payments from refinancing came from lower rates.

This paper puts the four measures on one footing. The resulting series show what the option was worth to each cohort of borrowers and how much each cohort gained from the rate declines that followed. The ex-ante series also serves as a benchmark for optimal exercise, against which actual behavior is imperfect (Keys et al., 2016), unequal (Gerardi et al., 2023), and hampered even at origination (Bhutta et al., 2026).

The rest of the paper calculates the four measures. Section 2 sets out the accounting identity behind them and the data. Section 3 presents the ex-ante option-value series and the reasons it varies

over time. Section 4 compares it with the market's price of the same option embedded in MBS spreads. Section 5 measures the gains households actually realized, the surprise component, and the gain net of the implied market price paid for the option. Section 6 measures who captured these gains. Section 7 concludes with implications. The appendix provides the formal model, derivations, robustness checks, and supplementary results.

2 Framework, Measurement, and Data

2.1 An accounting identity

A fixed-rate mortgage is an amortizing callable liability. Conceptually, the borrower owes a non-prepayable amortizing mortgage bond and holds an American-style right to repurchase it at its scheduled unpaid balance. She typically exercises that right by originating a new loan to repay the existing loan. When market rates fall, the market value of the borrower's above-market-rate liability rises above par, but the borrower extinguishes it at par. That right is a call option on the bond, struck at par.

A separately traded call would work the same way: exercise after a rate decline pays the bond's value above the exercise price, net of the option's cost, and under ordinary circumstances that payoff is a realized capital gain. Refinancing delivers the identical gain, only through extinguishing the liability at par instead of through a cash settlement. The gain is the mirror image of the prepayment risk mortgage investors are short. The aggregate payoff is a redistribution from lenders to borrowers, not a change in aggregate wealth.

The contract rate sets the monthly payment, but it overstates the borrower's expected financing cost because the borrower can replace the loan when rates fall. It can be decomposed as

$$\underbrace{m(0)}_{\text{contract rate}} = \underbrace{c^*}_{\text{option-adjusted cost rate}} + \underbrace{V}_{\text{option value}}, \quad (1)$$

where c^* is the constant rate that delivers the same expected present-value cost on a hypothetical no-option mortgage and $V = m(0) - c^*$ is the value the borrower expects to derive from the refinancing

option.¹

2.2 Value, price, and payoff

I measure the option four ways: what it is worth at origination, what the market charges for it, what it pays when borrowers refinance, and who received the payoff and who paid the charge. The first is the ex-ante option value V from equation (1), the gap between the contract rate $m(0)$ and the option-adjusted cost rate c^* . It measures what the option is worth at origination to a borrower who exercises it optimally. The second is the market price of the same option: the mortgage-backed-securities spread over Treasuries, net of its option-adjusted component. This price reflects what investors charge for bearing prepayment risk. It exceeds V in most years. The third is the realized payoff P^{realized} , the implied gain from refinancing households actually exercised over 1990–2024. At each refinance, I calculate this as the value of the retired liability minus the balance repaid at par and closing costs. The fourth measure, the net incidence, is who received the payoff and who paid the premium.

Table 1 defines each of these values, summarizes how they are measured, and lists the data sources from which they are derived.

2.3 Data

Interest-rate data come from the Federal Reserve Economic Data (FRED) database: the Freddie Mac Primary Mortgage Market Survey 30-year fixed rate (PMMS, series MORTGAGE30US) and Treasury constant-maturity yields at 11 maturities from 1-month through 30-year. I take data from the first business day of each month, January 1990 through June 2026. From these I extract the contract rate, the full yield curve, the 10-year-minus-2-year slope, trailing 36-month mortgage-rate volatility, and the mortgage–Treasury spread. The mortgage rate ranges from 2.67% (January 2021) to 10.56% (May 1990). Appendix Q describes the replication code and data.

¹Wedges common to the callable loan and the no-option counterfactual (credit risk, servicing, and the primary–secondary spread) cancel because they are held fixed across the two. Future variation in the all-in mortgage spread does affect the value of refinancing and is included in V . Appendix Table 16 reports the sensitivity to alternative spread processes.

Table 1: The refinancing option: value, price, payoff, and incidence.

	Ex-ante value (V)	Market price	Realized payoff	Net incidence
What it measures	Value of the option to a borrower who exercises it optimally; the gap between the contract rate and the option-adjusted cost rate (eq. (1))	What investors charge for bearing prepayment risk; the MBS spread net of its option-adjusted component	Contractual exercise payoff: market value of the retired liability minus par and closing costs (eq. (2)), discounted at the market mortgage rate at exercise	Who received the payoff and who paid the premium; each group's net position (Section 6.4)
Rate dynamics	Hull–White simulation; drift fit to the origination-date forward curve, realized trailing volatility	Investor prepayment model under risk-neutral valuation; option-implied volatility	None; the realized rate path	None; inherited from the payoff and price columns
Refinancing behavior	Agarwal et al. (2013) benchmark threshold, near-optimal (Appendix K.2)	Actual, as investors expect it	Actual refinancers	Actual; refinancers collect, every origination pays
Source and coverage	Model on FRED inputs; monthly, 1990–2026	Bloomberg current-coupon series; monthly, 1996–2026	HMDA, MBA, Freddie Mac loan level, SCF; annual, 1990–2024	HMDA, SCF, and the cohort ledger; 1990–2024
Headline estimate	Mean 29 bps and median 21 bps over 1990–2026. Mean 103 bps in 2022–24	Above V in most years. V averages about one-quarter of the secondary-market spread	Gross \$5.4 trillion over 1990–2024. Survival-weighted use value \$1.9 trillion. Constant 2026 dollars	Black borrowers hold 7.4% of mortgage debt and received 4.4% of the gross payoff and 3.6% of the net gain

Notes: The payoff column is the contractual exercise payoff; Section 5 reports the survival-weighted use value alongside it. Two derived quantities combine the columns. Net realized gain = payoff – market price: what borrowers gained over what they paid. Surprise = realized value – ex-ante value V : the part of the payoff not priced at origination. The surprise is measured for the 318 origination cohorts of 1990–2016 with a full ten-year realized window, on the rate-average basis, and compares an ex-ante mean of 41 bps with a realized mean of 88 bps. The net gain is measured for the closed cohorts originated 1990–2014, with payoffs of \$4.2 trillion less market premiums of \$0.9 trillion, or \$3.3 trillion. Sections 3–6 estimate each column; Appendix A details the payoff conventions.

The realized-gains analysis of Section 5 draws on four additional public sources. The distribution of realized refinancing flows comes from the Home Mortgage Disclosure Act (HMDA) loan-level records, which cover essentially the universe of mortgage originations. I use the pre-2018 Loan/Application Register files (harmonized across the 2004 change in race and ethnicity coding) together with the modern files for 2018 onward, which additionally report the contract rate, for every year 1990–2024. Aggregate refinance volumes are taken from the Mortgage Bankers Association and validated against HMDA.

Take-up, the probability that an in-the-money borrower actually refinances, is estimated from the Freddie Mac single-family loan-level dataset, a loan-month panel of repayment and prepayment outcomes (about three million sampled loan-months).

I use the Federal Reserve’s Survey of Consumer Finances (SCF) to understand the distribution of refinancing gains across households. Appendix A documents each source and its coverage.

2.4 Measurement

Computing the option-adjusted cost rate involves three ingredients: a rule for when to refinance, a model of future interest rates anchored to conditions at origination, and a simulation that averages over many possible rate paths. I use the closed-form refinancing threshold of Agarwal et al. (2013), a Hull–White one-factor term-structure model (Hull and White, 1990) recalibrated each month to the observed Treasury yield curve, and a Monte Carlo simulation that applies the threshold along 10,000 simulated rate paths at each origination month.

Refinancing is not as simple as it looks. A borrower should not refinance the moment the present-value savings cover the closing costs, because exercising the option today is costly, and it forecloses exercising it later at an even better rate. Agarwal et al. (2013) solve this stopping problem in closed form: refinance when the gap between the current rate and the market rate exceeds a threshold. The threshold rises with rate volatility and with closing costs, falls as less time remains on the loan, and sits far above the naive break-even (100 bps under the January 2024 calibration used here, about three times the naive level). Their rule is exactly optimal under their assumptions, but not under the rate process I simulate here, so I treat it as a benchmark. Appendix K.2 solves the exact

stopping problem and finds the benchmark understates the option value by 2–5 bps. Appendix B states the formula and replicates their published thresholds.

I model rate dynamics using the Hull–White model, calibrated each month so that simulated paths reprice the Treasury yield curve observed on that date and match the trailing volatility of 10-year yields. Simulated Treasury paths become mortgage-rate paths through an estimated pass-through coefficient on monthly Treasury changes, about 0.6, estimated on expanding windows to avoid look-ahead bias. I add a persistent stochastic mortgage–Treasury spread and anchor each path at the observed contract rate. Appendix C states the equations. The largest source of model uncertainty is the spread specification and I assess how alternative assumptions affect the result in Appendix Table 16.

The resulting series is a forward-curve benchmark rather than a valuation under either the physical or the risk-neutral measure. The drift is fit to the term structure observed at origination, as in risk-neutral pricing, so the mean path carries whatever term premia forward rates embed. The volatility is realized trailing volatility rather than option-implied. Cash flows are discounted at the observable 10-year Treasury yield, the market opportunity cost of funds, instead of an unobserved preference parameter, following the market-based discounting of the MBS-valuation literature (Stanton, 1995). The series is an expected-cost measure conditional on origination-date bond prices.

The simulation applies the threshold refinancing rule along each path. When simulated rates cross the threshold, the borrower refinances. She pays that era’s closing costs. Her rate resets to the market rate, and the loan re-amortizes over thirty years.

Refinanced loans carry the same option, and exercise can recur along a path, so the measured value is the value of access to refinancing over the borrower’s expected tenure, not the value of a single exercise embedded in the initial loan. Each refinancing resets the rate to the prevailing market rate on a callable loan, which embeds the charge for the next option.

I weight cash flows by the probability that the mortgage is still outstanding. The relevant horizon is the borrower’s expected loan life at origination, which was about 8.5 years in the early 1990s and rose to 12.4 years in the 2020s as homeowner mobility fell.

The option-adjusted cost rate is the constant rate on a hypothetical no-option mortgage that

costs the same in expected present value, inclusive of refinancing costs and terminal balances, as the contract rate paired with optimal refinancing. The option value is the gap between the two, expressed in basis points. Appendix F states the formal present-value-equivalence condition (Equation (A.9)). Each monthly estimate uses only information available at the origination date, and the Monte Carlo standard error is about ± 1 basis point at 10,000 paths (Appendix H).

2.5 Calibration

I calibrate four borrower-side parameters of the refinancing threshold (the mortgage balance, the effective tax rate on mortgage interest, the annual moving probability, and origination costs) to time-varying public data, so that thresholds and option values reflect the conditions borrowers actually faced at origination. All four moved substantially over the sample. The average government-sponsored enterprise (GSE) loan balance rose from roughly \$102,000 in 1990 to \$350,000 in 2024, while origination points fell from about 2% in the early 1990s to 0.5–0.7%. Homeowner mobility fell from about 9% to 4–5%, lengthening the survival-weighted expected loan life. The effective tax rate on mortgage interest dropped from roughly 0.15–0.18 to about 0.065 after the 2018 doubling of the standard deduction.

Larger balances and cheaper originations lower the threshold and raise the option value. Longer expected tenure raises the value. The tax change works modestly in the opposite direction. Appendix I documents each series and its sources. Appendix Table 20 shows that the time-varying calibration barely changes the full-sample mean but changes the decade pattern. Holding the discount rate and horizon fixed, it lowers the 1990s mean from 38 to 23 bps, because loans were small and points were high, and raises the 2020s mean from 47 to 55 bps.

3 The Value of the Refinancing Option, 1990–2026

3.1 The option-adjusted cost rate

Figure 1, Panel A, plots the 30-year fixed mortgage contract rate alongside the option-adjusted cost rate for each month in the sample. The gap between the two lines is the option value (Panel B).

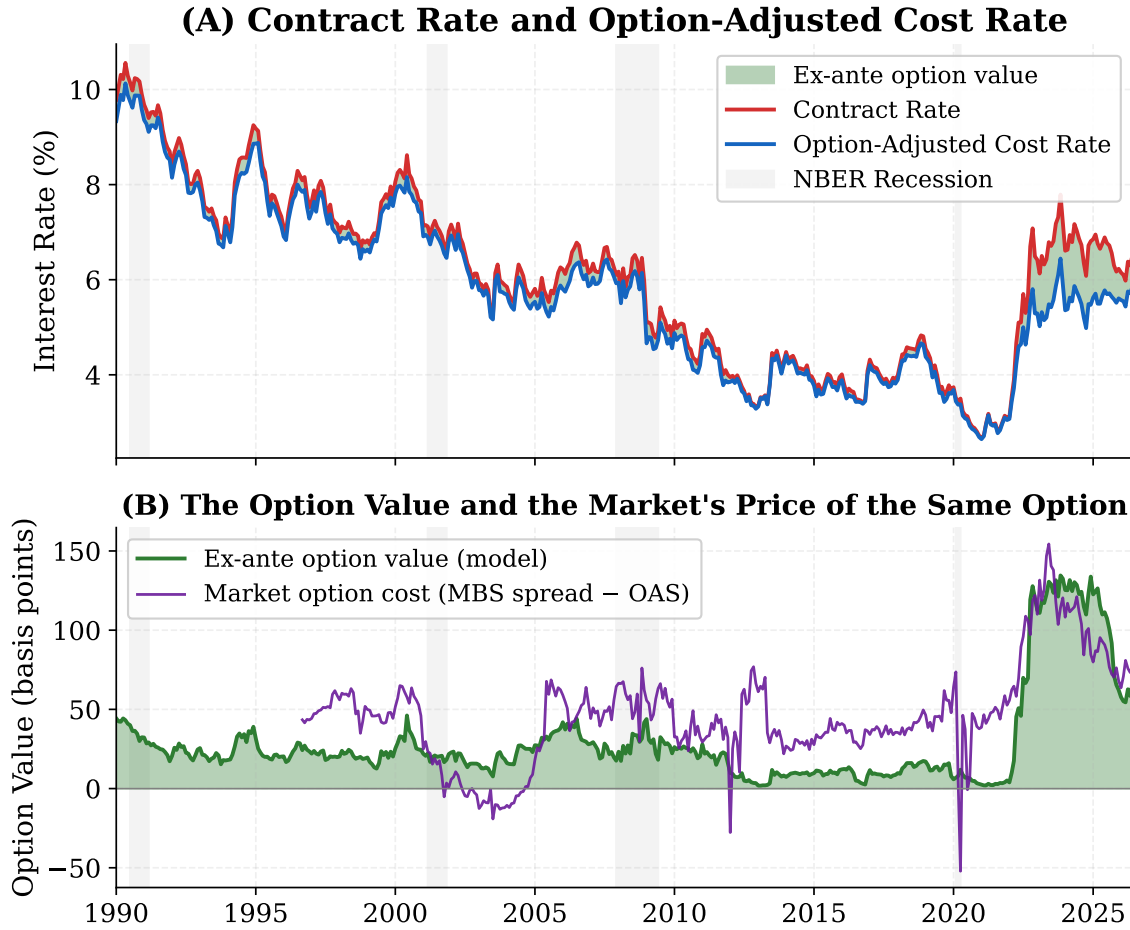


Figure 1: Contract rate, option-adjusted cost rate, and the market's price of the option. Panel A: the red line shows the 30-year fixed mortgage contract rate; the blue line shows the option-adjusted cost rate, accounting for optimal refinancing over the survival-weighted expected loan life (10,000 paths per month); the green shaded area between the two lines is the refinancing option value. Panel B: the same ex-ante option value (green, in basis points) against the market's price of the same option, the mortgage-backed-securities spread net of the option-adjusted spread (OAS), extracted from Bloomberg current-coupon yields (purple, September 1996 onward). The market series is negative in a few months when the OAS exceeds the MBS spread. Monthly, January 1990–June 2026. Gray bars indicate NBER recessions.

The option-adjusted cost rate usually moves closely with the contract rate, but the gap widened sharply in the 2020s. The option value averages 29 bps over the full sample, with a median of 21 bps (Table 2). The distribution is right-skewed. The 90th percentile is 60 bps, and the maximum is 135 bps, reached in November 2023 when the contract rate was 7.8%. High-rate environments produce especially large values: the option averages nearly six times as much when the contract rate exceeds 6% as when it is below 4% (Appendix Table 21). Recent borrowers had a much greater modeled chance of refinancing than borrowers who originated during the preceding low-rate period.

Table 2: Option value summary statistics by period (basis points)

	Full Sample	1990s	2000s	2010s	2020s	2022–24
Mean	29	24	26	12	69	103
Median	21	22	25	10	68	122
90th pct	60	35	37	23	129	132
Max	135	44	46	28	135	135
N (months)	438	120	120	120	78	36

Notes: Option value is $V = m(0) - c^*$ in basis points, where $m(0)$ is the contract rate and c^* is the option-adjusted cost rate (PV-equivalent measure; see Appendix F). Computed from 10,000 Monte Carlo paths per month over the survival-weighted expected loan life. Decades are defined by origination date (1990s = Jan 1990–Dec 1999, etc.); the 2022–24 column is a subset of the 2020s. Borrower parameters (M , τ , μ , points) are time-varying as described in Section 2.5; remaining parameters as in Appendix J.

Outcomes vary widely across simulated paths (Appendix Figure 16). In some paths the borrower never refinances and pays the full contract rate. In others, she refinances multiple times and achieves a rate well below the mean.

3.2 Why the option value varies

The option value depends on three features of the rate environment. Higher volatility generates more dispersion in future rate paths and so more refinancing opportunities. This is the standard result that volatility benefits the option holder. The contract-rate level matters too: a higher starting point leaves more room for rates to fall. In earlier high-rate decades, smaller loan balances and higher origination costs raised the refinancing threshold and offset that effect. The yield-curve slope enters negatively. Conditional on level and volatility, a steeper curve signals expected rate increases and reduces the chance that rates fall below the refinancing threshold.

A fourth ingredient, the mortgage–Treasury spread, matters at the end of the sample. The 2022–2024 spread ran roughly 80 bps above its historical average (Appendix Table 21), and the baseline model expects partial normalization. That expectation uses only information available at origination, since the average toward which spreads revert is estimated on data through each origination month (Appendix C). Removing spread mean reversion entirely leaves the 2020s decade mean essentially unchanged (Appendix Table 16). In the data, the highest option values occur where high rates meet flat or moderately sloped curves, as in the early 1990s and again in 2022–2024 (Appendix Figure 17).

The 2020s show the highest option values, a mean of 69 bps, reflecting elevated rates, high volatility, and longer expected borrower tenure. The 2000s and 1990s are moderate, with means of 26 and 24 bps. The 2010s are lowest, at 12 bps: a prolonged, low-rate, low-volatility era.

A borrower originating a mortgage in January 2024 at 6.61%, when volatility was elevated, the yield curve was slightly inverted, and the optimal threshold was 100 bps, would have expected to refinance about 2.6 times over the loan’s expected life, for an option value of 125 bps, or roughly \$28,000 in present value on that year’s average balance. An identical borrower in January 2021 borrowing at 2.67%, with volatility near its sample low, would have expected to refinance 0.15 times, and the option was worth about 2 bps: 87% of simulated paths never refinance in that environment, against 3% in January 2024.

The option has tended to be worth the most precisely when rates are highest and borrowers feel most burdened by their mortgage costs.²

3.3 Robustness

The level of the series depends most on the assumed behavior of the mortgage–Treasury spread, the largest source of model uncertainty. Alternative spread processes span 29–46 bps without changing the timing. None lowers the mean materially, and most raise it, so the headline sits at the conservative end of the range and the level is better read as a range than as a point estimate. The discount rate and the borrower horizon bracket the mean between 21 and 30 bps. The rate model, the threshold

²Appendix M reports the full calibration values for these scenarios; Appendix P adds a by-regime summary (Appendix Table 21), sample simulated rate paths, the refinancing-event distributions (Appendix Table 22), and box-plot and heatmap views of the full-sample distribution (Appendix Figures 16 and 17).

rule, the volatility calibration, and the discretization each move it by a few basis points. Appendix L collects the full set (Appendix Table 15).

4 The Market’s Price of the Same Option

The borrower-side option value has a natural counterpart: the charge mortgage investors build into MBS yields for bearing the same prepayment risk, the market option cost. I measure it as the current-coupon MBS spread over Treasuries net of the option-adjusted spread (OAS). The OAS comes from a dealer prepayment model, so the residual is a market-implied measure of the option’s cost rather than a quoted price. It is negative in a few months when the OAS exceeds the spread. That charge exceeded the borrower-side value in most years. Over the full sample, the borrower-side value averages roughly one-quarter of the secondary-market spread, ranging from about one-tenth in the 2010s to more than one-half in the 2020s.

Figure 2 plots the market option cost, extracted from Bloomberg current-coupon yields, against the model’s option value; Figure 1, Panel B, plots the two series over time. The two are positively correlated and the relationship is robust: the market option cost rises 0.71 bps per basis point of borrower-side option value (Newey–West $t = 8.11$, Panel A), and the broader secondary-market spread, which also contains the non-option OAS component, rises 0.59 bps (Newey–West $t = 4.78$). The relationship runs specifically through the secondary-market (prepayment) spread rather than the primary–secondary spread, consistent with investors pricing prepayment risk. Both series are persistent. Appendix O verifies that the level relationship is not a spurious-regression artifact and reports the construction, alternative constructions, and the full specification table.

Both measures rise when rates are high and volatile, but their levels differ. The market measure uses risk-neutral valuation and option-implied volatility, and it includes the compensation investors require for bearing prepayment risk. Each factor tends to raise the market price relative to the borrower-side benchmark. Imperfect refinancing works in the other direction: an option that borrowers fail to exercise is less costly to investors. The gap between the two series includes, and likely understates, the prepayment-risk premium documented by Gabaix et al. (2007) and

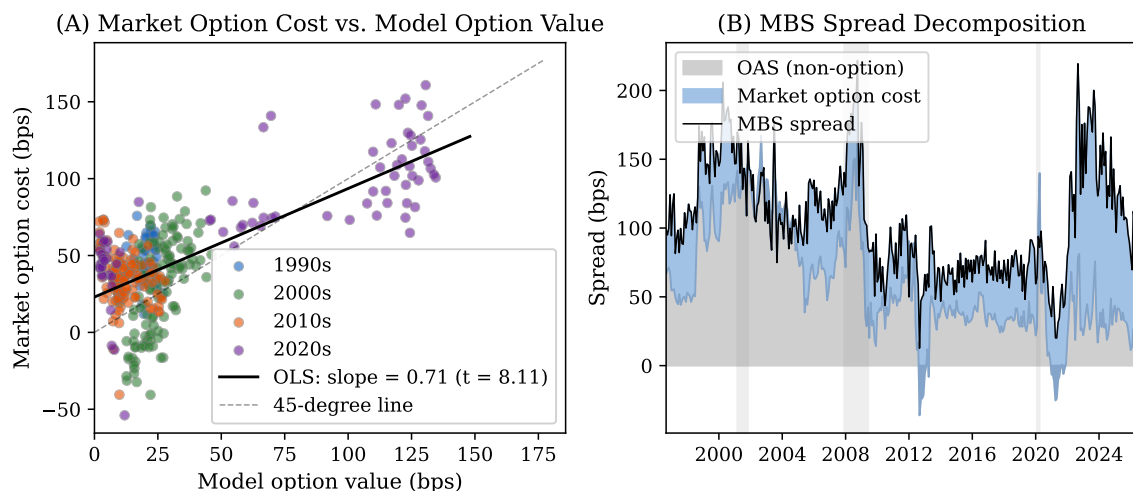


Figure 2: Borrower-side option value and MBS pricing. Panel A: market option cost (MBS spread minus OAS) vs. the headline model option value for 358 monthly observations (September 1996–June 2026), colored by decade, with OLS regression line and 45-degree line. Panel B: time-series decomposition of the MBS spread into OAS (gray, non-option component) and market option cost (blue, prepayment-related component); gray bands mark NBER recessions. The printed slope in Panel A is the market-option-cost row of Appendix Table 19, which reports every specification.

Boyarchenko et al. (2019). The charge is embedded in the contract rate of every loan, whether or not the borrower ever refinances. This value is used in Section 5 as the price side of the net gain.

5 The Realized Gains from Refinancing, 1990–2024

Rates fell for three decades after 1990, leading to waves of refinancing. Retiring above-market mortgages at par delivered about \$5.4 trillion in exercise payoffs between 1990 and 2024, in constant 2026 dollars. This section measures that gross gain and the net-of-price-at-origination gain.

5.1 Measuring the realized payoff

I measure the gain from a refinance as the payoff of the call the borrower exercises: at the moment of refinancing, the borrower retires an above-market liability at par (Section 2.1). Consider a refinance on remaining balance B , with n_{old} months left on the old loan and closing cost K . The per-dollar

payoff is the contractual exercise payoff

$$g^{\text{OT}} = \frac{MV_{\text{old}} - (B + K)}{B}, \quad MV_{\text{old}} = \sum_{m=1}^{n_{\text{old}}} P^{\text{old}} e^{-c_{\text{new}} m/12}, \quad (2)$$

where MV_{old} is the value of the old loan's remaining scheduled payments P^{old} discounted at the prevailing mortgage rate c_{new} , and the superscript marks the option-theoretic convention. The payoff subtracts the balance repaid at par and closing costs. The discount rate is the market mortgage rate observed at exercise. It is the cash settlement described in Section 2.1, the amount a separately traded call on the old loan would have paid at exercise. Discounting at c_{new} values the retired payments at the rate the household can actually borrow at, so MV_{old} is their replacement value to the household. The rate on an otherwise identical noncallable loan would be lower by the option premium, and discounting at that rate would raise the payoff, so the convention is conservative.³ The new loan is issued at market terms and is worth its balance, so stretching repayment back out to thirty years just moves cash flows across time at market prices.

Consider a household ten years into a 10% loan, with a \$250,000 remaining balance that refinances into a fresh thirty-year loan at 7%. The scheduled payment falls from \$2,413 to \$1,663, a saving of \$750 a month. Only \$475 of that comes from the lower rate. The other \$275 comes from stretching the remaining twenty years of repayment back out to thirty. That is a change in duration, not wealth. The household's gain is the market value of the retired loan's remaining twenty years of scheduled payments at the new rate, less the balance repaid at par and closing costs: about \$58,000, or 23 cents per dollar refinanced. The rate effect's share of the payment drop depends on how far rates fell. In 2020, rates were already lower and typical declines smaller, so only about \$88 of the average \$233 monthly payment reduction was the rate effect; the remaining \$145 was re-amortization.

The national flow in a year is the average payoff per dollar among the loans refinanced that year,

³Two other conventions answer different questions and are reported in Appendix Table 5: the survival-weighted use value, which discounts the monthly payment reduction over the period the old mortgage would probably have remained outstanding (eq. (A.1)), and a legacy fixed-horizon present value, computed over ten years at a 5% nominal discount rate.

times the dollars of old balances those loans retired:

$$G_t = g_t \times Q_t.$$

The per-dollar payoff g_t is eq. (2) averaged over the year's refinancers. From 1999 on I compute it loan by loan from Freddie Mac records, which report each loan's old rate, balance, and remaining term, with the new rate set to the market rate at payoff. Before 1999 there are no loan-level records, so I compute it for a loan of that year's median age using the year's rate decline; the two methods agree where they overlap (Appendix A.8). The volume Q_t is the old balances refinanced, from MBA and HMDA. A rate-and-term refinance counts in full. A cash-out refinance counts only its old balance, because the cash taken out is new borrowing at the new rate and earns no gain. Excluding cash-out loans entirely gives a floor of \$3.5 trillion, and crediting the full cash-out proceeds gives a ceiling of \$6.0 trillion. Appendix A.2 gives the full aggregation (eq. (A.3)) and shows the central estimate is stable across the plausible range of cash-out shares.

Borrowers paid for this option in their contract rates, so the payoff is a gross figure. Section 5.3 charges the premium back; because it is a running charge in the rate that the borrower pays only while the loan survives, it is converted to an upfront price as the survival-weighted present value of that charge over the loan's expected life (Appendix A.2).

5.2 Aggregate realized gains, 1990–2024

Figure 3 reports the realized payoffs year by year: the exercise payoff booked by the households that refinanced in each year, in the year of exercise, the same convention income statistics use to record capital gains in the year an asset is sold. The flow is a settlement value, not that year's cash saving. The value arrives as lower payments over the loan's remaining life.

The historical gains came in waves. The per-dollar payoff was about 21 cents in 1992–93, 11 cents in 2001–03, and 15 cents in 2009–13 and 2020–21 (Table 3). The 2009–13 wave, the era of the Home Affordable Refinance Program (HARP), was the largest, delivering roughly \$1.2 trillion. The 2001–03 boom produced about \$950 billion, and the 2020–21 pandemic boom another \$880 billion.

Each of these four largest waves comes to between 1.1 and 1.7% of contemporaneous GDP. The table also reports the smaller 1998 episode.

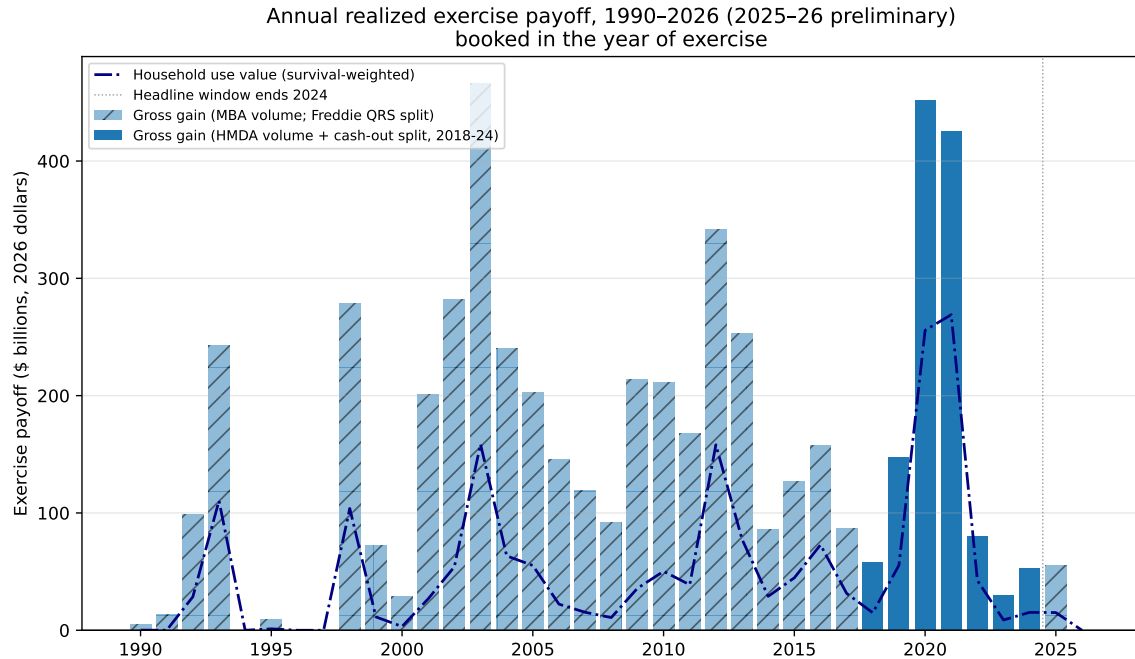


Figure 3: Annual realized exercise payoff, 1990–2026. Bars show the gross payoff of the households that refinanced in each year, on the existing-loan basis (rate-and-term loans in full plus the old-balance portion of cash-out refinances), booked in the year of exercise as the market value of each retired liability minus par and closing costs. Hatched bars use MBA volume with a modeled cash-out split, solid bars HMDA volume and split (2018–24). The dash-dotted line is the survival-weighted household use value of the same rate reductions. Constant 2026 dollars (PCE deflator). The 2025–26 bars are preliminary and are excluded from the 1990–2024 totals in the text.

The cumulative gross realized payoff since 1990 is about \$5.4 trillion (Figure 4). For scale, lenders originated about \$97 trillion of one-to-four-family mortgages over the same years (purchase and refinance, MBA estimates, in 2026 dollars), so the cumulative payoff comes to about 5.5 cents per dollar originated. Both the annual and the cumulative series represent the market value of the retired liability minus par and closing costs, in constant 2026 dollars. The survival-weighted use value, the same rate reductions discounted over the life the old loan would probably have had, is plotted alongside and totals about \$1.9 trillion.⁴

⁴Of the \$5.4 trillion gross headline, about \$648 billion comes from the modeled pre-1999 years; the remainder is measured from observed refinance volume and, from 1999 on, loan-level rates (Appendix A.12 reports representative-loan and fixed-horizon cross-checks).

The gain is, in a practical sense, a capital gain, and the resources it provides are income in the Haig–Simons sense. Refinancing lowers required interest payments so the household can consume more, save more, or both. An economically equivalent separately traded option would settle in cash and generate a taxable capital gain.

The refinancing gain is instead invisible to standard household statistics. It is not income for tax purposes, and Census money income counts only regular cash receipts. The gain does not show up as a jump in wealth at closing either: the old loan is recorded on the household balance sheet at par and discharged at par, and the new loan typically has the same balance. If the household withdraws equity, the added cash and added debt offset. The Survey of Consumer Finances and the Distributional Financial Accounts carry both loans at their unpaid balances and correctly show little change at exercise. However, those balance sheets do not reflect the option’s appreciation as rates fell. The national accounts are a partial exception: BEA deducts mortgage interest when it imputes the rental income of owner-occupants, so the savings raise measured income slowly over the remaining life of the loan. Otherwise the benefit shows up only later, as lower future payments, higher saving, or higher consumption.

Table 3: Realized refinancing capital gain by episode (headline series), in constant 2026 dollars. Volume and rates are rate-and-term refinances; the source column flags the data tier.

Episode	Volume	Share	Old rate	New rate	Gain/\$	Gross	Net of model V	Source
1992–93	455	0.476	9.92	7.63	0.2097	341	324	Modeled
1998	445	0.516	9.15	6.95	0.2016	279	266	Modeled
2001–03	3063	0.562	7.05	6.14	0.1133	949	893	GSE + MBA
2009–13	4774	0.805	6.41	4.38	0.1486	1187	1104	GSE + MBA
2020–21	3718	0.719	3.68	2.96	0.1451	878	847	GSE + HMDA

Notes: Volume = Q^{rt} , rate-and-term refinance volume (\$bn); Share = rate-and-term share of total refinance volume; Old/New rate = volume-weighted average coupon rates (%); Gain/\$ = g_t , the per-dollar exercise payoff (eq. (2)); Gross = G_t and Net of model V = G_t^{net} , cumulated over the episode (\$bn, eqs. (A.3)–(A.4)), on the headline existing-loan basis. The Volume column is in nominal dollars; Gross and Net are constant 2026 dollars (PCE). Gross applies g_t to existing-loan volume Q_t^{exist} , which exceeds the rate-and-term Volume column. Source: Modeled = PMMS-lag model gain with MBA volume (pre-1999); GSE + MBA = loan-level per-dollar gain with MBA volume; GSE + HMDA = loan-level gain with HMDA-era volume split. Generated by 08_realized_gains_figures.py.

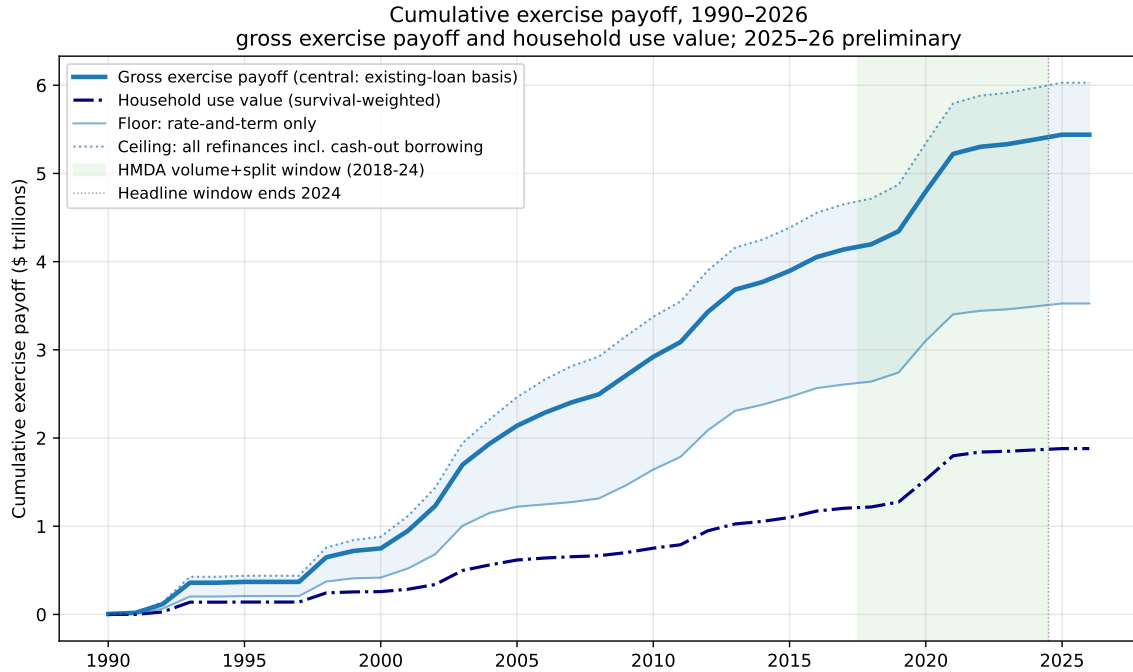


Figure 4: Cumulative realized exercise payoff, 1990–2026. The central estimate (solid) applies the per-dollar exercise payoff (eq. (2)) to all existing balances refinanced (rate-and-term loans in full plus the old-balance portion of cash-out refinances), reaching about \$5.4 trillion. The dash-dotted line is the survival-weighted household use value (\$1.9 trillion). The shaded band runs from the rate-and-term floor (\$3.5 trillion, cash-out excluded) to the ceiling that also credits the cash-out proceeds (\$6.0 trillion). Per-dollar payoffs are loan-level actual (Freddie, 1999+) and a median-age-calibrated model before; volume is MBA, validated against HMDA. All amounts are in constant 2026 dollars (PCE deflator). The 2025–26 observations are preliminary and are excluded from the 1990–2024 totals in the text.

5.3 The surprise and the net gain

The realized payoff dwarfs the ex-ante value because the post-1990 rate path delivered far more than origination-date forward curves priced in. I compare, cohort by cohort over 1990–2016 origination cohorts with full 10-year realized windows ($N = 318$), the option value at origination with the value realized under optimal exercise on the actual PMMS path: $V_t^{\text{realized}} = V_t^{\text{ex-ante}} + \text{Surprise}_t$. The ex-ante value averages about 41 bps. The realized optimal value averages about 88 bps. The surprise is roughly 47 bps (Appendix Figure 12).⁵ The ex-ante series was informative about the cross-section: cohorts originating in months with higher option values went on to realize systematically larger savings, with a correlation of 0.63 across cohorts. But nearly every cohort realized more than its ex-ante value, and little of the windfall was predictable: survey rate expectations and a random-walk benchmark together bound the predictable part at a small fraction of the gap, and the rest is path luck (Appendix N, Appendix Figure 13).

The net gain, after charging the embedded option price, was lower. All borrowers paid for the option, including those who did not refinance. Every originated loan, exercised or not, carried the premium in a contract rate above what an otherwise identical non-callable loan would have required. The net gain to households is therefore the payoff less the premium the whole cohort of originations paid. For each origination cohort, I assign each loan its origination-year premium, at the market price investors actually collected for bearing prepayment risk, and attribute realized payoffs back to the cohort that paid for them using the loan age at refinance (Appendix Table 6). For the cohorts closed by 2024 (originations through 2014, each observed over a full ten-year exercise window), \$42 trillion of originations, in nominal dollars, paid \$0.9 trillion in premiums and collected \$4.2 trillion in payoffs, both in constant 2026 dollars, for a net gain of \$3.3 trillion. Roughly a fifth of the gross payoff was, in effect, pre-paid at origination. The premium was small against a rate decline that far exceeded what origination-date prices anticipated, and it is small under every alternative convention. Charging only the loans that refinanced, or charging the model value V rather than the market price, each leaves households further ahead (Appendix Tables 1 and 6).

⁵Both sides of this comparison use the rate-average measure of financing cost from the cohort file of Appendix N. That is why the ex-ante mean of 41 bps exceeds the PV-equivalent headline of 29 bps; the two sides are measured consistently.

Households in aggregate collected far more than they paid for the option. Whether any individual household did depends on whether it exercised, and exercise was uneven.

6 Who Captured the Gains

Refinancing gains depended not only on timing but also on the ability and choice to refinance. Households differ in credit access, loan size, income, and attention (Keys et al., 2016; Andersen et al., 2020; Gerardi et al., 2023). To measure the resulting distribution of the gains from refinancing, I use HMDA to allocate the aggregate flows, Freddie Mac loan-level data to estimate take-up, and the SCF to compare the gains with mortgage holdings by income, wealth, and race.

6.1 The distribution of realized gains

Realized gains were concentrated among higher-income borrowers and were disproportionate across racial groups. Figure 5 allocates the aggregate gain G_t across groups by each group's share of that year's refinancing volume, $G_{g,t} = G_t \times (\text{RefiVol}_{g,t} / \text{RefiVol}_t)$, using actual HMDA loan-level flows for every year 1990–2024 (the data described in Section 2.3). This assigns every group the same per-dollar gain, so a group's share of the realized total equals its share of refinance volume. The resulting shares bundle take-up, application, approval, and loan size.⁶

By applicant income, the top quintile captured about 37.1% of the realized gain and the bottom quintile about 8.5%, a gradient of more than four to one. By race, white borrowers captured about 78.8% of the gain, Asian and other borrowers about 10.1%, Hispanic borrowers about 6.7%, and Black borrowers about 4.4%, below their share of mortgage debt and population. This puts the Black–white refinancing gap documented by Gerardi et al. (2023) in dollars. The gains are also geographically concentrated: California alone accounts for about 25% of the national total.

The same-per-dollar assumption is a good approximation. For 2018 onward, where HMDA reports the contract rate, I can replace the common per-dollar gain with each group's actual new

⁶All shares are dollar-volume weighted, and the allocation uses rate-and-term dollar shares for 2018 onward, when HMDA begins to distinguish cash-out from rate-and-term refinances. Reallocating by total-refinance shares moves no group's cumulative share by more than 1.6 percentage points, and because Black and Hispanic refinance dollars tilt toward cash-out, the disparities reported here are, if anything, slightly understated (Appendix A.12, Appendix Table 9).

rate; the resulting per-dollar gains are nearly identical across groups, about 5.8 cents per dollar for white refinancers in 2020 versus 6.1 cents for Black refinancers, because all groups refinance into the same market rates in a given year. For Black borrowers' share of the gains to catch up to their 7.4% share of mortgage debt, their per-dollar gains would have to run roughly 70% above white refinancers'. The loan-level rates put the two within a few percent of each other. Higher old note rates among minority refinancers (Gerardi et al., 2023) would narrow the gap by about two-fifths, raising the Black gain-to-debt ratio from 0.59 to 0.76, but would not close it (Appendix A.12).

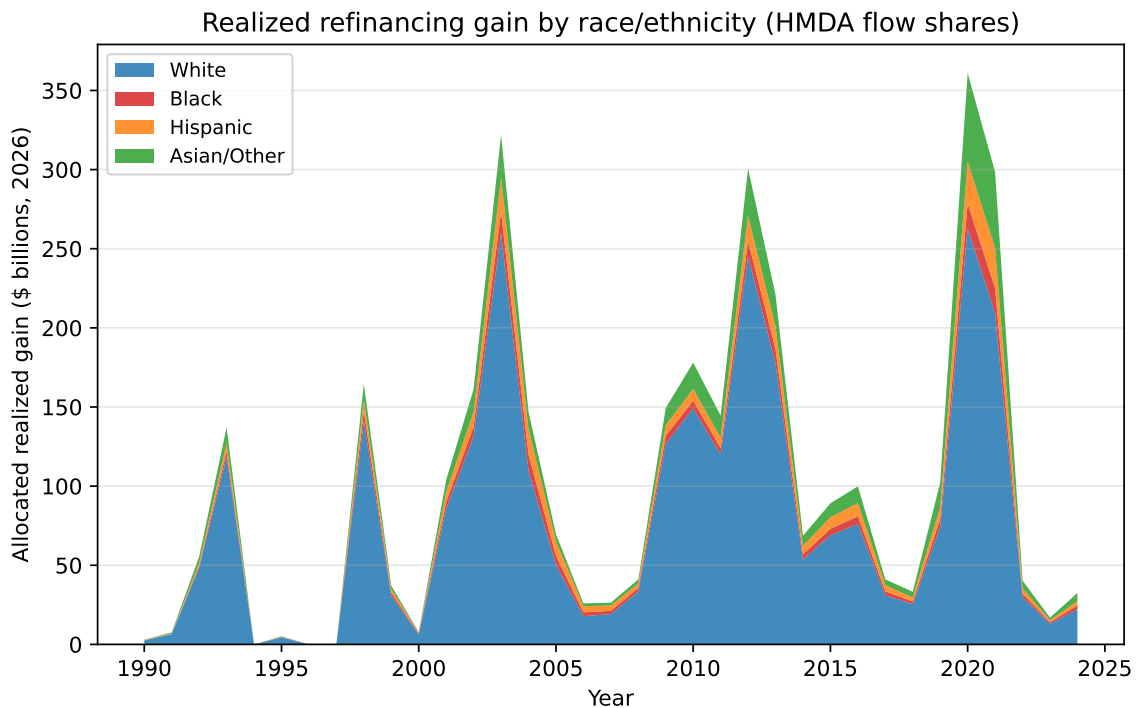


Figure 5: Realized refinancing gain by race/ethnicity, 1990–2024. Annual flows in billions of constant 2026 dollars, stacked by group, allocated from HMDA refinance volume shares for every year. Appendix Figure 4 shows each group's share of refinance volume over time.

The income gradient is stable across waves (Table 4 repeats the allocation wave by wave). In every major wave the top quintile collected at least a third of the dollars and the bottom quintile about a tenth or less. The racial distribution moved more. Black borrowers' share of wave gains was smallest in the early 1990s. It fell from 4.1% in 2001–03 to 3.3% in 2009–13, then reached its high of 4.8% in 2020–21, when Hispanic borrowers' share also peaked. Even the 2009–13 wave, which

included the Home Affordable Refinance Program, delivered Black borrowers a share below their 1990–2024 average.

Table 4: Group shares of the realized gain, by refinancing wave (percent).

Wave	Income quintile					Race/ethnicity			
	Q1	Q2	Q3	Q4	Q5	White	Black	Hispanic	Asian/Other
1992–93	5.4	18.8	11.3	28.0	36.5	86.7	2.3	3.6	7.3
1998	5.6	17.0	10.2	26.3	40.8	85.7	3.9	4.2	6.2
2001–03	7.9	14.1	17.7	21.0	39.4	81.2	4.1	6.3	8.3
2009–13	9.2	11.6	18.3	24.4	36.4	82.4	3.3	5.1	9.2
2020–21	10.5	12.6	18.7	24.9	33.4	71.6	4.8	7.9	15.7
1990–2024	8.5	13.0	17.3	24.0	37.1	78.8	4.4	6.7	10.1

Notes: Shares of each wave’s realized gain, in percent. The allocation is the one behind Figure 5: each year’s aggregate gain G_t is split across groups by that year’s HMDA refinance volume shares and summed over the wave years, so the 1990–2024 row reproduces the pooled shares quoted in the text. Income quintiles are count-weighted within year and built from \$10,000 income bins; a bin that straddles a quintile boundary is assigned whole, so adjacent quintile shares are coarse in the early 1990s. Race shares exclude loans with race withheld. Wave dollar totals appear in Table 3. Generated by 27_wave_distribution.py.

6.2 Gains relative to debt

A group’s share of the gains is not by itself evidence of unequal exercise, because groups differ in how much mortgage debt they hold. Table 5 reports three shares side by side: each group’s share of households with a mortgage, its share of mortgage debt, and its share of the realized gains. The final column divides gain share by debt share. The ratio equals one when a group’s gains are proportional to its debt. Because gains are allocated by refinance volume, the ratio is equivalently the group’s refinancing flow per dollar of mortgage debt relative to the aggregate.

For Black households the shares fall at each step. Black households are 9.9% of households with a mortgage, but they hold only 7.4% of mortgage debt, because their average balances are smaller, and they received just 4.4% of the gains. White borrowers’ gains were roughly proportional to their debt (1.01). The Black ratio is 0.59: per dollar of mortgage debt, Black borrowers’ refinancing flow ran about 40% below proportional. Hispanic borrowers are close to proportional (0.93), and Asian and other borrowers are well above it (1.38), consistent with refinance volume concentrated in higher-balance markets. Had Black borrowers received the realized gain in proportion to their 7.4%

share of mortgage debt, rather than their 4.4% share of the gain, they would have collected roughly \$162 billion more of the gross payoff over 1990–2024.

The income rows work the same way. The bottom quintile is 19.5% of households with a mortgage but holds only 10.5% of the debt, so much of its small gain share reflects fewer and smaller mortgages. Even so, its gain share falls below its debt share: the bottom-quintile ratio is 0.82, against 1.11 in the fourth quintile. That gap is consistent with the take-up and denial evidence below, though the ratio itself cannot separate exercise from application and approval. The ratios are robust to building the debt denominator year by year, their ordering survives the old-coupon adjustment, and the remaining cross-group patterns reflect volume and loan composition rather than per-dollar gains (Appendix A.12). Allocating the realized flow across SCF households independently reproduces the regressive income pattern and a Black gain-to-debt ratio of 0.66, corroborating the HMDA-based 0.59.

Table 5: Who captured the realized gains: group shares of mortgage-holding households, mortgage debt, and realized gains, 1990–2024.

	Share of			Gain share ÷ debt share
	Households with a mortgage	Mortgage debt	Realized gains	
<i>By race/ethnicity</i>				
White	78.5%	78.0%	78.8%	1.01
Asian/other	4.4%	7.4%	10.1%	1.38
Hispanic	7.2%	7.2%	6.7%	0.93
Black	9.9%	7.4%	4.4%	0.59
<i>By income quintile</i>				
Q1 (lowest)	19.5%	10.5%	8.5%	0.82
Q2	19.9%	13.6%	13.0%	0.96
Q3	19.9%	16.9%	17.3%	1.02
Q4	20.4%	21.6%	24.0%	1.11
Q5 (highest)	20.4%	37.3%	37.1%	0.99

Notes: Each row reports the group’s share of households with a mortgage (population-weighted count, not dollar-weighted), its share of mortgage debt outstanding, and its share of the realized gains. Gain shares are allocated from HMDA refinance-origination dollar volumes, 1990–2024. Mortgage-debt and household shares are from the Survey of Consumer Finances, pooled over the 1989–2022 waves (population weights, five implicates); race uses the wave-correct public-file coding (x6809 for 1998+, x5909 for 1989–95), with Asian and other/multiple-race responses reported together. Per-wave shares are benchmarked against Census/AHS homeownership-by-race envelopes in `scf_share_diagnostics.csv`. The final column is the ratio of the group’s gain share to its mortgage-debt share: one means gains proportional to debt, below one less than proportional. Generated by `20_denominators.py` (config: `scf_capture`).

6.3 Take-up and access

Two frictions help explain the uneven distribution: take-up and approval. Take-up is the probability that an in-the-money borrower actually exercises the option. A discrete-time hazard model on the Freddie Mac loan-month panel puts cumulative take-up over a twelve-month in-the-money window at about 32% for borrowers with FICO scores below 660, rising to 39% at 740–780. Take-up also rises with loan balance, from about 24% at a \$29,000 balance to 41% at \$373,000. The balance pattern is consistent with a fixed component of refinancing costs: small-balance borrowers refinance the least (Appendix Figure 2).

The second friction is approval. Pooling the roughly 185 million refinance applications acted upon in HMDA over 1990–2024, Black applicants were denied at 41.1%, about twice the 20.5% rate for white applicants. These are unadjusted rates. Bhutta et al. (2025) find that observable risk factors account for most of the gap in adjusted comparisons, but whatever its source, a denied application is a gain not captured. The GSE samples carry no borrower race or income, so the racial and income gaps here are sourced from HMDA and the SCF. Appendix A.12 reports the hazard specification and incentive-response curve and the denial-rate history (Appendix Figures 2 and 3, Appendix Table 8).

6.4 Who paid for the option: net incidence

One reason the incidence of the gain is uneven is that refinancing rates and success varied across groups. Another is that all borrowers paid for the option but only some exercised it. Agency securitization pools prepayment risk, and the premium embedded in the mortgage rate reflects the market’s expectation of pool-average exercise. Every borrower in the pool pays roughly that pool-average premium per dollar of debt, regardless of their own likelihood of refinancing. When exercise differs across groups, pooled pricing adds a cross-subsidy on top of the borrower–investor transfer: a group that refinances less than average still pays the full premium but collects less than the pool-average payoff, while a group that refinances more pays the same premium and collects more (Fisher et al., 2024; Berger et al., 2024).

Combining that ledger with the group shares of Table 5 gives each group’s net position (Appendix Table 7). I allocate payoffs by gain share and premiums by each group’s share of mortgage debt,

the same denominator Table 5 uses. Every group came out ahead, because the realized rate decline was large relative to what anyone paid: over the closed cohorts white borrowers netted \$2612 billion, Black borrowers \$118 billion, the bottom income quintile \$263 billion, and the top quintile \$1224 billion. But Black borrowers' share of the net gain, 3.6%, sits below their 4.4% share of the gross payoff, which is itself well below their 7.4% share of mortgage debt. Appendix Table 7 reports the same accounting with premiums following origination flow rather than the debt stock, and with the model value charged in place of the market price. The ordering is unchanged in all four.

Netting reduces the gains of lower-exercising groups. Premiums scale with debt held while payoffs scale with exercise, so the net distribution is at least as unequal as the gross one. But the same logic operates across individuals: non-refinancers paid for optionality that refinancers used, so these group means are a lower bound on the person-level inequality of the net incidence.

7 Conclusion

This paper measures four things about the refinancing option in a fixed-rate mortgage: what it was worth at origination, what the market charged for it, what it paid when borrowers exercised it, and who collected the payoff. The option was worth 29 basis points at origination on average, and far more during the 2022–2024 rate spike. The charge for the same option embedded in MBS yields was larger in most years. Exercise payoffs came to about \$5.4 trillion over 1990–2024, most of it a surprise, and, for the cohorts originated through 2014, \$3.3 trillion net of the premium every originated loan paid. The dollars went disproportionately to higher-income borrowers, and Black borrowers received a smaller share of the gains than their share of debt.

One implication of this analysis is that the contract rate is not the loan's expected long-run cost. It sets the payment on the current loan, and the borrower can replace that loan when rates fall. Headline-rate comparisons overstated the rise in expected financing cost during 2022–2023, because the refinancing option is most valuable precisely when rates are high. Existing consumer mortgage tools omit that distinction entirely.⁷

⁷The *New York Times* Upshot calculator applies a single user-entered mortgage rate to the entire analysis horizon. Zillow's methodology document assumes "the chosen constant interest rate and loan duration of 30 years." Bankrate

A second implication is that comparisons of fixed- and adjustable-rate mortgages based on initial rates miss the same asymmetry. An adjustable-rate loan passes rate declines through to the borrower automatically, but it passes increases through as well. The fixed-rate borrower is protected if rates rise and can still capture declines by refinancing, at a cost and subject to approval (Campbell and Cocco, 2003; Koijen et al., 2009). The relevant comparison is therefore the fixed loan's option-adjusted cost rate against the expected path of the adjustable rate, not one initial rate against the other. Rate risk, caps, and refinancing constraints would enter a full comparison and are beyond this paper.

The ex-ante values should be read as optimal-exercise upper bounds because real borrowers face inattention, inertia, and credit constraints (Keys et al., 2016; Gerardi et al., 2023).

The realized-payoff estimates put a dollar magnitude on the transfer from mortgage investors to households produced by the refinancing channel of monetary policy (Berger et al., 2021; Eichenbaum et al., 2022). They also show how unevenly that transfer was distributed over three decades of falling rates. That transfer is significant, particularly because it tends to arise in times of economic distress, but it is not apparent in household survey measures of income and wealth.

acknowledges the omission explicitly: “the ability to refinance a mortgage for a lower rate . . . was not factored in.” Calculators from Fidelity, SmartAsset, Schwab, and AARP do the same. Refinance calculators are a separate, reactive tool: they ask whether to refinance now, not what the mortgage is expected to cost at origination.

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The appendix has four parts. The first covers data and methodology for the realized-gains analysis (Appendix A). The second covers the ex-ante model: the refinancing threshold (Appendix B), the rate and spread model (Appendix C), the Monte Carlo engine and cost-rate definition (Appendix F), and the borrower calibration (Appendix I). The third covers verification and robustness: model validation (Appendix K), the collected robustness checks (Appendix L), the calibration reference values (Appendix M), and the cohort-level comparison of ex-ante and realized values with the surprise decomposition (Appendix N). The fourth covers supplementary evidence: the MBS spread analysis (Appendix O) and additional ex-ante exhibits (Appendix P).

A Data and Methodology for Realized Refinancing Capital Gains

This appendix documents the realized refinancing capital-gain series of Sections 5 and 6. It covers the quantities estimated, the per-dollar gain formula, every data source and its coverage, the assumptions and imputations and the period each applies to, the cross-checks used to validate the series, the limitations, and the code that reproduces it. The headline magnitudes for 1994–2024 rest on observed data. Every place a value is instead modeled, held, or proxied is stated explicitly.

A.1 Quantities estimated

I construct four related quantities and keep them conceptually separate.

1. **The realized capital-gain flow G_t** (headline): the present-value gain captured by borrowers who refinanced an existing loan in year t (rate-and-term loans in full plus the old-balance portion of cash-out refinances), aggregated to a national annual series, 1990–2026.
2. **The ex-ante/ex-post “surprise”**: the difference between the option value priced at origination and the value realized along the actual rate path, which isolates how much of the realized gain was unanticipated.
3. **The distribution of G_t across households**: who *earned* the realized gain, measured by income, race, and geography from HMDA loan-level flows (1990–2024); by wealth rank and

birth cohort from the Survey of Consumer Finances (SCF); with take-up frictions estimated from the GSE loan-month panel.

4. **The unrealized lock-in stock** (companion): the point-in-time gap between the unpaid balance B and the market value of outstanding mortgage liabilities L^{callable} (both defined in Appendix A.7), from the SCF; about \$2.2 trillion in the 2022 wave, in 2026 dollars.

The flow (1) and the stock (4) are distinct quantities (a cumulated payoff versus a current mark) and are never added together.

A.2 The per-dollar gain

Consider a refinance from old coupon c_{old} to new coupon c_{new} on balance B , with n_{old} months remaining on the old loan. The borrower refinances into a fresh $n_{\text{new}} = 360$ -month loan at cost $K = \text{fixed} + \text{points} \times B$. The headline per-dollar payoff is the contractual exercise payoff g^{OT} of eq. (2) in the main text: the market value of the old loan's remaining scheduled payments, discounted at the prevailing mortgage rate, minus the balance repaid at par and closing costs. It has no discount-rate or horizon parameter: it is exactly the buy-back-at-par settlement value of Section 2.1. The new loan is issued at market terms and is worth its balance, so no feature of the new loan enters the payoff.

The survival-weighted use value (g^{SL} , for survival-weighted life) instead prices the rate reduction over the life the mortgage actually has:

$$g^{\text{SL}} = \frac{1}{B} \left[\sum_{m=1}^{n_{\text{old}}} S_m (P^{\text{old}} - P^{\text{new}}) e^{-c_{\text{new}} m/12} - \sum_{m=1}^{n_{\text{old}}} (S_{m-1} - S_m) (B_m^{\text{new}} - B_m^{\text{old}}) e^{-c_{\text{new}} m/12} - K \right], \quad (\text{A.1})$$

where S_m is the probability the loan survives m months (the hazard-implied curve of Appendix A, by refinance year). The payment savings accrue while the loan lives. The re-amortized balance difference is settled when the loan terminates; loans still alive at n_{old} settle the remaining difference then. All terms are discounted at the market rate.

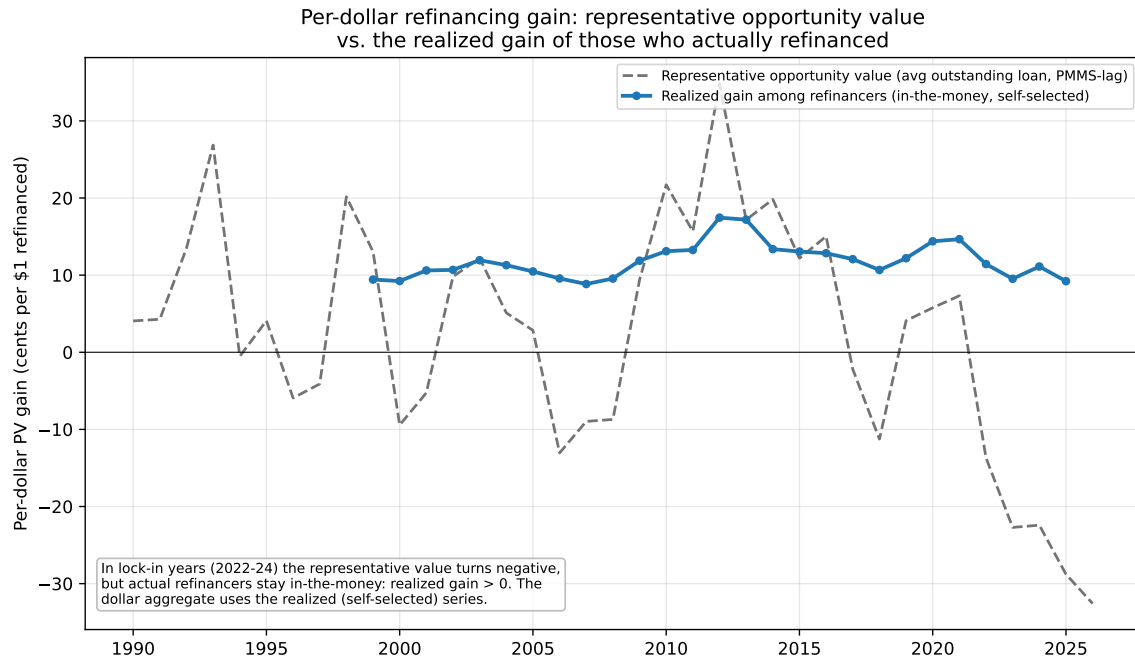
The legacy fixed-horizon convention is the *terminal-balance-adjusted* present value

$$g^{\text{TBA}} = \frac{1}{B} \left[\underbrace{\sum_m P^{\text{old}} d^m + B_H^{\text{old}} d^H}_{\text{PV of keeping the old loan to } H} - \underbrace{\left(\sum_m P^{\text{new}} d^m + K + B_H^{\text{new}} d^H \right)}_{\text{PV of the new loan}} \right], \quad (\text{A.2})$$

where P is the scheduled monthly payment and B_H the balance remaining at horizon H . The discount factor $d^m = e^{-\rho^{\text{nom}} m/12}$ is continuous, with ρ^{nom} fixed at 5% in the baseline and varied in the scenario fan below. The cash flows are nominal, so ρ^{nom} is a nominal discount rate; the superscript distinguishes it from the real rate ρ of the ex-ante appendices, which is also fixed at 5% in the legacy configurations. Including both terminal balances matters: it charges the borrower for the higher residual principal on the re-amortized new loan ($B_H^{\text{new}} > B_H^{\text{old}}$), which neutralizes the maturity-extension that would otherwise inflate a raw payment-savings measure.

A naive annuity measure g^{simple} , which omits the terminal-balance term, is included only to quantify the maturity-extension bias. Baseline parameters for the legacy measure are $\rho^{\text{nom}} = 5\%$, $H = 120$ months, costs of 1 point plus \$2,000, and a representative balance $B = \$250,000$. A scenario fan varies $\rho^{\text{nom}} \in \{3, 5, 7\}\%$, $H \in \{60, 120, 360\}$, the loan age, and costs. The ex-ante half of the paper discounts nominal cash flows at the observable nominal Treasury yield $r_d(t)$. The realized-gains series instead holds ρ^{nom} fixed, so a single rate prices gains across refinance years. This choice does not matter much: the $\rho^{\text{nom}} \in \{3, 7\}\%$ rows of the scenario fan and the market- and Treasury-discount rows of Appendix Table 5 show the totals are insensitive to it. Households that refinance repeatedly appear once per event. The gains telescope: each event's gain is measured from the coupon set by the previous one, so chained refinances sum to approximately the gain from the full rate decline rather than double-counting it.

The per-dollar series combines two distinct measures (Appendix Figure 1). One is a representative opportunity value for the average outstanding loan; it turns negative in lock-in years (2022–24), when the typical borrower is out of the money. The other is the realized gain among refinancers, which stays positive throughout because the households that refinance are positively selected from being in the money. The dollar aggregate uses the latter, so a lock-in environment never shows up as a negative realized gain.



Appendix Figure 1: Two distinct per-dollar series, in dollars of gain per dollar of balance refinanced; annual, 1990–2024. The *representative opportunity value* marks the average outstanding loan to market and turns negative in lock-in years (2022–24) when the typical borrower is out of the money; the *realized gain among refinancers* (GSE loan-level, 1999+) stays positive throughout because households that refinance are positively selected into being in the money. The aggregate flow G_t uses the latter.

The aggregate flow, on the headline existing-loan basis, is

$$G_t = \max(g_t, 0) \times Q_t^{\text{exist}}, \quad Q_t^{\text{exist}} = Q_t^{\text{rt}} + (1 - q_t) Q_t^{\text{co}}, \quad (\text{A.3})$$

where g_t is the balance-weighted per-loan exercise payoff g^{OT} among loans refinanced in year t (loan-level for 1999–2024, modeled before; Appendix A.4). Write Q_t^{tot} for total refinance origination volume (written RefiVol_t in Section 6.1) and s_t^{co} for the cash-out *incidence*, its share of that volume. Then $Q_t^{\text{rt}} = (1 - s_t^{\text{co}})Q_t^{\text{tot}}$ is rate-and-term volume and $Q_t^{\text{co}} = s_t^{\text{co}}Q_t^{\text{tot}}$ is cash-out volume. q_t is the cashed-out share of a cash-out loan's new balance, so $(1 - q_t)Q_t^{\text{co}}$ is the old-balance portion of cash-out refinances. The rate-and-term floor $\max(g_t, 0)Q_t^{\text{rt}}$ excludes cash-out entirely; the ceiling $\max(g_t, 0)Q_t^{\text{tot}}$ also credits the cash-out proceeds. The aggregate realized payoff of Section 2.2 is $P^{\text{realized}} = \sum_t G_t$. Every dollar figure is expressed in constant 2026 dollars. Each year's G_t is a present value in that refinance year's dollars; I scale it by the ratio of the 2026 to the year- t PCE price index before cumulating, so the 1990–2024 total is a single constant-dollar magnitude. The floor $\max(g_t, 0)$ encodes self-selection: rate-and-term refinancing essentially vanishes when it is not in the money, so years with a negative model gain, such as the rising rates of 2022–2024, contribute approximately zero realized gain rather than a negative number.

Net of the premium: two bases. Equation (A.3) gives the *gross* realized payoff. A callable mortgage carries a higher contract rate than an otherwise-identical non-callable loan, so the borrower pays a premium for the refinancing option. The net gain is the payoff minus that premium, and there are two ways to charge it.

The first basis is the market price: what borrowers actually paid, in their contract rate, for the option. Netting the market price yields a net gain in the ordinary sense of a return net of price. The second basis is the model value V , which is not a price but the option's expected value under optimal exercise. Netting V instead measures the payoff in excess of what the option was expected to deliver at origination, the dollar counterpart of the cohort-level surprise of Section 5.3. Because the market price exceeds V in most years, netting against V also bounds the net gain from above.

Both bases are converted to a per-dollar present value the same way: the per-dollar PV of a rate

reduction equal to the basis, on a fresh 30-year loan at origination, by origination year, matched to each refinance year through the loan's median age:

$$g_t^{\text{net}} = g_t - p_{t-a_t}, \quad G_t^{\text{net}} = \mathbf{1}\{g_t > 0\} g_t^{\text{net}} Q_t^{\text{exist}}, \quad (\text{A.4})$$

where p_y is the per-dollar basis for origination year y and a_t is the median age of loans refinanced in t ; the rate-and-term floor replaces Q_t^{exist} with Q_t^{rt} . The gate $\mathbf{1}\{g_t > 0\}$ is the same selection floor as in eq. (A.3): it is set by the sign of the gross gain, so years with no rate-and-term activity contribute zero rather than a negative net. This flow netting charges each refinancing loan its own cohort's premium, the convention by which realized capital gains are netted against the seller's basis, so it can be read year by year. That distinguishes it from the cohort ledger of Section 5.3, which also counts the premiums of loans that never refinanced.

The model value nets out about 6 percent of the gross, so the cumulative falls from roughly \$5.4 to \$5.1 trillion; the market price nets out about 11 percent. The gap reflects how far the realized post-1990 rate decline exceeded what was priced ex ante: most of the realized gain is an unanticipated windfall. The section's numbers form one consistent set. The market price is the cost basis. The realized PV reduction is the payoff. Their difference is the net gain, and the payoff in excess of the model value is the windfall.

The market basis. The market basis is the market option cost: the OAS-netted prepayment component of the secondary spread (Appendix O), converted to a per-dollar present value with the same machinery as p_y . Appendix Table 1 reports both nettings. In most years with OAS data, the market basis *exceeds* the model basis, because the market price embeds a prepayment-risk premium on top of the forward-curve-implied option value. Netting it gives a net gain of about \$4.8 trillion (ratio backfill before 1997; \$4.9 trillion under a zero pre-1997 basis), against the \$5.1 trillion upper bound from the model value. The gross payoff is unaffected either way.

A.3 Data sources

Appendix Table 2 lists each data source, its coverage, and its role in the analysis.

Appendix Table 1: Net realized gain under the two premium bases, 1990–2024 (constant 2026 dollars).

Premium basis	Net gain (existing-loan, \$tn)	Net, rate-and-term floor (\$tn)
Market price, ratio backfill pre-1997	4.8	–
Market price, zero basis pre-1997	4.9	–
Model value V (upper bound)	5.1	3.3

Notes: The gross payoff is the contractual exercise payoff (eq. (2)). The market basis is the OAS-netted prepayment component of the secondary spread (current-coupon yield minus the 10-year Treasury, minus the option-adjusted spread; Bloomberg, monthly from September 1996), the running charge borrowers actually paid for the option; negative months floored at zero. Both bases are converted from a running rate charge to an upfront dollar price as the survival-weighted present value of the rate differential over the loan’s expected life, because borrowers pay the running premium only while the loan survives; capitalizing the same charge over the full term or a fixed horizon would overstate the price. Pre-1997 months have no OAS data and are bracketed by two rules: the 1997–99 average market-to-model ratio applied to the model value, and a zero basis. The rate-and-term floor is computed only on the model basis, so the market rows show no floor entry. The model basis nets out the paper’s ex-ante option value (forward-curve-implied, optimal exercise). It is not a price; netting it measures the payoff in excess of expected value and, because the market price exceeds the model value in most years, bounds the net gain from above. Generated by 08_realized_gains_figures.py.

Appendix Table 2: Data sources. All are public; the GSE loan-level data require a free click-through license. MBA estimates are themselves model-based but are the standard national series and are benchmarked to HMDA.

Series	Source	Coverage	Role
Refinance \$ volume	MBA Historical Origination Estimates	1990–2026 (annual)	Headline national denominator
validation	HMDA loan-level (LAR; CFPB historic; API)	1990–2024	Volume cross-check
Distribution of flows	HMDA loan-level (race/income/geography)	1990–2024	Who earned the gain (primary)
Take-up frictions	Freddie loan-month panel (hazard)	1999–2024	Exercise probability by FICO/LTV/balance
Cash-out share	Freddie Quarterly Refinance Report (Total US)	1994–2017 (annual)	Rate-term/cash-out split
modern split	HMDA (loan-purpose 31 vs 32)	2018–2024	Split (purpose field)
Per-dollar gain	Freddie Single-Family Loan-Level	1999–2024	Actual borrower rates/balances/terms
Median loan age	Freddie Quarterly Refinance Report	1994–2016	Old-rate lag in the model gain
Mortgage rate (PMMS)	Freddie PMMS via FRED (MORTGAGE30US)	1971–	Old/new rate path
Household detail	SCF full public microdata	1989–2022 (triennial)	Stock + who-earned-it distribution
Mortgage debt outstanding	FRED HHMSDODNS	quarterly	Scaling / SCF coverage check
External anchor	NY Fed refinance-boom estimates (Haughwout et al., 2023)	2020–21	External magnitude check

Volume. The national denominator is the MBA refinance-origination series, used for every year so the aggregate has no source discontinuity. As an independent check I sum originated refinance loans (loan purpose = refinancing, action = originated) directly from the HMDA Loan/Application Register files; the HMDA total tracks MBA within roughly 10–20% in every overlapping year.

Cash-out split. The headline existing-loan basis credits a cash-out refinance only for its existing-loan component: the rate reduction on the old balance is a genuine refinancing gain, while the cash-out proceeds are home-equity extraction (new borrowing at the new rate) and earn no gain (eq. (A.3)); the rate-and-term floor excludes cash-out loans entirely. The split comes from HMDA’s loan-purpose field for 2018–2024 (which distinguishes refinancing from cash-out refinancing) and from the Freddie Quarterly Refinance Report’s annual “Total US” cash-out incidence for 1994–2017. Freddie defines cash-out as a new loan balance at least 5% above the refinanced loan’s unpaid balance, on conventional prime matched refinances; this differs from HMDA’s purpose-field definition, creating a modest definitional seam at 2018 that I flag and treat as a robustness margin. Each cash-out loan’s old balance is imputed as the new loan minus the dollars cashed out; the Freddie average amount cashed out and average balance imply a cashed-out share of a cash-out loan’s new balance averaging $q_t \approx 0.23$, and the aggregate is stable across $q_t \in [0.15, 0.35]$, within a few tenths of a trillion of the \$5.4 trillion headline. The one approximation: cash-out borrowers, who may refinance when less in the money on rate, are assigned the rate-and-term refinancers’ per-dollar gain, so the central estimate leans toward the upper part of the firm \$3.5 trillion floor.

Per-dollar gain. For 1999–2024 the payoff in (2) uses each loan’s *actual* note rate, balance, and remaining term from the Freddie Single-Family Loan-Level data; the new rate is the prevailing PMMS at the payoff month (a market proxy, validated against the realized new rate for the HARP cohort). For 1990–1998, where no public loan-level data exist, the gain is a model value: the old rate is PMMS lagged by the median loan age and the new rate is the contemporaneous PMMS.

A.4 Layer 1: the aggregate flow

The per-dollar gain g_t is the balance-weighted actual gain from the loan-level data where available (1999–2024) and the model gain otherwise (1990–1998). I multiply it by existing-loan volume Q_t^{exist} per (A.3) and cumulate the result. A scenario fan brackets the series under the parameter grid above.

A.5 Layer 2: loan-level micro (GSE)

From the Freddie Single-Family Loan-Level sample I identify voluntary payoffs (zero-balance code 01) on fixed-rate loans. The data do not say whether a payoff was a refinance or a home sale. I attribute payoffs to refinancing with a rate-gap rule: a payoff counts as a refinance if the loan was in the money by at least 75 basis points (note rate minus prevailing PMMS) when it was paid off. Home sales occur roughly independently of the rate gap. For attributed refinances I compute (2) and (A.1) at the payoff date, using the loan’s actual balance and remaining term, and report the balance-weighted per-dollar payoff by payoff year and by FICO and LTV bucket. I also report how the attributed share moves with the threshold (0, 50, 75, 100 bps).

Take-up hazard. To replace the hard rate-gap threshold with an estimated take-up elasticity, I build a discrete-time hazard panel from the same loan-level samples. Positive-balance loan-months are the at-risk spell-months; the event is a voluntary payoff (zero-balance code 01). I attach the event to the *last* positive-balance month, so its covariates are not read off the zero-balance record.

I case-control sample the at-risk months: all events, all in-the-money months, and a weighted subsample of the rest. I then fit a logit of the event on a spline in the refinancing incentive, FICO, LTV, the CLTV gap, log balance, a loan-age spline, a burnout proxy, and calendar-year and state fixed effects, with the incentive spline interacted with log balance. The refi-attributed hazard is the fitted hazard net of the deep-out-of-the-money (move) baseline. Twelve-month take-up is the cumulative refi-attributed exercise probability, tabulated by FICO, LTV, balance decile, loan age, and state.

Estimated take-up rises with FICO through the 740–780 range and rises with loan balance; the response to the incentive rises steeply as the option moves into the money, peaks near 1.5 percentage

points, and declines at deeper levels, a burnout pattern. The incentive-by-balance interaction matters for the balance profile. Small balances have a several-fold higher deep-out-of-the-money payoff rate. Without the interaction, that background rate loads onto the refinancing attribution and inverts the balance gradient. Raw event rates and fits stratified by balance both show take-up rising with balance at a common incentive.

Because the public GSE samples carry no race or income, this layer supplies the friction structure, not the racial or income gap.

A.6 Layer 3: ex-ante vs ex-post surprise

For each origination cohort, I compare the option value priced at origination—the model’s ex-ante value—with the value realized under optimal exercise along the actual PMMS path. The gap is the “path-luck” surprise. This layer reuses the paper’s existing threshold and Monte Carlo machinery. It adds no new data.

A.7 Layer 4: who earned the gain, and the lock-in stock (SCF)

Mark-to-market. For every SCF mortgage, I value the liability as $L^{\text{callable}} = \min\{\text{PV}, B + K\}$, where PV is the present value of the remaining scheduled payments at the survey-date PMMS, using the household’s reported coupon, balance, and remaining term. Hidden wealth is $B - L^{\text{callable}}$. The min operator is the refinancing option itself: a borrower never lets the liability’s market value exceed the cost of refinancing it away. Aggregated by wave, this is the lock-in stock. Broken out by age, race, and income quintile, it gives the stock’s distribution. I benchmark total SCF mortgage debt against FRED HHMSDODNS; coverage rises from 0.67 in 1989 to 0.91 in 2022.

Who earned the flow (HMDA, primary). I allocate the realized flow by race, income, and geography from actual HMDA refinance-volume shares, $G_{g,t} = G_t \times (\text{RefiVol}_{g,t}/\text{RefiVol}_t)$, for every year 1990–2024, with no interpolation. I harmonize three coding eras: the pre-2004 files (a single combined race field, code 4 = Hispanic); the 2004–2017 files (separate applicant ethnicity and race, code 4 = Native Hawaiian); and the 2018+ modern files (derived race/ethnicity, plus the contract

rate, which I use to refine each group’s new rate). Withheld-race volume is reported separately and excluded from the share denominator. Income groups are per-year count-weighted quintiles. The pre-2007 files use deflate64 compression, which I stream with a pure-Python decoder. I map state FIPS codes to postal abbreviations so geography aligns across eras.

Who earned the flow (SCF, wealth rank and cross-check). I identify recent refinancers using the first-mortgage refinance flag, restricting to loans obtained within three years of the survey. For these households I take the refinanced balance and characteristics and apply the year’s per-dollar gain to allocate the realized flow across income quintiles, race, and birth cohort. This exercise corroborates the HMDA flows independently, and it is the source for the *wealth*-rank dimension that HMDA lacks. SCF variables used: coupon (X816), balance (X805), term (X806/X815), year obtained (X802), refinance flag (X7137), age (X14), race (X6809), income (X5729), and the analysis weight (X42001). All statistics use the five imputation implicates and the survey weight.

A.8 Assumptions and imputations

Appendix Table 3 audits what is observed and what is modeled, period by period. The headline 1994–2024 magnitudes rest on observed volume and an observed cash-out split. Only two modeled inputs touch that window: the new-rate proxy in the loan-level gain, which is validated against the HARP cohort, and the refinance/move attribution rule.

Appendix Table 3: Observed vs. modeled inputs by period.

Input	1990–1993	1994–1998	1999–2024
Refi volume	MBA (HMDA-checked)	MBA (HMDA-checked)	MBA (HMDA-checked)
Cash-out share	held at 1994 *	Freddie (observed)	Freddie / HMDA (observed)
Old rate	PMMS lag (model)	PMMS lag (model)	loan-level (actual)
New rate	PMMS (model)	PMMS (model)	PMMS at payoff (proxy) [†]
Median loan age	held at 1994 (2.7y)	Freddie (observed)	loan-level (actual)
Refi vs. move	—	—	75 bp rate-gap rule

Notes: *No public, consistently defined cash-out split exists before 1994 (Freddie’s matched-refinance series begins in 1994); for 1990–1993 the cash-out split is held at its 1994 value, so those years enter the headline as a modeled component rather than an independent measurement. [†]The actual new rate is observed only for the HARP cohort (2009–2016), against which the PMMS proxy is validated.

A few more assumptions enter, all spanned in sensitivity checks. The discount rate ρ^{nom} , the

horizon H , the new-loan term (a fresh 30-year), and refinancing costs are modeling parameters covered by the scenario fan. The representative balance affects only the per-dollar cost share, not the aggregate, which scales with observed dollar volume. The pre-1999 model gain is not free-form. Fed the observed median loan age, it reproduces the loan-level estimate where the two overlap (2003: 0.075 model versus 0.074 loan-level), so the early-period series is calibrated to the micro data rather than assumed.

A.9 Validation and reconciliation

- **Volume:** HMDA-summed refinance originations match the MBA series within ~10–20% annually, 1990–1999 and 2018–2024.
- **Per-dollar gain:** the model gain (with observed median age) and the loan-level actual gain agree where they overlap (2003: 0.075 vs. 0.074; 2020–2021: 0.064 model vs. 0.065–0.067 loan-level).
- **Cash-out split:** the Freddie annual series matches the Freddie matched-refinance research-note values (1998: 0.484 vs. 0.486; 2003: 0.363 vs. 0.366; 2006: 0.868 vs. 0.867).
- **External anchor:** for 2020–2021 the New York Fed estimates that the nine million rate refinances of the pandemic boom cut annual housing costs by about \$24 billion (Haughwout et al., 2023). Capitalized over the survival-weighted expected life, that flow comes to roughly \$220–250 billion in then-current dollars. The survival-weighted use value I estimate for the episode, about \$430 billion then-current (\$525 billion in 2026 dollars), is larger mainly because it also credits the rate reduction on the old balances of cash-out refinances, which the Fed’s rate-refinance count excludes, and covers all of calendar 2020–2021. The two are the same order of magnitude (flow vs. present value).
- **SCF coverage:** total SCF mortgage debt is 0.67–0.91 of FRED HHMSDODNS, the expected survey-coverage range.
- **Self-selection:** in lock-in years where the representative model gain is near zero, the loan-level

data confirm that the households who *did* refinance were in the money with positive gains.

A.10 Limitations

No public loan-level mortgage data exist before 1999, so the pre-1999 per-dollar gain is necessarily a calibrated model rather than a measurement. No public, consistently defined cash-out/rate-term split exists before 1994 either; for 1990–1993 I hold the split at its 1994 value, a modeled component of the headline flagged in Appendix Table 3. Except for the HARP cohort, the new rate in the loan-level gain is a market (PMMS) proxy, because the public GSE data do not link a paid-off loan to its replacement. And the refinance/move attribution is a rule, not an observed field.

The Freddie loan-month panel carries no borrower race or income, so the distributional shares come from HMDA volumes and the SCF instead of the take-up model. MBA volume is itself a model-based estimate, benchmarked to HMDA. Because the SCF is triennial, the who-earned-it allocation draws on recent refinancers in each wave; it is not a continuous flow. The cash-out definition differs, too, between the Freddie (5%-balance) and HMDA (loan-purpose) sources: the pre-2018 group allocation pools rate-and-term with cash-out volume, a bias bounded at roughly half a percentage point of the Black and Hispanic cumulative shares and at most 1.6 percentage points for any group (Appendix Table 9). Finally, the realized gain is a transfer from mortgage investors to borrowers, not an addition to aggregate wealth.

The net-incidence allocations of Section 6.4 are approximations as well. The payoff side applies gain shares measured over the full window to closed-cohort aggregates. The premium side inherits the market-price proxy of Section 5.1, and cohort attribution uses the median loan age. The flow allocation reported in Appendix Table 7 removes the debt-stock proxy and the pooled timing on the premium side. The exercise takes pooled pricing as given, rather than modeling how group-specific pricing would change rates.

A.11 Reproducibility

The analysis is a self-contained pipeline that reuses the paper’s valuation code and writes only its own outputs. It fetches the external data (FRED, HMDA API), parses the HMDA LAR files, the Freddie

loan-level samples, the Freddie Quarterly Refinance Report, and the SCF, computes the per-dollar gain primitives, and then builds the aggregate flow, the loan-level micro results, the surprise, and the SCF stock and flow attribution, ending with the figures and tables. A single orchestrator script regenerates every figure and table from the public inputs. Each intermediate series is written to a documented CSV.

A.12 Take-Up Hazard and Robustness Exhibits

This section collects the take-up and robustness exhibits. Appendix Figure 2 reports the estimated take-up hazard from the Freddie Mac loan-month panel. Appendix Table 4 brackets the cumulative realized gain under the representative-model per-dollar measure of Section 5. Appendix Table 5 recomputes the cumulative total on a single representative-loan construction under each of the paper's conventions: the contractual exercise payoff (the headline), its cash-out-inclusive variant, the survival-weighted use value, and the legacy fixed-horizon present values. Appendix Table 6 reports the origination-cohort ledger that charges every originated loan the option premium once, and Appendix Table 8 the refinance denial rates by race summarized in Section 6.3.

Representative-loan cross-checks. Appendix Table 5 checks the cumulative totals against a representative-loan construction. There the exercise payoff is \$3.0 trillion and the representative-loan use value is \$1.1 trillion. The legacy fixed-horizon present-value conventions give less still: \$0.9 trillion at a five-year horizon and \$2.0 trillion at ten years, discounted at the Treasury rate. A deliberately conservative alternative replaces the loan-level rates with a representative loan and varies the discount rate, benefit horizon, closing costs, and loan age over a wide grid. Across that grid the cumulative gain runs from about \$0.9 trillion to \$3.0 trillion, and it is large under every assumption (Appendix Table 4). The loan-level headline of \$5.4 trillion sits above that conservative range by construction, because it uses each loan's actual rates and the existing-loan volume basis rather than representative-loan inputs on the rate-and-term floor.

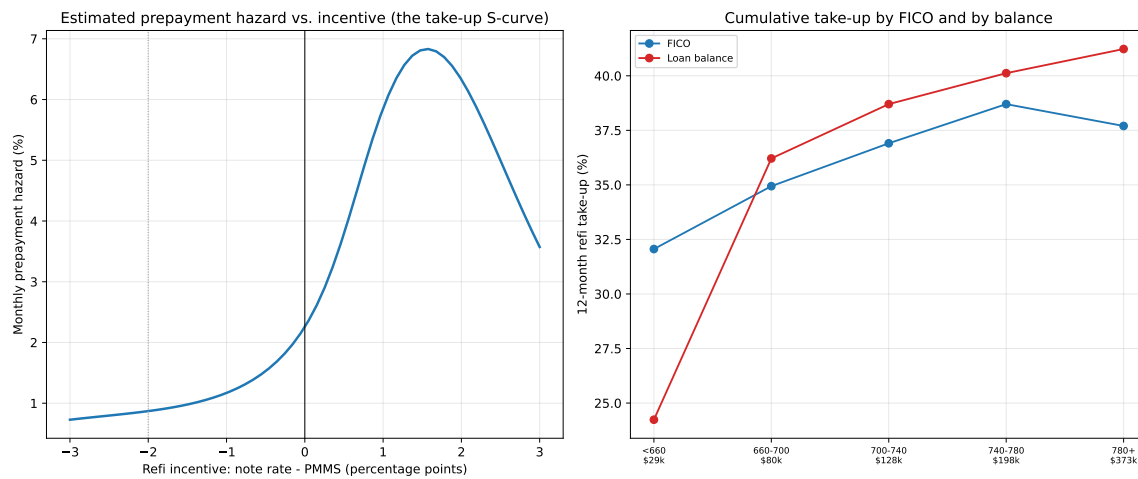
Capture-ratio robustness and composition. The gain-to-debt ratios of Section 6.2 survive two robustness checks. Building the mortgage-debt denominator year by year, using gain-weighted

annual shares interpolated across SCF waves rather than the pooled share, leaves the Black ratio at 0.60 (against 0.59 pooled) and the ordering unchanged. The old-coupon channel moves the ratio more. The 2018–2024 rate check cannot observe higher old note rates among minority refinancers (Gerardi et al., 2023); the SCF can. Among in-the-money loans, Black borrowers’ coupons run 30 to 90 basis points above white borrowers’ in every wave since 1989, enough to raise their per-dollar gain by roughly 25%. Building group-specific per-dollar gains from those coupons raises the Black ratio from 0.59 to 0.76. That adjustment is an upper bound, because coupons accumulate in the in-the-money stock precisely when their holders do not refinance; it narrows the gap by about two-fifths but does not close it.

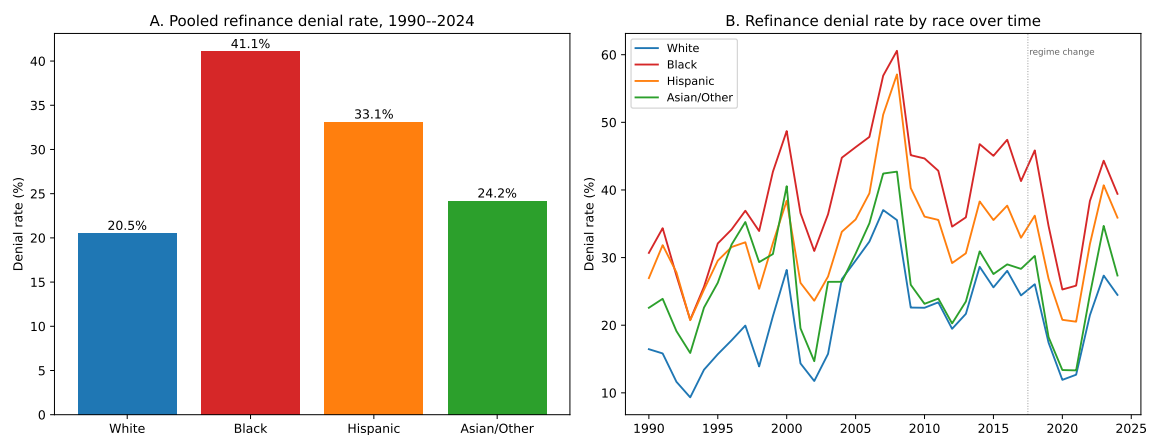
Two cross-group patterns are composition, not per-dollar gains. Asian and other borrowers’ share of the gains runs above their share of debt (1.38): their refinance volume tilts least toward cash-out—about 27% of refinance dollars, versus 37–40% for other groups (Appendix Table 9)—so more of it counts toward the rate-and-term gain, and their mortgage debt is concentrated in recent, high-balance coastal-market cohorts, originated at rates that made the 2019–21 wave especially exercisable. Their per-dollar gains when refinancing are no larger than other groups’. And the income gradient peaks at the fourth quintile (1.11), not the top (0.99): top-quintile debt tilts toward jumbo and adjustable-rate products whose refinances fall partly outside the conventional rate-and-term universe measured here, so the top-quintile ratio is sensitive to sample coverage and is best read as descriptive. Neither pattern speaks to the Black–white gap, which Gerardi et al. (2023) attribute to exercise differences conditional on being in the money.

The hazard regresses voluntary prepayment on the refinancing incentive (a spline), FICO, loan-to-value, loan balance, loan age, burnout, and calendar and geographic controls. The in-the-money excess over the deep-out-of-the-money baseline is attributed to refinancing. Take-up responds strongly to the incentive. The estimated incentive-response curve replaces the paper’s earlier hard threshold, and there is a seasoning hump around three years. The balance profile requires letting the incentive response vary with balance: small balances carry a several-fold higher background payoff rate from moves and payoffs of small remaining balances, and a specification that holds the incentive response common loads that background rate onto the refinancing attribution. The denial

gap persists in the modern reporting regime (Appendix Table 8). The SCF cross-sectional take-up measure—the fraction of each group’s in-the-money balance that refinanced, population-weighted over the five imputation implicates—is kept as a robustness exhibit. The GSE loan-month hazard is the primary take-up measure.



Appendix Figure 2: Estimated prepayment-hazard “take-up” response to the refinancing incentive (left) and cumulative twelve-month take-up by FICO and by loan balance (right), from the Freddie Mac loan-month panel. The left panel’s hazard rises steeply as the incentive turns positive, peaks near 1.5 percentage points in the money, and declines at deeper levels.



Appendix Figure 3: Refinance denial rates by race, 1990–2024 (HMDA LAR; the 2018 reporting-regime change is marked). Panel A: pooled denial rate by race. Panel B: the denial rate by race over time. Denial rate = denied / (originated + denied) among refinance applications with race reported. Appendix Table 8 reports the two reporting regimes separately.

Appendix Table 4: Sensitivity of the cumulative realized refinancing capital gain under the conservative representative-model per-dollar measure, 1990–2024 (\$ billions, constant 2026 dollars).

Variation	Cumulative gain (\$bn)
Baseline (terminal-balance-adjusted gain, 5% discount)	1878
Market discount (prevailing mortgage rate)	1873
All refinances, rate effect only (incl. cash-out)	2830
Option-theoretic gain (market value minus par)	3022
Naive annuity (no terminal-balance adjustment)	2277
Discount rate 3%	2052
Discount rate 7%	1720
Horizon 5 years	934
Horizon full term (30 years)	3048
Closing costs 0.5% + \$2,000	1992
Closing costs 1.5% + \$3,000	1673
Median loan age 2.5 years	1361
Median loan age 5.0 years	1951

Notes: Each row is the cumulative realized gain $\sum_t G_t$, 1990–2024, under one variation of the per-dollar gain measure, discount rate, horizon, closing costs, or median loan age of Appendix A.2. The baseline row here is the terminal-balance-adjusted measure (eq. (A.2)); the option-theoretic row (market value minus par, eq. (2)) is the paper’s headline payoff convention on this representative construction, and the naive-annuity row (no terminal-balance adjustment) is reported only to quantify the maturity-extension bias. This representative-model series is deliberately conservative and sits below the headline loan-level series of Figure 4; see Section 5.2. Generated by 03_layer1_aggregate.py.

Appendix Table 5: Cumulative realized gain 1990–2024 under alternative horizon, discounting, and loan-life conventions (\$ billions, constant 2026 dollars; representative-loan construction, rate-and-term volume).

Convention	Cumulative gain (\$bn)
Contractual exercise payoff: full-term market value minus par	3022
Cash-out inclusive (exercise payoff on all refs)	4514
Empirical loan life, market discount (survival-weighted use value)	1121
Empirical loan life, Treasury discount	1156
Fixed 10-year horizon, 5% nominal discount (legacy baseline)	1878
Fixed 10-year horizon, market discount	1873
Fixed 10-year horizon, Treasury discount	2017
Fixed 5-year horizon, 5% nominal discount	934

Notes: Every row uses the representative-loan construction (PMMS rate pair at the median loan age, rate-and-term volume unless noted), so no row matches the loan-level headline series, which sits above its convention’s row here because actual refinancers are in-the-money selected. The first row is the paper’s headline payoff convention (eq. (2)), the market value of the extinguished liability minus par and costs, with no discount-rate or horizon parameter. The empirical-loan-life rows measure the household use value instead, weighting each month’s saving by the probability the loan survives to it (hazard-implied curve where available). The fixed-horizon rows are the legacy present-value conventions; the ten-year 5% nominal row matches the baseline of Appendix Table 4. Generated by 16_waterfall.py.

Appendix Table 6: Origination-cohort ledger: option premium charged at origination vs. realized exercise payoffs, constant 2026 dollars.

Cohorts	Originations (\$tn)	Premium charged (\$tn)		Payoffs (\$tn)	Net of premium (\$tn)	
		Model	Market		Model V (upper bound)	Market (net gain)
Closed cohorts, 1990–2014	42	0.6	0.9	4.2	3.7	3.3
All cohorts, 1990–2024	66	1.0	1.8	5.3	4.3	3.4

Notes: Every originated loan (purchase and refinance; MBA origination estimates) is charged the origination-year option premium once, converted to an upfront dollar price over the loan’s expected life; realized existing-loan payoffs (the exercise payoff, eq. (2)) are attributed to origination cohorts by the median loan age at refinance. The origination column is in nominal dollars; premium, payoff, and net columns are constant 2026 dollars. Closed cohorts have at least a ten-year observed exercise window through 2024; the all-cohort row charges recent cohorts the full premium while their exercise window is still open, so it understates their eventual net gain. MBA totals include adjustable-rate originations, which carry little prepayment-option premium but are charged the full premium here, so both net columns are conservative. Netting the model value *V* is an upper bound on the net gain, not a price; the market column is the net gain. Payoffs attributed to pre-1990 cohorts are excluded. Generated by 17_cohort_net.py.

Appendix Table 7: Net incidence of the refinancing option by group: payoffs collected less premiums paid under two premium allocations, closed cohorts 1990–2014 (\$ billions, constant 2026 dollars).

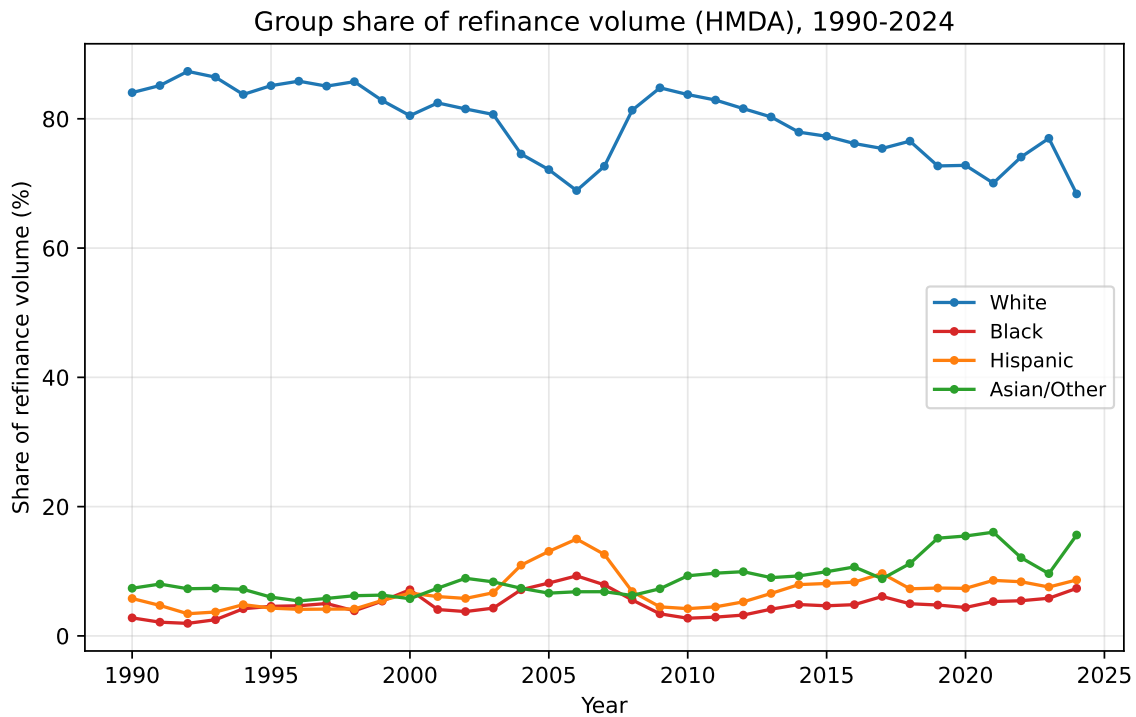
Group	Gain share (%)	Debt share (%)	Flow share (%)	Payoff (\$bn)	Net of model <i>V</i> (upper bound)		Net, market premium	
					Debt alloc	Flow alloc	Debt alloc	Flow alloc
White	78.8	78.0	79.3	3,331	+2,896	+2,896	+2,612	+2,600
Black	4.4	7.4	5.2	186	+145	+155	+118	+138
Hispanic	6.7	7.2	7.8	283	+243	+235	+217	+211
Asian/Other	10.1	7.4	7.7	427	+386	+384	+359	+356
Income Q1	8.5	10.5	6.9	359	+301	+320	+263	+295
Income Q2	13.0	13.6	13.4	550	+474	+475	+424	+426
Income Q3	17.3	16.9	16.7	731	+637	+637	+576	+578
Income Q4	24.0	21.6	23.9	1,015	+894	+882	+815	+794
Income Q5	37.1	37.3	39.1	1,568	+1,360	+1,351	+1,224	+1,208

Notes: Payoffs allocate the closed-cohort attributed exercise payoffs (Appendix Table 6) by each group’s HMDA gain share. Premiums are allocated two ways. The debt allocation charges each group its pooled SCF mortgage-debt share of the closed-cohort premium (model \$557bn, market \$922bn). The flow allocation charges each cohort year’s premium by that year’s group share of HMDA purchase-plus-refinance origination dollars, so premium exposure follows originations rather than the debt stock; the flow-share column reports the charge-weighted average of those annual shares. Income quintiles are within-year quintiles of purchase and refinance applicants respectively. Both allocations reflect pooled pricing: agency TBA execution charges every borrower in the pool the pool-average prepayment-option premium per dollar, regardless of the borrower’s own exercise propensity. Model and Market columns net the model-optimal and market-priced premium bases respectively. Netting the model value *V* is an upper bound on the net gain, not a price; the market columns are the net gain. Within-group heterogeneity (individual non-refinancers subsidizing refinancers inside every group) makes the person-level incidence more unequal than these group means. Generated by 26_net_capture.py.

Appendix Table 8: Refinance denial rates by race, 1990–2024 (HMDA).

	Applications (mn)	1990–2024	1990–2017	2018–2024
White	142.8	20.5%	21.2%	16.0%
Black	14.1	41.1%	42.5%	32.7%
Hispanic	15.9	33.1%	34.6%	25.8%
Asian/other	11.8	24.2%	26.2%	16.9%

Notes: Refinance applications acted upon—originated or denied—by race, from the HMDA Loan/Application Register: full-record local archives for 1990–2017 (loan purpose = refinance) and the CFPB data-browser API for 2018–2024 (purposes 31+32, matching the legacy definition; purpose-31-only rates are within 1 pp). The 2018–24 universe excludes open-end lines of credit and reverse mortgages, which the pre-2018 LAR did not report; race/ethnicity uses the same harmonized mapping as the volume-share analysis (Hispanic ethnicity takes precedence). Denial rate = denied / (originated + denied); race-not-reported applications excluded. The 2018–24 sub-period is reported separately because the reporting regime changed in 2018. Generated by 25_denials_combined.py.



Appendix Figure 4: Each group’s share of HMDA refinance dollar volume over time, 1990–2024; the companion to Figure 5’s allocated-gain levels.

Cash-out composition detail. The headline aggregate is on the existing-loan basis (rate-and-term loans in full plus the old-balance portion of cash-out refinances), and for 2018 onward the group allocation uses purpose-31 (rate-and-term) dollar shares; before 2018 HMDA does not distinguish purpose within refinances, so shares come from all refinance volume. Appendix Table 9 compares the two bases on 2018–24 data: reallocating G_t by total-refinance shares moves no group’s cumulative share by more than 1.6 percentage points. Black and Hispanic borrowers tilt toward cash-out (about 40% of their refinance dollars, versus 37% for white and 27% for Asian/other borrowers), so an all-refinance allocation slightly *overstates* their share of the rate-and-term gain. Applying the 2018–24 propensity gaps to the Freddie-era aggregate cash-out share (47% of 1994–2017 volume) bounds the pre-2018 overstatement of the Black and Hispanic allocated shares at roughly 10% of those shares (~0.5 percentage points of the total), so the main-text disparities are, if anything, slightly understated.

Appendix Table 9: Cash-out composition robustness: allocation of the 2018–24 rate-and-term gain under alternative share bases.

	Total-share alloc.	Rate-term alloc.	Δ share	Cash-out propensity
White	73.0%	72.0%	-1.07 pp	37%
Asian/other	13.6%	15.2%	+1.58 pp	27%
Hispanic	8.2%	7.9%	-0.37 pp	40%
Black	5.1%	5.0%	-0.14 pp	40%

Notes: HMDA 2018–24, dollar-volume weighted. “Total-share” allocates the annual rate-and-term gain G_t by each group’s share of all refinance volume (purposes 31+32), the only basis available before 2018; “rate-term” uses purpose-31 volume only (the basis actually used for 2018+). Cash-out propensity is the purpose-32 dollar share of each group’s total refinance volume. Freddie-era aggregate cash-out share (1994–2017 mean): 47%. Generated by 23_cashout_composition.py.

B Optimal Refinancing Threshold

The borrower holds a mortgage with current rate i and remaining balance M . Refinancing incurs a tax-adjusted cost κ . The mortgage rate follows a Brownian motion with volatility σ . The borrower has a real discount rate ρ (fixed at 0.05 in this appendix’s replication; the headline series uses the observable ρ_t), faces exogenous repayment at rate λ , and has a marginal tax rate τ .

The optimal threshold from Agarwal et al. (2013) Theorem 2 is:

$$x^* = \frac{1}{\psi} \left[\phi + W(-e^{-\phi}) \right], \quad (\text{A.5})$$

where $W(\cdot)$ is the Lambert- W function and:

$$\begin{aligned} \psi &= \frac{\sqrt{2(\rho + \lambda)}}{\sigma}, \\ \phi &= 1 + \psi(\rho + \lambda) \frac{C(M)}{M}, \\ \frac{C(M)}{M} &= \frac{\kappa/M}{1 - \tau}. \end{aligned}$$

The exogenous repayment rate λ is calibrated following Appendix C of Agarwal et al. (2013):

$$\lambda = \mu + \frac{i_0}{e^{i_0 \Gamma} - 1} + \pi,$$

where μ is the annual moving probability, i_0 is the original nominal rate, Γ is the remaining term, and π is inflation.

The tax-adjusted closing cost κ follows Appendix A of Agarwal et al. (2013):

$$\kappa = F + fM \left(1 - \frac{\tau}{\theta + r_d} \left[\frac{1 - e^{-(\theta + r_d)T_m}}{T_m} \cdot \frac{r_d}{\theta + r_d} + \theta \right] \right),$$

where F is the fixed closing cost, f is the points fraction, $\theta = \mu + 0.10$, $r_d = \rho + \pi$, and T_m is the mortgage term in years.

Using baseline parameters, the exact threshold reproduces Table 1 of Agarwal et al. (2013) (Appendix Table 10):

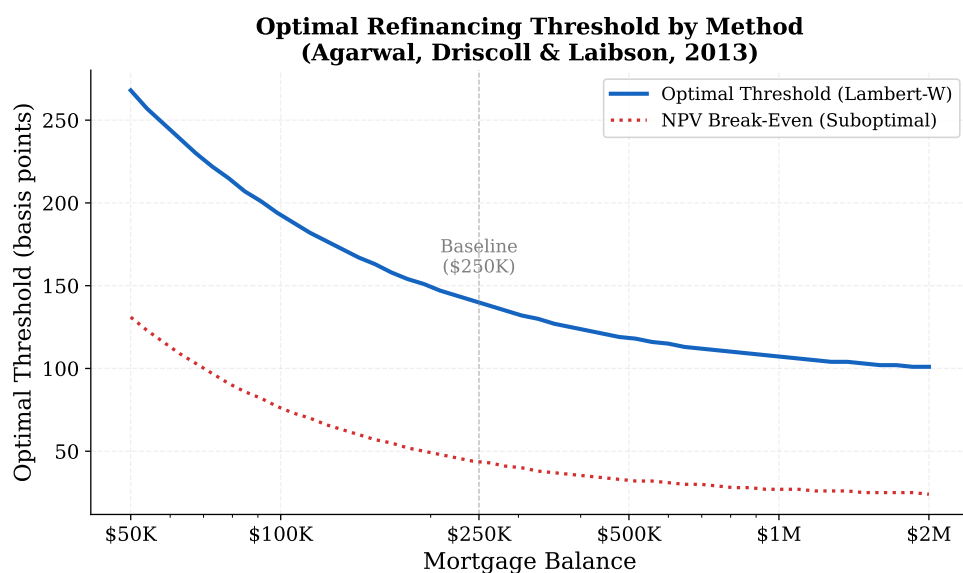
Appendix Figure 5 illustrates how both thresholds vary with mortgage balance. The gap between the exact threshold and the NPV break-even is the “option value of waiting”: for a \$250,000 mortgage, the exact threshold is 139 bps versus an NPV break-even of 44 bps, more than three times the naive level. Both thresholds decrease with mortgage size because fixed closing costs become proportionally smaller relative to the balance. With the January 2024 headline calibration

Appendix Table 10: Replication of Agarwal et al. (2013) Table 1: Optimal Refinancing Thresholds

Balance	Exact (bps)	NPV (bps)	Paper (bps)
\$100,000	193	76	193
\$250,000	139	44	139
\$500,000	118	33	118
\$1,000,000	107	27	107

Notes: Exact = Lambert-W closed-form threshold (Equation (A.5)). NPV = naive break-even threshold ignoring the option value of waiting, computed using the same tax-adjusted closing costs as the exact formula. Paper = exact threshold values reported in Table 1 of Agarwal et al. (2013). All thresholds in basis points; baseline parameters: $\rho = 0.05$, $\sigma = 0.0109$, $\tau = 0.28$, $\mu = 0.10$, closing costs = 1% + \$2,000.

(Section 2.5), the threshold falls to 100 bps because the larger balance and lower origination costs reduce per-dollar closing costs; setting $\Gamma = 30$ (instead of 25) has negligible effect on option values, and setting $\tau = 0$ (a non-itemizer) modestly increases the option value.



Appendix Figure 5: The exact optimal threshold (blue solid) is always well above the NPV break-even (red dotted) because the optimal rule accounts for the option value of waiting. Both thresholds decrease with mortgage size because fixed costs become proportionally smaller. Baseline parameters as in text.

C Term-Structure and Spread Dynamics

This appendix states the interest-rate and mortgage-spread model summarized in Section 2.4.

I model the short rate using the Hull–White one-factor model (Hull and White, 1990):

$$dr(t) = [\theta(t) - \alpha r(t)] dt + \sigma_{\text{HW}} dW(t), \quad (\text{A.6})$$

where α is the mean-reversion speed, σ_{HW} is the short-rate volatility, and $\theta(t)$ is calibrated analytically to fit the initial Treasury yield curve. This ensures that the model prices all observed bonds correctly at time zero, so simulated rate paths are consistent with the term structure observed at each origination date.

Mortgage rates along each simulated path are derived from model-implied 10-year Treasury yields using the Hull–White analytical bond pricing formula (Appendix E). At each time step, I compute the model-implied 10-year zero-coupon bond price $P(t, t + 10)$ and invert it to obtain the yield $y(t, 10) = -\frac{1}{10} \ln P(t, t + 10)$. The resulting yield path carries the full term-structure dynamics of the Hull–White model: mean reversion, forward-rate drift, and stochastic volatility.

Mortgage rates do not move one-for-one with Treasury yields. Regressing monthly changes in the 30-year PMMS rate on changes in the 10-year Treasury yield (post-1990, expanding window) gives a pass-through coefficient $\beta \approx 0.60$: a 100 bps decline in Treasury yields translates into roughly a 60 bps decline in mortgage rates. Part of this attenuation is mechanical survey smoothing. An error-correction specification that imposes the long-run one-for-one relationship between the two levels is reported with the spread-model robustness checks (Appendix Table 16).

The residual, the mortgage–Treasury spread, is persistent and stochastic. I model it as an AR(1) process estimated from the time series of $u_t \equiv \text{PMMS}_t - \beta \cdot \text{DGS10}_t$:

$$u_t = c + \varphi u_{t-1} + \sigma_\varepsilon \eta_t, \quad \eta_t \sim N(0, 1), \quad (\text{A.7})$$

$\varphi = 0.97$ is held fixed. The innovation standard deviation σ_ε is vol-targeted: calibrated each month so that total simulated mortgage-rate volatility matches the observed trailing 36-month PMMS

volatility (Appendix K.1). The vol-targeted σ_ε is close to the legacy expanding-window estimate of roughly 16 bps per month in most periods; it diverges mainly in the 2020s, when the $\beta < 1$ pass-through compressed the Treasury-driven component of mortgage-rate movements. The intercept c pulls the spread back toward its historical average rather than anchoring it permanently at the origination-date level, and that average is estimated recursively. Like β and σ_ε , the intercept and the implied long-run mean $c/(1 - \varphi)$ come from an expanding window ending at the origination month, so no spread parameter uses information from after the origination date.

The mortgage rate along each simulated path is then:

$$m(t) = m(0) + \beta[y(t, 10) - y(0, 10)] + (u_t - u_0), \quad (\text{A.8})$$

where $u_0 = m(0) - \beta \cdot y(0, 10)$ is the initial spread residual. The path starts at the observed contract rate $m(0)$ and moves with Treasury yields, damped by $\beta < 1$, while the spread term (eq. (A.7)) lets it wander around its long-run mean. The short-rate volatility σ_{HW} is calibrated to the volatility of the 10-year Treasury yield alone, not total mortgage-rate volatility, so the Hull–White model supplies only the pure interest-rate component; the AR(1) process absorbs the rest.

For each month in the sample, the model is calibrated to the observed yield curve (a zero-coupon curve bootstrapped from up to 11 constant-maturity Treasury quotes and fit with a cubic spline), with $\alpha = 0.05$ and short-rate volatility σ_{HW} calibrated analytically so that the model-implied volatility of 10-year Treasury yields matches the observed trailing 36-month estimate. Full calibration details follow in Appendix D; Appendix K plots the mean simulated path against the forward curve implied by the initial term structure and verifies the volatility calibration.

The Hull–White drift is calibrated to the observed forward curve. The Agarwal et al. (2013) threshold and the discounting use the observable real 10-year Treasury yield ρ_t in place of a time-preference parameter, following the market-based discounting of the MBS-valuation literature (Stanton, 1995): the rate dynamics match the term structure at origination, and the discounting reflects the market opportunity cost of funds.

D Hull–White Calibration

The time-dependent drift $\theta(t)$ in Equation (A.6) is:

$$\theta(t) = \frac{\partial f}{\partial t}(0, t) + \alpha f(0, t) + \frac{\sigma_{\text{HW}}^2}{2\alpha} (1 - e^{-2\alpha t}),$$

with $f(0, t)$ the instantaneous forward rate derived from observed Treasury yields.

For each month, the zero-coupon curve is bootstrapped from the observed constant-maturity Treasury quotes at up to 11 maturities. Constant-maturity yields are par-equivalent, semiannual-coupon yields, not zero-coupon rates. Sub-six-month bill quotes are converted directly (bond-equivalent to continuous compounding); coupon-bearing maturities are bootstrapped on a semiannual payment grid, which reprices every observed par bond exactly. A natural cubic spline is then fit through the resulting zero yields.

Beyond the last observed maturity (30 years, or 20 years where the 30-year series is unavailable), the curve is extrapolated at a constant forward rate equal to the average forward over the trailing ten-year segment. Simulated bond prices reach 40-year maturities under the survival-weighted horizon, but the extrapolated segment enters only through bond prices at simulation dates beyond year 20, which carry little discounted, survival-weighted weight: the entire curve-construction change, extrapolation included, moves the mean option value by 0.6 basis points. The 20-year constant-maturity series is masked during its 1987–September 1993 discontinuation, when the data source carries a stale forward-filled quote. An earlier version of this paper instead treated the par quotes directly as zero yields; that approximation raises the mean option value by 0.6 basis points (Appendix L.1).

The instantaneous forward rate is computed by numerical differentiation: $f(0, T) \approx [(T + \Delta T) \cdot y(T + \Delta T) - T \cdot y(T)] / \Delta T$ with $\Delta T = 0.01$ years. The mean-reversion speed is fixed at $\alpha = 0.05$, a half-life of about 14 years. The short-rate volatility σ_{HW} is calibrated so that the model-implied annualized volatility of 10-year Treasury yields matches the observed trailing 36-month estimate $\hat{\sigma}_T$. Using the Hull–White bond-pricing formula, the annualized volatility of the model-implied 10-year yield is $\sigma_{\text{HW}} \cdot B(10)/10$, where $B(T) = (1 - e^{-\alpha T})/\alpha$ (equal to $B(0, T)$ in the notation of

Appendix E). Setting this equal to $\hat{\sigma}_T$ gives $\sigma_{\text{HW}} = \hat{\sigma}_T \cdot 10/B(10)$; for $\alpha = 0.05$, the multiplier is $10/7.87 \approx 1.27$. Paths are generated using Euler discretization with monthly time steps, fully vectorized over 10,000 paths.

E Model-Implied 10-Year Yield

The Hull–White model admits closed-form zero-coupon bond prices. The price at time t of a bond maturing at T is:

$$P(t, T) = A(t, T) \exp[-B(t, T) r(t)],$$

where

$$B(t, T) = \frac{1 - e^{-\alpha(T-t)}}{\alpha},$$

$$\ln A(t, T) = \ln \frac{P^M(0, T)}{P^M(0, t)} + B(t, T) f(0, t) - \frac{\sigma_{\text{HW}}^2}{4\alpha} (1 - e^{-2\alpha t}) B(t, T)^2,$$

with $P^M(0, T)$ the market-observed discount factor and $f(0, t)$ the instantaneous forward rate.

The model-implied 10-year yield at simulation step t is:

$$y(t, 10) = -\frac{1}{10} \ln P(t, t + 10).$$

Feeding this into Equation (A.8) carries the same mean reversion, forward-rate drift, and stochastic volatility into the mortgage rate—not just the current short rate plus a constant.

F Simulation and the PV-Equivalence Condition

This appendix details the Monte Carlo simulation and the formal definition of the option-adjusted cost rate summarized in Section 2.4.

For each month in the sample, I simulate 10,000 monthly short-rate paths over a 30-year horizon, convert each to a mortgage rate path using the bond pricing formula, and apply the Agarwal et al. (2013) threshold at each monthly step. I do not truncate at an arbitrary fixed horizon. Instead, I

weight each month's cash flows by the probability that the mortgage is still outstanding—the survival function $S(t) = e^{-\mu_{\text{tot}} t}$ implied by the exogenous (non-refinancing) termination hazard μ_{tot} —so the effective horizon is the borrower's expected loan life. (In the discrete equation below, $S_k = S(k/12)$.) The relevant hazard is the rate at which the loan terminates without refinancing, since refinancing resets the rate but keeps the borrower in the mortgage.

The model-implied expected life rises from about 8.5 years in the early 1990s to 12.4 years in the 2020s as homeowner mobility falls (Appendix P, Appendix Figure 15), between the seven-year MBS weighted-average-life convention, an industry rule of thumb, and the sixteen-year tenure implied by mobility alone. Appendix H confirms that results are well converged by 10,000 paths (Monte Carlo standard error $\approx \pm 1$ bp). A fixed seven-year life lowers the option value and a thirteen-year life raises it (Appendix Figure 9, Appendix Table 17), but neither changes the qualitative time-series pattern. The borrower refinances when her current rate exceeds the market rate by at least x^* : her rate resets to the market rate, she pays closing costs out of pocket, and the mortgage term resets to 30 years with a new amortization schedule.

For each simulated path, the borrower's actual monthly cash flows are computed: at origination (and after each refinance), the monthly payment is recomputed for a fresh 30-year amortization at the new rate on the outstanding balance, and closing costs are recorded as additional cash flows. For the cost-rate calculation, nominal mortgage cash flows are discounted at the observable nominal 10-year Treasury yield $r_d(t) = y_{10}^{\text{Tsy}}(t)$ prevailing at origination, in place of a subjective rate; the real discount rate entering the Agarwal et al. (2013) threshold is correspondingly the real 10-year Treasury yield $\rho_t = y_{10}^{\text{Tsy}}(t) - \mathbb{E}_t[\pi_{10}]$ (Cleveland Fed expected inflation), so that $\rho_t + \mathbb{E}_t[\pi_{10}] = r_d(t)$. The parameters π , τ , and μ continue to enter the Agarwal et al. (2013) threshold as in the refinancing rule, with ρ replaced by the observable ρ_t . Because the threshold formula requires a positive real discount rate, ρ_t is floored at 0.25%; the floor binds in 52 of the 438 months (2011–2013 and 2020–2021, when the real ten-year yield reaches -0.8%), and removing it entirely changes the option value in those months by at most 0.2 basis points, so the floor is a numerical safeguard rather than a substantive assumption. The threshold itself is evaluated at the loan's origination state (initial balance, contract rate, and remaining term) and is not recomputed as the balance amortizes or after

a refinance resets the contract; a dynamic variant that recomputes the threshold each month from the live loan state is reported with the robustness checks (Appendix L).

Formally, the option-adjusted cost rate c^* is the constant annual mortgage rate such that the survival-weighted present value of payments and termination payoffs on a hypothetical no-option mortgage at c^* equals the mean survival-weighted present value of actual payments, refinancing costs, and termination payoffs across all simulated paths:

$$\begin{aligned} & \sum_{k=1}^T S_{k-1} \text{PMT}_k(c^*) e^{-r_d k/12} + \sum_{k=1}^T (S_{k-1} - S_k) B_k(c^*) e^{-r_d k/12} + S_T B_T(c^*) e^{-r_d T/12} \\ &= \mathbb{E} \left[\sum_{k=1}^T S_{k-1} \text{CF}_k(\omega) e^{-r_d k/12} + \sum_{k=1}^T (S_{k-1} - S_k) B_k(\omega) e^{-r_d k/12} + S_T B_T(\omega) e^{-r_d T/12} \right], \end{aligned} \quad (\text{A.9})$$

where $\text{PMT}_k(c)$ is the scheduled monthly payment under the counterfactual no-option mortgage, $B_k(\cdot)$ is the outstanding balance at month k , and $\text{CF}_k(\omega)$ is the borrower's actual cash flow at month k along path ω , including any refinancing costs. $S_k = e^{-\mu_{\text{tot}} k/12}$ is the probability that the loan is still outstanding after k months: the month- k payment is weighted by survival to the start of the month, and with probability $S_{k-1} - S_k$ the loan terminates and its balance is settled. T is the 30-year horizon in months, and $r_d = y_{10}^{\text{Tsy}}$ is the nominal Treasury discount.

I solve this numerically via bisection. Both sides are survival-weighted and discounted at the same observable nominal Treasury rate, so this is a PV-equivalence condition, not an IRR. The option-adjusted cost rate is the single fixed rate at which the borrower would be indifferent between (a) paying the contract rate $m(0)$ with the option to refinance optimally and (b) paying a constant rate c^* with no option. The option value is then $V = m(0) - c^*$, expressed in basis points. Appendix G reports a simpler rate-average measure and the legacy payment-only PV equivalent as comparisons.

All expectations are conditioned on the current yield curve and volatility estimate. The Hull–White model is calibrated to the observed term structure, so the option-adjusted cost rate reflects forward-curve-consistent rate dynamics conditioned on current bond prices.

G Rate-Average Comparison

The primary metric throughout the paper is the terminal-balance-adjusted PV-equivalent option-adjusted cost rate c^* (Equation (A.9)), which accounts for payments, refinancing costs, amortization resets, and the remaining balance at the horizon. As a simpler alternative, I compute a *rate-average* measure: the discount-weighted average of the coupon rates actually paid along each path,

$$c^{\text{RA}} = \frac{\mathbb{E} \left[\int_0^H m_s(\omega) e^{-\rho s} ds \right]}{\int_0^H e^{-\rho s} ds},$$

where $m_s(\omega)$ is the mortgage rate paid at time s along path ω and ρ is the borrower's discount rate (a fixed rate here, in contrast to the observable ρ_t of the headline configuration). This measure captures the direct interest-rate savings from refinancing. Closing costs enter only indirectly, through the optimal threshold; they are not included as separate cash flows.

Across the full 438-month sample, the rate-average and terminal-balance-adjusted PV measures are highly correlated, but their levels can differ materially, and the gap widens in high-option-value scenarios. For January 2024, the rate-average measure gives an option value of 180 bps; the terminal-balance-adjusted PV-equivalent gives 125 bps. The gap comes from amortization: each refinancing resets the schedule to a fresh 30 years, which temporarily raises the interest share of early payments. The rate-average measure just weights the rate paid at each point in time by the discount factor, so it misses this mechanical effect. The PV-equivalent catches it, along with the full cash-flow consequences of refinancing—closing costs, amortization resets, and the remaining loan balance at the horizon—which is why it is the preferred measure.

H Monte Carlo Convergence

To verify convergence, I ran the simulation for October 2023 (contract rate 7.31%) with varying path counts (Appendix Table 11):

Results are well converged by 10,000 paths, with the Monte Carlo standard error approximately ± 1 bps.

Appendix Table 11: Monte Carlo Convergence (*fixed_borrower* configuration, October 2023)

Paths	Mean Effective Rate (%)	Option Value (bps)
1,000	6.41	90
5,000	6.42	89
10,000	6.42	89
50,000	6.42	89

Notes: Simulation uses the October 2023 yield curve (contract rate 7.31%). The mean rate shown is the terminal-balance-adjusted PV-equivalent option-adjusted cost rate. Option value = contract rate minus this mean rate, in basis points. Fixed borrower parameters as in Appendix J (“Fixed” column).

I Historical Borrower Calibration

This appendix documents the time-varying borrower calibration summarized in Section 2.5. The Agarwal et al. (2013) threshold depends on four borrower-side parameters that changed substantially over the sample period: the mortgage balance M , the effective tax rate on mortgage interest τ , the annual moving probability μ , and origination costs (points plus fixed costs). I calibrate each to publicly available time-varying data instead of fixing them at representative-borrower values, so the threshold and option value reflect the conditions borrowers actually faced at origination.

The average GSE loan balance, from the FHFA Monthly Interest Rate Survey spliced with the National Mortgage Database, rose from roughly \$102,000 in 1990 to \$350,000 in 2024. A larger balance reduces the per-dollar impact of fixed closing costs and lowers the refinancing threshold, making the option more valuable.

Origination fees and discount points come from the FRED series MORTPTS30US, which reports the combined fee lenders charge as a percentage of the loan amount.⁸ Points declined from about 2% in the early 1990s to 0.5–0.7% after 2005. Lower origination costs reduce the threshold and raise the option value. The survey contract rate is quoted conditional on these points, so the initial points are already embedded in $m(0)$; they re-enter only as out-of-pocket costs at future refinances. The option value therefore measures the cost of exercising the option, not the cost of originating the loan—it is distinct from the APR.

⁸This series is paired by construction with the PMMS: the reported contract rate is conditional on the borrower paying the reported points. The series was discontinued in November 2022 when Freddie Mac changed the PMMS methodology; post-2022 values are extrapolated at the 2022 average ($\approx 0.7\%$).

The annual moving probability μ is measured by the Census CPS/ASEC homeowner mobility rate: the fraction of persons in owner-occupied housing who report having moved in the preceding year.⁹ Homeowner mobility fell from about 9% in the early 1990s to 4–5% in recent years. After adding the non-relocation hazard δ and survival-weighting the cash flows, this implies expected loan life rose from about 8.5 years in the early 1990s to 12.4 years in the 2020s (Appendix P, Appendix Figure 15). A lower μ raises the option value: a longer expected stay gives the borrower more time to encounter a favorable refinancing opportunity.

The 2023 CPS rate of 4.1% partly reflects the “lock-in” of existing low-rate borrowers (Batzer et al., 2024). A new high-rate borrower who is not locked in would move sooner, shortening expected life toward the seven-year case in Appendix Table 17 and lowering the option value. The headline, which uses observed mobility, is if anything an upper bound for such a borrower.

The effective marginal tax rate on mortgage interest is computed as $\tau_{\text{eff}}(t) = \Pr(\text{itemize} \mid \text{mortgage}, t) \times \text{MTR}_{\text{itemizers}}(t)$, combining NBER TAXSIM marginal rates (Feenberg and Coutts, 1993) with IRS SOI itemization data. Before the Tax Cuts and Jobs Act, roughly 73–78% of mortgage holders itemized at marginal rates of 19–23% (from this TAXSIM/SOI construction; on the tax treatment of owner-occupied housing see Poterba and Sinai, 2008), yielding $\tau_{\text{eff}} \approx 0.15$ –0.18. After the TCJA doubled the standard deduction in 2018, only about 27% itemize, yielding $\tau_{\text{eff}} \approx 0.065$. A lower tax rate reduces the tax benefit of mortgage interest and raises the effective cost of refinancing, modestly increasing the threshold.

At each future refinancing event within a simulation path, the borrower is assumed to face the same per-dollar origination cost and fixed closing cost (\$2,000) as at the original origination. The discount rate is the observable real 10-year Treasury yield (Appendix F), time-varying over the sample; the remaining constants are inflation ($\pi = 0.03$, which enters only the threshold’s tax term) and remaining term ($\Gamma = 25$). Appendix J reports the full parameter set, and Appendix Table 20 quantifies the contribution of each time-varying parameter.

⁹The CPS series measures relocations directly; I add a constant $\delta = 0.025$ to capture the remaining non-refinancing terminations (in-place payoff, default, death) so that $\mu_{\text{tot}} = \mu_t + \delta$ is the full exogenous hazard of leaving the mortgage in the Agarwal et al. (2013) framework. This same hazard sets the survival-weighted horizon of Appendix F. See Molloy et al. (2011) for trends; Frost (2020) and Batzer et al. (2024) for recent declines.

J Baseline Parameters

Several exhibits report results under named model configurations. Appendix Table 12 defines each configuration once; table notes elsewhere refer to these names rather than re-describing parameters. Appendix Table 13 reports the baseline parameter values.

Appendix Table 12: Configuration glossary

Configuration	Definition
<i>headline</i>	Time-varying borrower calibration (balance, tax rate, mobility, and points from CPS/SCF/GSE series); observable real 10-year Treasury discount ρ_t ; nominal-Treasury PV discount; survival-weighted horizon with hazard $\mu_t + \delta$ ($\delta = 0.025$); PV-equivalent measure; 10,000 paths; January 1990–June 2026. The configuration behind every headline number.
<i>calibration_only</i>	Time-varying borrower calibration as in <i>headline</i> , but with the legacy subjective discount $\rho = 0.05$, PV discount $\rho + \pi = 0.08$, and a fixed 10-year horizon. Isolates the calibration channel from the discount and horizon changes.
<i>fixed_borrower</i>	Legacy baseline: constant borrower parameters ($\rho = 0.05$, $\tau = 0.28$, $\mu = 0.10$, $\Gamma = 25$, \$250K balance), fixed 10-year horizon. Matches Agarwal et al. (2013)-style calibrations.
<i>backtest_window</i>	The <i>headline</i> configuration restricted to 1990–2016 origination cohorts with full 10-year realized windows ($N = 318$)—the cohorts for which the realized-path comparison is feasible.

Appendix Table 13: Baseline Parameters

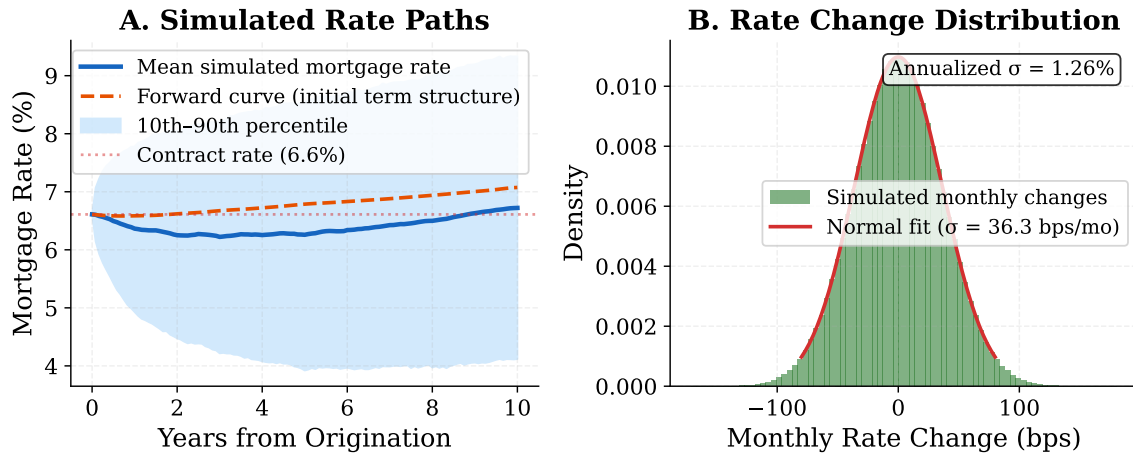
Parameter	Description	Fixed	Time-varying
ρ_t	Real 10-year Treasury yield	—	−0.8–4.9%
δ	Non-relocation hazard add-on	0.025	—
τ	Effective tax rate	0.28	0.065–0.18
μ	Annual moving probability	0.10	0.04–0.09
π	Inflation rate	0.03	—
Γ	Remaining mortgage term (years)	25	—
i_0	Original nominal rate	Contract rate	
M	Mortgage balance	\$250K	\$102K–\$365K
f	Refinancing points	1%	0.5–2.0%
F	Fixed closing cost	\$2,000	—
α	Hull–White mean reversion	0.05	—
N_{sim}	Simulation paths	10,000	
Horizon	Survival-weighted expected life	8.3–13.1 years (output)	

Notes: The “Fixed” column shows values used for the Agarwal et al. (2013) replication (Appendix B) and the sensitivity analysis (Appendix Figure 9). The “Time-varying” column shows the range of values used in the main historical analysis, calibrated to data as described in Section 2.5. The M and Horizon ranges are full-sample min–max, so they need not match the decade means quoted in the text. A “—” indicates that the parameter is held constant. Sensitivity to each parameter is examined in Appendix L.2.

K Model Validation

Appendix Figure 6 presents two diagnostic checks on the Hull–White simulation.

Model Validation: Hull-White Simulation Diagnostics



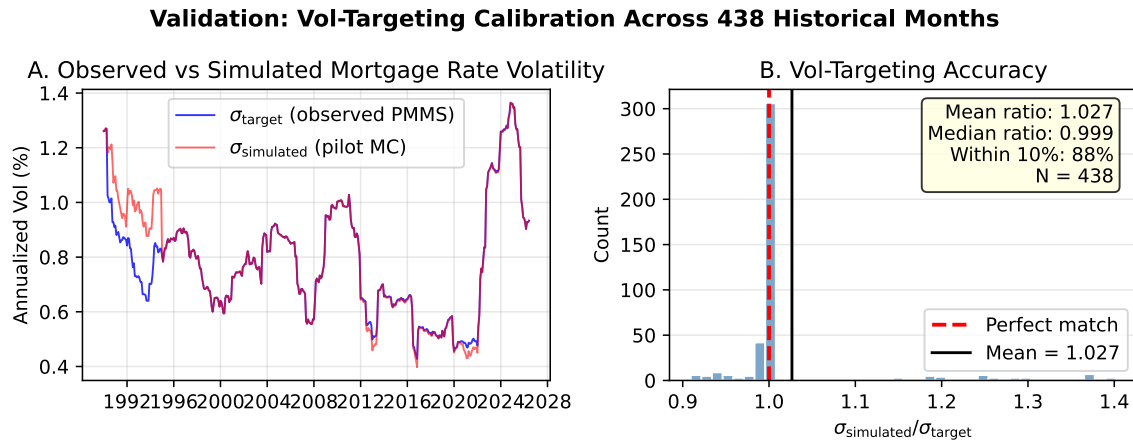
Appendix Figure 6: *Left:* Mean simulated mortgage rate path (blue) with the 10th–90th percentile fan, anchored at the January 2024 contract rate (6.61%). The dashed orange line shows the forward 10-year mortgage rate implied by the initial term structure, computed as $-\frac{1}{10} \ln[P(0, t+10)/P(0, t)] + s$. Two offsetting forces move the mean path relative to the forward curve: a risk-neutral convexity adjustment pushes it above, and mean reversion of the AR(1) spread from its elevated January 2024 level pulls it below; the net gap at year 10 is about –35 bps. *Right:* Distribution of simulated monthly mortgage rate changes across all 10,000 paths over a 10-year horizon. The histogram closely matches a normal distribution, consistent with the Gaussian diffusion assumption, and the printed annualized volatility (1.26%) equals the calibrated target.

The left panel shows the mean simulated path anchored at the observed contract rate. Two forces move it relative to the forward curve implied by the initial term structure: risk-neutral convexity pushes the mean above the forwards, and mean reversion of the elevated January 2024 spread pulls it down, leaving the mean about 35 bps below the forward curve at year 10. The 10th–90th percentile fan shows how much uncertainty builds up in rate paths over a 10-year horizon.

The right panel shows the distribution of monthly rate changes, which is close to Gaussian, consistent with the Hull–White diffusion assumption. The printed annualized volatility of the simulated changes, 1.26%, matches the trailing 36-month calibration target; Appendix K.1 verifies the match month by month across the full sample.

K.1 Volatility Calibration Validation

Vol-targeting calibrates the AR(1) spread innovation variance so that total simulated mortgage rate volatility matches the observed trailing 36-month PMMS volatility. Appendix Figure 7 checks the calibration across all 438 historical months. For each month I simulate 5,000 pilot paths, discard the first 36 months of each path as AR(1) burn-in, and compare the annualized standard deviation of simulated monthly mortgage rate changes to the target.

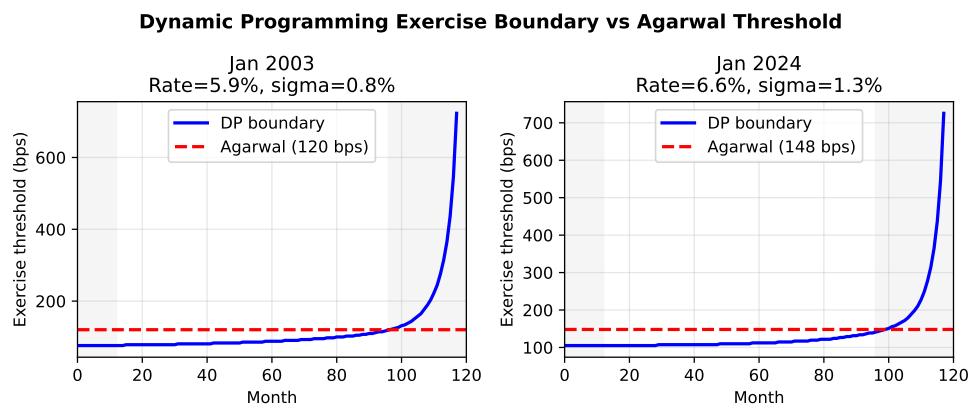


Appendix Figure 7: Panel A: time series of target mortgage rate volatility (trailing 36-month PMMS) versus simulated mortgage rate volatility from 5,000 pilot paths. Panel B: distribution of the ratio $\sigma_{\text{sim}}/\sigma_{\text{target}}$ across all 438 months. Median ratio is 1.00; 88% of months fall within 10% of unity.

The median ratio of simulated to target volatility is 1.00, with 88% of months falling within 10% of the target. Post-2000, the calibration is essentially exact (decade mean ratios of 1.00, 0.99, and 0.98 for the 2000s, 2010s, and 2020s respectively). The 1990s show modest over-simulation (mean ratio 1.12) because the expanding-window pass-through estimate $\hat{\beta}$ equals 1.0 in the earliest months when insufficient data are available to estimate $\beta < 1$; with a full pass-through there is no spread component to vol-target. Forcing $\beta = 0.60$ for the first 36 months (consistent with the full-sample estimate) lowers the 1990s decade mean option value by 14 bps but changes the full-sample mean by less than 4 bps.

K.2 Optimal Threshold Verification

The Agarwal et al. (2013) closed-form threshold is a constant approximation to the time-varying boundary that solves the borrower’s stopping problem. To check it, I solve the discrete dynamic programming (DP) problem directly: backward induction with a Gaussian transition matrix, on a grid of 500 rate-drop values (−400 to +800 bps) over 120 monthly time steps.



Appendix Figure 8: DP exercise boundary (solid, time-varying) versus Agarwal constant threshold (dashed) for two representative dates. The DP boundary is lower in the interior (months 12–96) and rises near the terminal date as the remaining horizon shrinks; gray shading marks the first 12 and last 24 months, where endpoint effects dominate. The Agarwal threshold is shown under the *fixed_borrower* configuration (Appendix Table 12).

In the interior of the horizon (away from endpoints), the DP boundary sits about 30–35 bps below the Agarwal threshold: the Agarwal rule is conservative, triggering refinancing slightly later than optimal. Near the terminal date the DP boundary rises, since the shrinking horizon reduces the option value of waiting—the “squeeze” visible in Appendix Figure 8. A constant threshold set at the DP time-averaged boundary raises the option value by about 2 bps for January 2003 (low-rate environment) and 5 bps for January 2024 (high-rate environment). The Agarwal closed-form threshold is a reasonable approximation, and it slightly understates the option value.

K.3 Alternative Rate Model: CIR++

The Hull–White model allows negative interest rates, which could in principle distort simulated rate dynamics and the option value, especially when rates are high. To check sensitivity to the

rate process, I implement a CIR++ (shifted Cox–Ingersoll–Ross) alternative: closed-form bond pricing plus a deterministic shift function $\xi(t)$ that matches the observed yield curve exactly. CIR++ enforces positive short rates and a square-root volatility structure. I recompute the option value for every month of the sample at the headline calibration, feeding CIR++-generated rate paths through the same simulation engine, survival-weighted horizon, and discounting as the headline series, so the two models differ only in the rate process. Appendix Table 14 summarizes the comparison.

Appendix Table 14: Hull–White vs. CIR++ option values, full sample

Statistic (438 months, headline calibration)	Value
Correlation of the HW and CIR++ option-value series	0.997
Mean absolute monthly difference	2.9 bps
Largest single-month difference	15.2 bps
Mean 2022–24 difference ($ \text{CIR++} - \text{HW} $)	1.5 bps

Notes: CIR++ supplies the model-implied rate paths, which are then run through the same Monte Carlo engine, calibration, survival-weighted horizon, and discounting as the Hull–White headline, so the two models differ only in the rate process; the Hull–White column reproduces the headline series month for month. Option value is the terminal-balance-adjusted PV measure, 10,000 paths per month, 1990–2026. CIR++ enforces positive short rates. Generated by `12_cir_alternative.py -full`.

Across all 438 months the two option-value series are nearly identical: they correlate 0.997, with a mean absolute monthly difference of 2.9 bps. CIR++, which rules out the negative rates the Gaussian model permits, produces marginally *lower* values—by 1.5 bps on average over 2022–2024, and only a few basis points even in the highest-rate months. The Gaussian rate assumption does not inflate the headline, and any difference is well inside the spread-model uncertainty (Appendix Table 16). The choice of rate process does not matter for the headline series.

K.4 Mechanical and Discretization Checks

I verify three discretization choices with spot checks at representative dates. Replacing the 10-year Treasury (DGS10) with the 7-year (DGS7) as the rate anchor changes option values by 0.0 bps across three test dates—the yield curve fitting interpolates identically. Weekly time steps ($dt = 1/52$) instead of monthly ($dt = 1/12$) move option values by at most 1.6 bps, so monthly discretization is adequate. Swapping the 36-month baseline for 24-month or 60-month trailing volatility windows shifts option values by ± 5 bps at individual dates, but the qualitative pattern does not change.

L Robustness

This appendix collects the paper’s robustness evidence in one place; Appendix Table 15 is the summary index.

The option value behaves as option pricing predicts. It rises with rate volatility, the strongest driver, and with the rate level, the loan balance, and the expected horizon. It falls with closing costs and with exercise frictions. These qualitative patterns follow from option logic.

The more serious concern is whether the level of the series depends on modeling choices. The largest source of uncertainty is the assumed behavior of the mortgage–Treasury spread, followed by the discount rate, the borrower horizon, and the rate-model calibration. One-at-a-time sensitivity curves appear in Appendix Figure 9. The plausible spread-model variants span 29–46 bps under the headline configuration, with cross-month correlations of 0.91 or higher; the deliberately extreme fixed- β stress tests stay at 35–37 bps at lower correlations (Appendix Table 16). The discount and horizon assumptions bracket the headline between 21 and 30 bps (Appendix Table 17). The headline sits at the conservative end of the spread range. The expanding-window pass-through uses only information available at origination, at the cost of attenuating β ; error-correction variants that correct the attenuation directly give 39–46 bps. The rate-model, threshold, volatility, and discretization checks (Appendix K) each move the value by only a few basis points.

L.1 Full-Sample Specification Reruns

Four structural concerns are checked by rerunning the entire 438-month simulation under the alternative specification and comparing month by month (cross-month correlations 0.999 or higher in every case). First, the refinancing threshold: the baseline evaluates the Agarwal et al. (2013) threshold at the loan’s origination state, but the economically relevant cost-to-balance ratio rises as the balance amortizes and the contract resets after each refinance. Recomputing the threshold every month from each path’s live state (current balance, current contract rate, and remaining term) raises the balance-weighted average threshold from 106 to 116 basis points and lowers simulated refinancing counts by about 18%, but moves the option value by 0.1 basis points on average (at most

Appendix Table 15: Other Robustness Checks

Concern	Test (Location)	Impact
Gaussian rate model	CIR++ alternative (App. K.3)	Mean $ \Delta $ = 2.9 bps
Threshold accuracy	Discrete DP verification (App. K.2)	OV understated 2–5 bps
Static threshold	Dynamic per-path recomputation, full-sample rerun (App. L.1)	Mean $ \Delta $ = 0.1 bps; refis –18%
Treasury curve construction	Legacy par-as-zero vs. baseline bootstrap, full-sample rerun (App. L.1)	Mean Δ = +0.6 bps
Real-rate floor	Floor removed, full-sample rerun (App. L.1)	≤ 0.2 bps in 52 binding months
Information timing	Calibration series lagged 12m, full-sample rerun (App. L.1)	Mean Δ = –0.7 bps
Vol calibration	Sim. vs realized vol (App. K.1)	Median ratio 1.00
Parameter sensitivity	Sensitivity curves at Jan 2024 (App. L.2)	Volatility grid 33–208 bps; other parameters within ± 30 bps
Tenor and timestep	Spot checks (App. K.4)	< 2 bps
Volatility window	24m vs 36m vs 60m (App. K.4)	± 5 bps at individual dates

Notes: “Mean $|\Delta|$ ” is the mean absolute difference in option value between the alternative and baseline models. Full details in each referenced appendix.

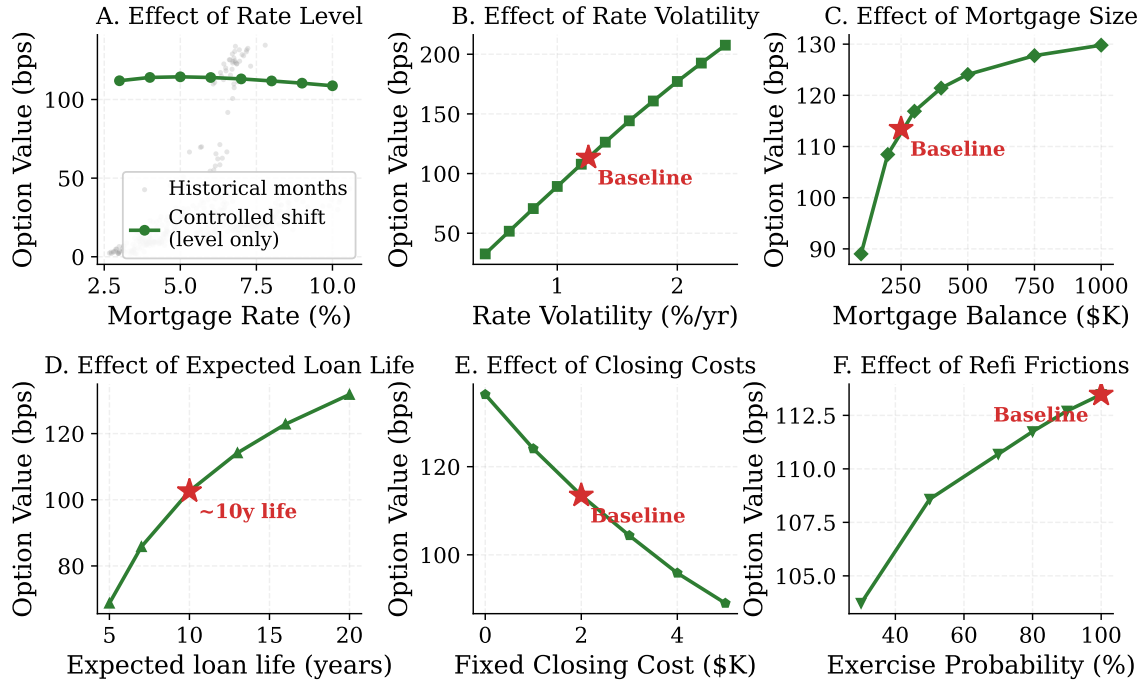
0.5 in any month): the value is insensitive because the marginal refinances that a higher threshold delays carry small gains. Second, the Treasury curve: the baseline bootstraps a zero-coupon curve from the constant-maturity par quotes (semiannual coupons, flat-forward extrapolation beyond the last observed maturity); treating the par quotes directly as zero yields, the approximation in an earlier version of this paper, shifts the mean option value by +0.6 basis points. Third, the 0.25% floor on the real discount rate binds in 52 months and matters at most 0.2 basis points. Fourth, information timing: the annual borrower-calibration series are interpolated around mid-year anchors, which lets early months of a year draw partially on that year's annual value; lagging every calibration series by twelve months, so that no month uses a value not yet observable, shifts the mean by −0.7 basis points. A related data correction is applied in the baseline itself: the 20-year constant-maturity series was discontinued from 1987 to September 1993, and the forward-filled stale quote is masked over that window rather than fitted (this matters up to nearly 30 basis points in single months of 1990–1993).

L.2 Parameter Sensitivity

Appendix Figure 9 shows how the option value varies with six key parameters, each varied individually from baseline.

Holding volatility, the yield curve's shape, and the borrower inputs at baseline, the contract-rate level alone moves the option value within a narrow band, about 109–114 bps across contract rates of 3–10% (Panel A): the strong association between option value and rate level in the historical series, shown in gray in the panel and in Appendix Figure 16, runs through the volatility and calibration conditions that high-rate months carry, not the level in isolation. Rate volatility (Panel B) has the strongest effect, mirroring the standard result from option pricing that volatility benefits the option holder. The exercise scales volatility in the simulated rate process and in the refinancing threshold together, holding the Treasury and spread mix fixed. Higher volatility raises the waiting threshold, but the added value of rate variability dominates, and the option value rises from about 33 basis points at the lowest volatility shown (0.4% per year) to about 208 at the highest (2.4%). Larger mortgage balances (Panel C) reduce the proportional impact of fixed closing costs, lowering the

Sensitivity of Prepayment Option Value



Appendix Figure 9: Sensitivity of the option value at the January 2024 calibration. All panels use a representative borrower (\$250K balance, $\tau = 0.28$, 1% points) with the headline discounting and survival-weighted horizon on the January 2024 yield curve (mortgage-rate volatility 1.26%); the resulting baseline of about 113 bps sits between the *calibration_only* and *headline* configurations (Appendix Table 12). Panel A: rate level; the solid line shifts the contract rate and yield curve together, holding every other input fixed, and the gray points are the 438 historical months for contrast. Panel B: rate volatility, varied consistently in the simulated rate process and in the refinancing threshold. Panel C: mortgage balance. Panel D: horizon. Panel E: closing costs (the fixed component). Panel F: exercise probability (refinancing frictions). Option value computed using the PV-equivalent measure (option-adjusted cost rate).

threshold and increasing the option value. Longer expected lives (Panel D) provide more time for favorable rate movements, though the marginal benefit diminishes as discounting and the falling survival probability reduce the weight on distant savings. Higher closing costs (Panel E) raise the refinancing threshold; doubling the fixed cost component lowers the option value by about 15%.

Panel F introduces behavioral frictions by allowing the borrower to exercise with probability $p < 1$ each time the threshold is crossed. Because crossings recur, halving the per-opportunity exercise probability trims the option value only modestly at this calibration; the larger losses documented by Keys et al. (2016) correspond to sustained inattention and ineligibility, which a per-opportunity coin flip does not capture.

A complementary check scales the analytically calibrated Hull–White volatility σ_{HW} by 0.7 and 1.5 at the January 2024 calibration: the option value shifts moderately in the expected direction, confirming that the volatility calibration matters quantitatively while the qualitative pattern—value increasing in volatility and rate level—is robust.

L.3 Spread Model Specification

The baseline uses expanding-window $\hat{\beta}$ with vol-targeting because it (i) uses only information available at origination, (ii) matches realized mortgage-rate volatility by construction, and (iii) avoids parameter choices that are inconsistent with early-sample data (where the estimated pass-through is near 1.0). Appendix Table 16 reports the variants in two panels, each run across all 438 historical months with 10,000 paths per month: Panel A reruns every variant under the headline configuration (Appendix Table 12), so the robustness evidence covers the series the paper actually reports; Panel B preserves the calibration-only rows from earlier drafts for continuity. Under the headline configuration, Student- $t(5)$ innovations leave the mean essentially unchanged at 29 bps (vol-targeting pins the innovation variance, so tail shape barely matters); the random-walk and regime-switching variants raise it to 40 and 38 bps; and the error-correction variants—which address the downward bias in the monthly-changes pass-through directly—give 39 bps with the long-run coefficient estimated ($\hat{b} \approx 0.93$) and 46 bps with $b = 1$ imposed, with the divergence concentrated in the 2010s.

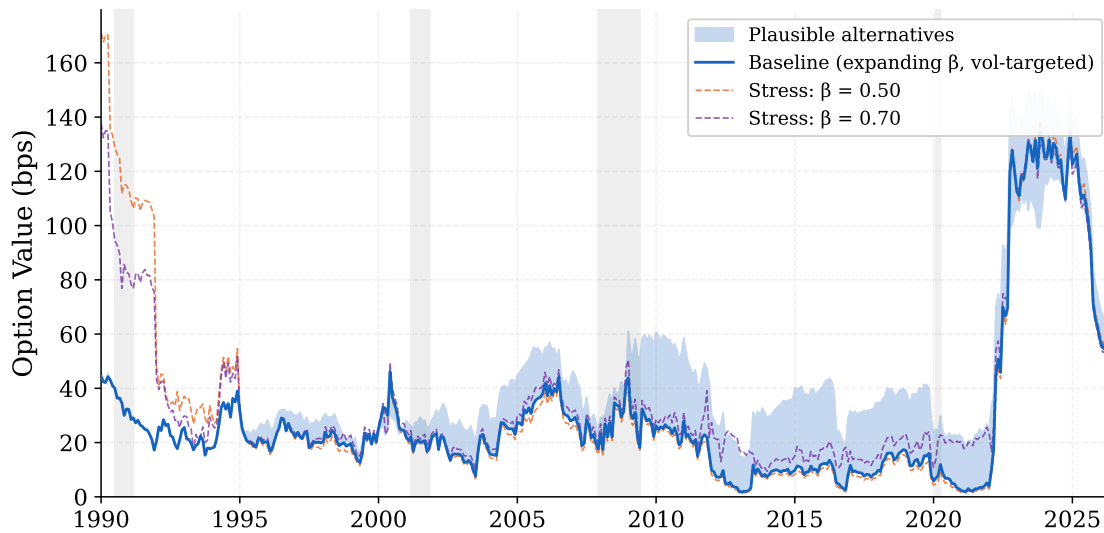
Appendix Figure 10 displays the time series, with the shaded band spanning the plausible

Appendix Table 16: Spread-model robustness (438 months, 10,000 paths each)

Specification	Mean	Median	Corr	1990s	2000s	2010s	2020s
<i>Panel A: headline configuration</i>							
Headline (expanding β, vol-targeted)	29	21	1.000	24	26	12	69
Student- $t(5)$ innovations	29	21	1.000	24	26	12	69
Random-walk spread ($\varphi = 1$)	40	32	0.922	27	37	36	72
Regime-switching spread mean	38	27	0.953	24	26	33	84
Error-correction pass-through ($\hat{b}$ estimated)	39	30	0.948	29	34	25	85
Error-correction pass-through ($b = 1$ imposed)	46	36	0.910	29	45	35	90
Fixed $\beta = 0.50$ (stress)	35	21	0.824	46	24	11	70
Fixed $\beta = 0.70$ (stress)	37	24	0.895	40	28	19	74
<i>Panel B: calibration-only configuration (for continuity with earlier drafts)</i>							
Calibration-only baseline (legacy discount, 10y)	24	18	1.000	23	21	8	54
Student- $t(5)$ innovations	26	19	0.915	38	19	5	47
Random-walk spread ($\varphi = 1$)	32	28	0.880	40	27	19	45
Regime-switching spread mean	31	22	0.944	38	19	17	57
Fixed $\beta = 0.50$ (stress)	34	18	0.538	72	18	4	47
Fixed $\beta = 0.70$ (stress)	34	21	0.620	62	21	9	49

Notes: Mean and median PV-equivalent option value in basis points; Corr = correlation with the panel's baseline series across 438 months; decade columns are decade means. Panel A runs every spread variant under the *headline* configuration (Appendix Table 12); its first row is the headline series itself. Panel B reruns the same variants under the *calibration-only* configuration (legacy subjective discount, fixed 10-year horizon), preserving the rows reported in earlier drafts. All runs use 10,000 Monte Carlo paths per month and the time-varying borrower calibration. Generated by analysis/make_spread_table.py.

alternatives and the stress tests shown as dashed lines.



Appendix Figure 10: Baseline option value (solid blue) with shaded band spanning the plausible spread specifications of Appendix Table 16, Panel A. Dashed lines show the two fixed- β stress tests. Recession periods shaded in gray.

Among the plausible alternatives, Student- $t(5)$ innovations leave the series essentially unchanged (correlation 1.000)—vol-targeting pins the innovation variance, so the option value is driven by variance rather than tail shape. The random-walk and regime-mean variants account for the upper end of the plausible range, with much of the divergence in the 2010s, where additional spread persistence or regime switching raises the decade mean from 12 to 33–36 bps. All plausible specifications preserve the broad time-series pattern (correlations of 0.91 or higher).

The fixed- β stress tests are deliberately extreme: they force a constant pass-through that is inconsistent with the early-sample data, where expanding OLS yields $\hat{\beta} \approx 1.0$. The result is a divergence concentrated in the 1990s, where the fixed $\beta < 1$ generates an artificially large spread component and raises the decade mean from 24 to 40–46 bps, but only modest differences post-2000. Even under these unfavorable assumptions the full-sample mean stays at 35–37 bps, within the range spanned by the plausible variants, though at the cost of a visibly distorted time path (correlations of 0.82–0.90).

L.4 Discount Rate and Horizon

The two assumptions replaced in this paper, the discount rate and the horizon, move the mean within a bounded range (Appendix Table 17). Moving from the fixed ten-year horizon to the empirical survival-weighted life adds about 2 bps, and the observable real-Treasury discount adds about 3 more; a fixed seven-year life gives 21 bps and a thirteen-year life 30 bps, bracketing the headline.

Appendix Table 17: Discount Rate and Horizon Robustness

Specification	PV-equiv OV (bps)	Rate-avg OV (bps)	Exp. life (years)
Baseline ($\rho = 5\%$, fixed 10y)	22	35	10.0
Empirical life, subjective discount	24	40	10.5
Fixed 7y life (MBS WAL)	21	39	6.9
Fixed 10y life	26	45	9.5
Headline (real disc., survival-weighted)	27	47	10.5
Fixed 13y life	30	51	12.8
CPS mobility, no bump ($\sim 16y$)	30	51	13.4

Notes: Mean option value (basis points) over 1990–2024 under alternative discount-rate and expected-life assumptions, varied as labeled in each row. The headline uses the observable real 10-year Treasury discount and a survival-weighted horizon with hazard $\mu_t + \delta$ ($\delta = 0.025$). The full-sample (through June 2026) headline mean is 29 bps; this window is 1990–2024 for comparability across rows.

M Key Calibration Reference Values

Appendix Table 18 reports reference values for three representative scenarios that appear throughout the paper. The high-rate scenario (6.61% contract rate on the January 2024 yield curve) is used in the worked example and sensitivity analysis. The low-rate scenario (January 2021) illustrates the small option value at cyclical rate troughs. The October 2023 scenario demonstrates behavior at the highest recently observed contract rates.

N Historical Signal Content

Months with higher predicted option values also exhibit greater realized savings from refinancing, indicating that the measure carries signal about future refinancing opportunities. Appendix Figure 11

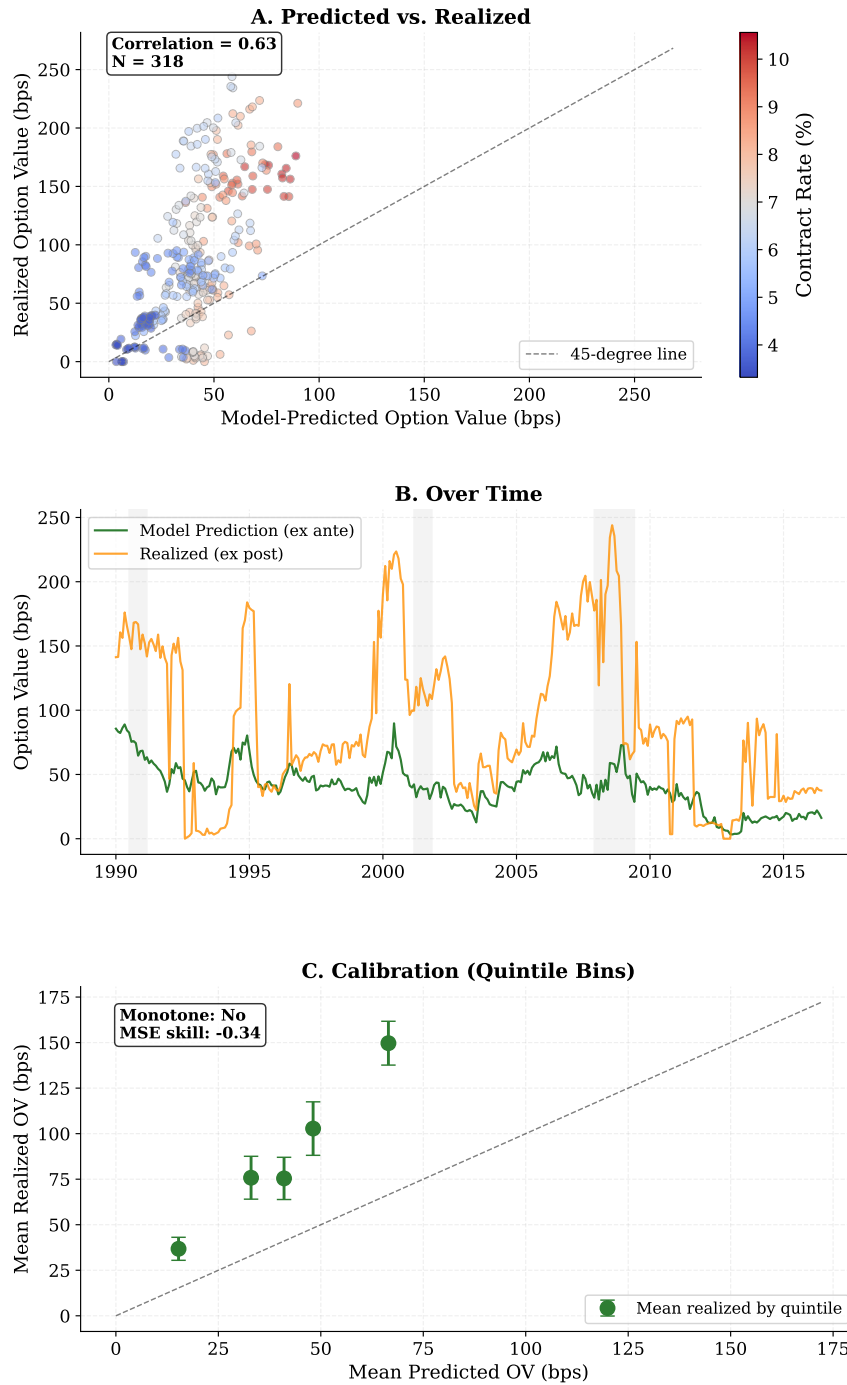
Appendix Table 18: Calibration Reference Values for Representative Scenarios

	6.61%	2.67%	7.31%
	(Jan 2024 YC)	(Jan 2021 YC)	(Oct 2023 YC)
Contract rate	6.61%	2.67%	7.31%
Balance	\$345K	\$288K	\$343K
Eff. rate (rate-avg)	4.81%	2.62%	5.54%
Option value (rate-avg, bps)	180	5	177
Eff. rate (PV-equiv.)	5.36%	2.65%	6.10%
Option value (PV-equiv., bps)	125	2	121
Mean refinancings	2.63	0.15	2.86
Threshold (bps)	100	63	95
Paths	10,000	10,000	10,000

Notes: Each column uses the yield curve, contract rate, and time-varying borrower parameters (Section 2.5) observed on the indicated date; all columns use the *headline* configuration (Appendix Table 12). Both effective rates are the option-adjusted cost rate c^* under the indicated measure, so option value = $m(0) - c^*$. Eff. rate (rate-avg) = discount-weighted average coupon rate paid under optimal refinancing. Eff. rate (PV-equiv.) = constant mortgage rate whose discounted payments plus terminal balance, discounted at the nominal 10-year Treasury yield $r_d = y_{10}^{\text{Ts}}$ (Appendix F), equal the mean discounted actual cash flows plus simulated terminal balance. Option value = contract rate minus effective rate, in basis points. Mean refinancings = average number of refinancing events per path. Threshold = Agarwal et al. (2013) optimal refinancing threshold for the indicated date's calibration.

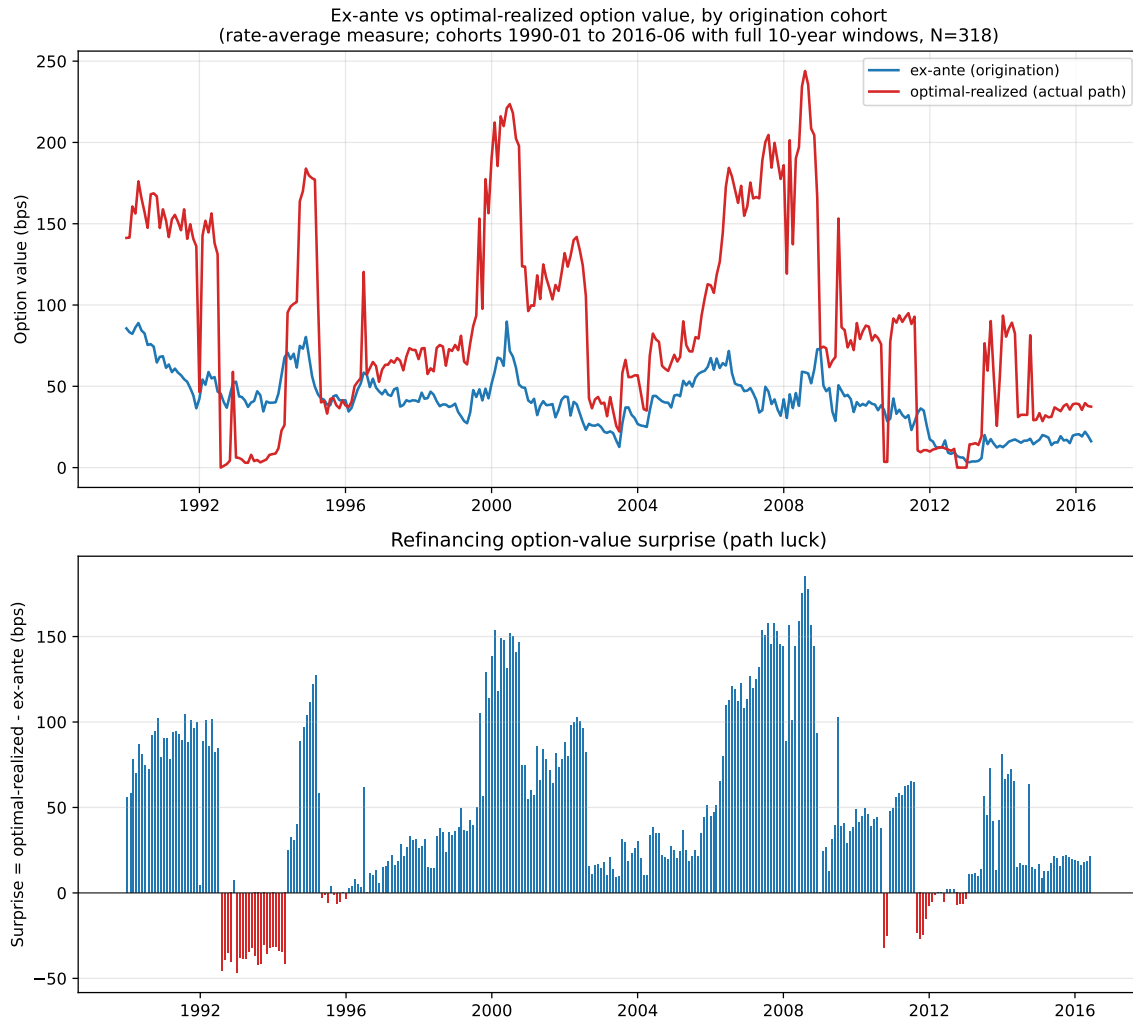
compares the model's ex-ante predicted option value (computed using only information available at origination) with the ex-post realized option value (computed by applying the same optimal refinancing rule to the actual path of mortgage rates over the subsequent 10 years). The sample is restricted to 1990–2016 origination cohorts with full 10-year realized windows ($N = 318$)—the origination months for which a full 10-year realized window is available. Both axes use the rate-average measure: the realized value from a single observed path is a discounted average rate by construction, so the predicted value uses the matching rate-average column rather than the PV-equivalent measure of the headline series. This sample is the same cohort series as the surprise decomposition of Section 5, built from a single file (`surprise_by_cohort.csv`).

The primary result is a correlation of 0.63 between predicted and realized option values: months with higher model-predicted option values experience substantially greater realized savings. The model systematically underpredicts (predicted mean 41 bps versus realized 88 bps, the same numbers reported in Section 5), and the MSE skill score is negative (-0.34), meaning the model's point predictions are somewhat less accurate than simply predicting the sample mean. But point-prediction



Appendix Figure 11: Panel A: each point is one origination month, colored by the contract rate (colorbar); x-axis is the model-predicted option value, y-axis is the realized value from actual rate paths. Panel B: time series of predicted versus realized values; gray bands mark NBER recessions. Panel C: binned calibration plot, mean realized option value by quintile of predicted, with 95% confidence intervals. Sample restricted to 1990–2016 origination cohorts with full 10-year realized windows ($N = 318$). Both axes use the rate-average measure, matching the realized single-path quantity; the series is identical to the surprise decomposition of Section 5 (source: `surprise_by_cohort.csv`).

accuracy is not the relevant criterion; what matters is whether the measure identifies high-versus-low option-value environments, and the correlation confirms that it does. Appendix Figure 12 plots the two cohort series and the surprise gap directly.



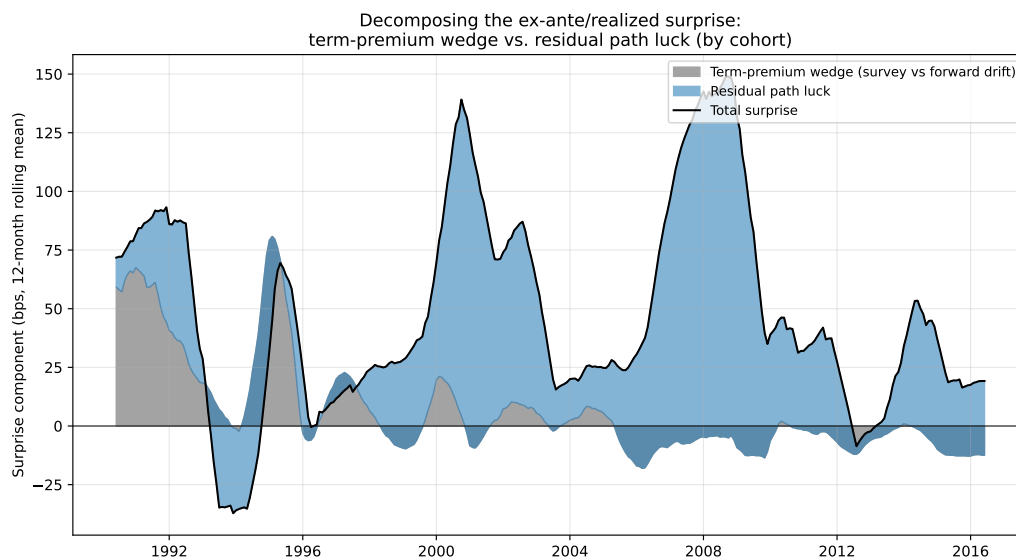
Appendix Figure 12: Ex-ante option value at origination versus the value realized under optimal exercise on the actual rate path, by cohort; the gap is the surprise, plotted directly in the lower panel. 1990–2016 origination cohorts with full 10-year realized windows ($N = 318$); rate-average measure on both series. Reuses the paper’s threshold and Monte-Carlo machinery; no new data (source: `surprise_by_cohort.csv`).

The source of the systematic underprediction is economically intuitive. The sample spans a period of extraordinary secular decline in interest rates, from roughly 10% in early 1990 to 2.67% in early 2021, punctuated by aggressive Federal Reserve interventions (the 2001 easing

cycle, quantitative easing after 2008, the near-zero-rate era of 2020–2021). The Hull–White model, calibrated to the yield curve at each origination date, generates paths that mean-revert around the forward curve observed at that date. It cannot anticipate that rates will continue declining for decades, nor can it predict policy regime changes or financial crises. Consequently, a borrower originating at 8% in 1992 appeared to have modest option value according to the model (which expected rates to fluctuate near 8%), but in reality benefited from the secular decline to 5–6%, a windfall the forward curve never priced in. The pattern is starkest in the 1990s and 2000s, when realized refinancing opportunities were much larger than the forward-curve-consistent model values at origination.

The gap admits a formal decomposition. The Hull–White drift is fit to the forward curve, which embeds term premia; a borrower with survey-consistent expectations would have anticipated part of the realized decline. Rerunning the ex-ante value for every cohort under two alternative drift benchmarks, holding the volatility calibration and all random draws fixed, splits the 47-bp surprise into a predictable term-premium wedge and residual path luck. Replacing the forward-curve drift with the SPF long-range 10-year Treasury expectation (linear convergence within a year; pre-1992 cohorts backfilled with the first 1992 survey) moves the mean ex-ante value by only 6 bps. Even a random-walk drift, which discards all forward-curve information, moves it by 19 bps, an upper bound on the predictable component that leaves at least three-fifths of the surprise as residual path luck; the point estimate is 41 bps. Appendix Figure 13 plots the split by cohort.

Because the model value is computed using only information available at origination, the gap between the realized and predicted option values measures the ex-post refinancing windfall generated by the realized interest-rate path. In the 1990–2016 sample, realized values substantially exceed conditional model values because the subsequent period delivered a large downward rate surprise, not because the origination-date measure is the wrong ex-ante measure. Thus $V_t^{\text{realized}} - V_t^{\text{ex-ante}}$ can be interpreted as a borrower-side option-value surprise: the component of refinancing value that was not priced by the origination-date yield curve. This quantity is distinct from the ex-ante option-adjusted cost rate, but it is economically meaningful as a measure of the unexpected transfer to borrowers with callable fixed-rate debt, subject to the important caveat that actual realized gains depend on refinancing take-up, credit access, and borrower attention.



Appendix Figure 13: Decomposing the ex-ante/realized surprise into the term-premium wedge (survey-expectations vs forward-curve drift) and residual path luck, by origination cohort (1990–2016 origination cohorts with full 10-year realized windows ($N = 318$); rate-average measure; 12-month rolling means). The wedge is computed by rerunning the ex-ante value with the SPF long-range 10-year Treasury expectation in place of the forward-curve drift, holding the volatility calibration and random draws fixed; identity term premium + path luck = surprise holds cohort by cohort. The two components are drawn as overlapping shaded areas, not stacked. Source: `surprise_decomposition.csv`.

This limitation is inherent to any conditional model: the option values computed here reflect forward-curve-consistent rate dynamics at origination, not what subsequently transpired. A hypothetical sample with equal periods of rising and falling rates would exhibit less systematic bias. For a prospective borrower, the relevant quantity is not whether the model predicted the future correctly but the conditional expectation the model provides: given what is known today about the yield curve and volatility, what is the expected benefit of the refinancing option.

O Borrower-Side Option Value and MBS Pricing

This appendix connects the borrower-side option value to the prepayment component priced in MBS markets: the construction of the spread series, one specification table containing every regression the paper runs, the time-series exhibits, and the interpretation caveats.

O.1 Construction

The total mortgage–Treasury spread, the gap between the primary mortgage rate (PMMS) and the 10-year Treasury yield, bundles secondary-market pricing of prepayment risk, originator margins and capacity constraints, credit risk, liquidity premia, and the effects of Federal Reserve MBS purchases (Boyarchenko et al., 2019). Because only the secondary-market component should directly reflect prepayment-option pricing, regressing the total spread on the option value is an apples-to-oranges test.

Following Fuster et al. (2013), I decompose the total spread into two components:

$$\underbrace{\text{PMMS} - 10\text{y Treasury}}_{\text{Total spread}} = \underbrace{(\text{CC yield} - 10\text{y Treasury})}_{\text{Secondary-market spread}} + \underbrace{(\text{PMMS} - \text{CC yield})}_{\text{Primary-secondary spread}}$$

where the current-coupon (CC) yield is the Fannie Mae 30-year MBS yield (Bloomberg: MT-GEFNCL). The secondary-market spread reflects compensation demanded by MBS investors for prepayment risk, duration, and convexity; the primary–secondary spread reflects originator profits, guarantee fees, and capacity constraints.

To construct the dependent variable, I decompose the MBS spread—the gap between the current-coupon Fannie Mae MBS pass-through yield (MTGEFNCL) and the 10-year Treasury—into the Bloomberg option-adjusted spread (OAS), which captures non-option factors, and the residual market option cost:¹⁰

$$\text{market option cost}(t) = \underbrace{(y_{\text{MBS}}(t) - y_{10}(t))}_{\text{MBS spread}} - \underbrace{\text{OAS}(t)}_{\text{non-option}} .$$

Two benchmark caveats attach to this construction. The Bloomberg OAS series with the longest history (MOASFNCL, from September 1996) is measured against the swap curve while the MBS-spread leg is measured against the 10-year Treasury, so the production market-option-cost series carries an uncontrolled swap–Treasury basis; and the spread leg compares an amortizing, negatively-convex security against a single 10-year point rather than a duration-matched curve position. Appendix Table 19 therefore adds two benchmark-consistent constructions on their available samples: the same current-coupon spread net of the Treasury-benchmarked OAS (NOASFNCL, from 2006), which keeps both legs on the Treasury curve and yields a similar positive slope, and the Bloomberg US MBS Index zero-volatility spread minus its Treasury OAS (from 2000), which is fully curve-based. The index construction prices the outstanding coupon stock rather than the current coupon. After 2022 the stock is deep out of the money, so its option cost falls exactly when the new-loan option value rises; the negative slope reflects that different asset rather than a contradiction of the current-coupon rows.

As a cross-check on the decomposition, the primary–secondary spread correlates 0.65 with the New York Fed’s Originator Profit and Unmeasured Cost (OPUC) series (Fuster et al., 2013) over the 389 overlapping months.

¹⁰The current-coupon yield is the pass-through rate to investors, net of the guarantee fee and servicing strip. It prices a hypothetical par-valued 30-year Fannie Mae TBA, not a portfolio of seasoned loans, making it directly comparable to the model’s new-origination option value.

O.2 Specifications

Appendix Table 19 reports every spread series regressed on the headline option value; each slope the paper has ever quoted appears here, once. The option value loads on the secondary-market spread (the preferred specification, slope 0.59, Newey–West $t = 4.78$, $R^2 = 0.22$) and not on the primary–secondary spread, consistent with investors pricing prepayment risk specifically; the OAS itself is a placebo and carries little option signal; and netting the OAS out of the secondary spread isolates a market option cost whose co-movement with the model is strongest of all, the same series used as the market cost basis in Appendix Table 1. Adding the VIX and changes in Federal Reserve MBS holdings as controls roughly doubles the explained variance while leaving the option-value slope unchanged, consistent with risk appetite and the Fed footprint as complementary spread drivers (Boyarchenko et al., 2019). The slope is not a deep behavioral parameter or a literal exercise probability: it blends actual exercise behavior with differences in rate models, horizon definitions, and the gap between risk-neutral and physical measures.¹¹

Because both the option value and the spread series are persistent, the level specifications raise the spurious-regression concern. An augmented Dickey–Fuller test rejects a unit root in the secondary spread itself ($p = 0.02$), so the dependent variable is persistent but mean-reverting, and deviations from the fitted level relationship close at -0.20 per month (Newey–West $t = -4.92$), while short-run monthly changes are uninformative, as expected when the relationship holds in levels. A formal Engle–Granger test does not reject the null of no cointegration ($p = 0.158$), so the relationship should be read as robust co-movement between two mean-reverting series rather than as a cointegrating one.

¹¹The Bloomberg OAS uses a proprietary prepayment model under the risk-neutral measure; the model option value uses Hull–White dynamics with a forward-curve drift and realized rather than implied volatility. The slope therefore captures the net effect of suboptimal exercise (which lowers the market cost), model differences, and any prepayment-risk premium (which raises it). The slope falls as the assumed expected life lengthens, because a longer life raises the model option value while the market cost is fixed; it remains positive and strongly significant across the expected-life range in Appendix Table 17.

Appendix Table 19: Spread series regressed on the borrower-side option value

Dependent variable	Slope	NW t	R^2	N	Sample
Total spread (PMMS – 10y Treasury)	0.85	8.11	0.38	438	1990–2026
Secondary-market spread (CC – 10y Treasury)	0.59	4.78	0.22	438	1990–2026
Primary–secondary spread (PMMS – CC)	0.26	1.82	0.03	438	1990–2026
OAS (placebo: non-option factors)	–0.12	–1.00	0.01	358	1996–2026
Market option cost (secondary spread – OAS)	0.71	8.11	0.47	358	1996–2026
Market option cost, Treasury OAS (secondary – Tsy OAS)	0.53	4.33	0.27	240	2006–2026
Market option cost, index ZV – Tsy OAS (US MBS Index)	–0.14	–2.78	0.08	309	2000–2026
Secondary spread, VIX + Δ Fed MBS controls	0.60	5.71	0.46	282	2003–2026

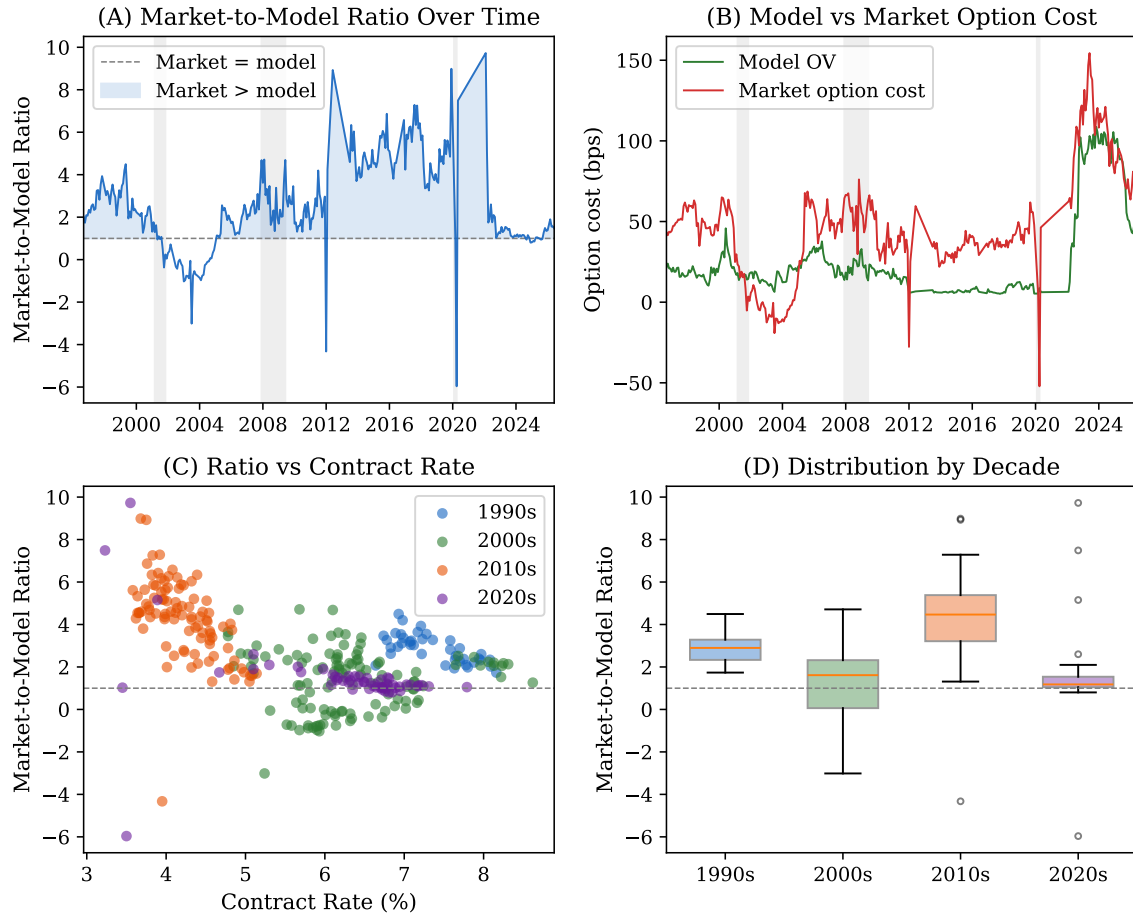
Notes: Monthly levels regressions of each spread series (bps) on the headline PV-equivalent option value (bps); Newey–West standard errors, 12 lags. The bold row is the preferred specification (the secondary-market spread is where prepayment risk is priced; its slope is the value quoted in the text). The OAS row is a placebo: the option-adjusted spread should and does carry little option signal. The market-option-cost row (spread net of OAS) isolates the prepayment component and is the basis series used for the cost-basis analysis (Appendix Table 1); its sample starts with OAS coverage (September 1996), and its OAS leg is benchmarked to swaps while the spread leg is benchmarked to the 10-year Treasury. The two benchmark-consistent rows put both legs on the Treasury curve: the Treasury-OAS current-coupon construction (available from 2006) and the Bloomberg US MBS Index zero-volatility-spread-minus-Treasury-OAS construction (available from 2000), which is also curve-based rather than measured against the single 10-year point. The index row prices the outstanding coupon stock rather than the current coupon: when rates rise the stock moves deep out of the money and its option cost falls exactly when the new-loan option value rises, so its negative slope reflects a different asset, not a contradiction of the current-coupon rows. The controls row adds the VIX level and the monthly change in Federal Reserve MBS holdings. Generated by `analysis/make_mbs_table.py`.

O.3 Time-Series Exhibits

The preferred prepayment series (the market option cost against the headline model value, and the decomposition of the MBS spread into its OAS and option components) are plotted in Figure 2 in the main text. Appendix Figure 14 provides additional detail on the market-to-model ratio. The ratio divides the market option cost by the calibration-only model value (Appendix Table 12). Panel A plots the ratio of market option cost to the model option value over time. Panel B overlays the model option value and the market option cost, illustrating their co-movement. Panel C shows the ratio against the contract rate level. Panel D reports the distribution by decade.

O.4 Interpretation and Caveats

The positive market relationship is consistent with independent evidence from the behavioral re-financing literature. Keys et al. (2016) find that many unconstrained borrowers who face a clear refinancing incentive nevertheless fail to exercise, while Gerardi et al. (2023) document racial dis-



Appendix Figure 14: Market-to-model option-cost ratio. Panel A: market option cost / model option value over time, with ratio = 1 line (dashed). Values above 1.0 (shaded) reflect periods where the market prices more prepayment risk than the model predicts. Panel B: model option value and market option cost time series. Panel C: ratio vs. contract rate, colored by decade. Panel D: box plots of the ratio by decade. The model value is the *calibration_only* configuration (Appendix Table 12); the ratio is unstable in months where the model value is near zero.

parities that imply lower effective exercise rates for minority borrowers, and Berger et al. (2025) show that even small inattention frictions generate large welfare losses from delayed exercise. Those frictions lower both the option's value to borrowers and its cost to investors, so they push the market's charge below the optimal-exercise benchmark and attenuate the slope. They do not explain why the charge exceeds the benchmark in most years; that excess must come from risk-neutral valuation, implied volatility, and the prepayment-risk premium (Appendix Table 1).

The 23-basis-point intercept is a measurement artifact rather than an economic wedge. The market option cost is computed as the nominal spread to the 10-year Treasury minus the OAS, but the OAS is fitted to the full Treasury curve. The mismatch between a single-point benchmark and a curve-based computation generates a positive residual that varies with curve shape.¹²

The comparison has two additional limitations. The model's rate dynamics are a hybrid: the drift is fit to the forward curve, as in risk-neutral pricing, while the volatility is realized trailing volatility rather than option-implied, and the discounting is the borrower-side Treasury rate. MBS prices embed implied volatility and a prepayment-risk premium, so the two valuations need not coincide exactly; the gap includes the prepayment-risk-premium wedge documented in the net-gain accounting (Appendix Table 1). In addition, the 10-year Treasury is an imperfect duration benchmark for 30-year mortgages; a duration-matched benchmark would sharpen the decomposition and is left to future work.

P Supplementary Ex-Ante Exhibits

This appendix collects exhibits that support the ex-ante analysis of Section 3 but are not essential to the main narrative: the decomposition of the time-varying calibration and the expected-loan-life input series, additional views of the option-value distribution including the by-regime summary, sample simulated rate paths, and the full refinancing-event distribution.

¹²Replacing the 10-year benchmark with the 7-year or 20-year Treasury shifts the intercept substantially (52 and –33 bps, respectively) while leaving the slope in the range 0.46–0.90, confirming that the intercept is benchmark-dependent but the positive marginal relationship is robust.

P.1 Time-Varying Calibration Decomposition

Appendix Table 20 quantifies the contribution of each time-varying parameter by sequentially adding them to a fixed representative-borrower baseline (\$250,000 balance, $\tau = 0.28$, $\mu = 0.10$, points = 1%). Holding the discount rate and horizon fixed to isolate the calibration channel, the full-sample mean is little changed by time-varying calibration (24 vs. 25 bps), though the time-series pattern shifts; the headline series of Section 3 additionally applies the observable real-Treasury discount and the survival-weighted horizon, which raise the mean to 29 bps. The 1990s option value falls from 38 to 23 bps, driven by smaller loan balances and higher origination costs: in the early 1990s, the average GSE loan was roughly \$105,000 with 2% in points, making refinancing less cost-effective. The 2020s rise from 47 to 55 bps, reflecting lower origination costs and larger balances that together reduce the refinancing threshold. Mobility and the effective tax rate each contribute roughly 1 bp to the full-sample mean.

Appendix Table 20: Counterfactual Decomposition of Time-Varying Calibration

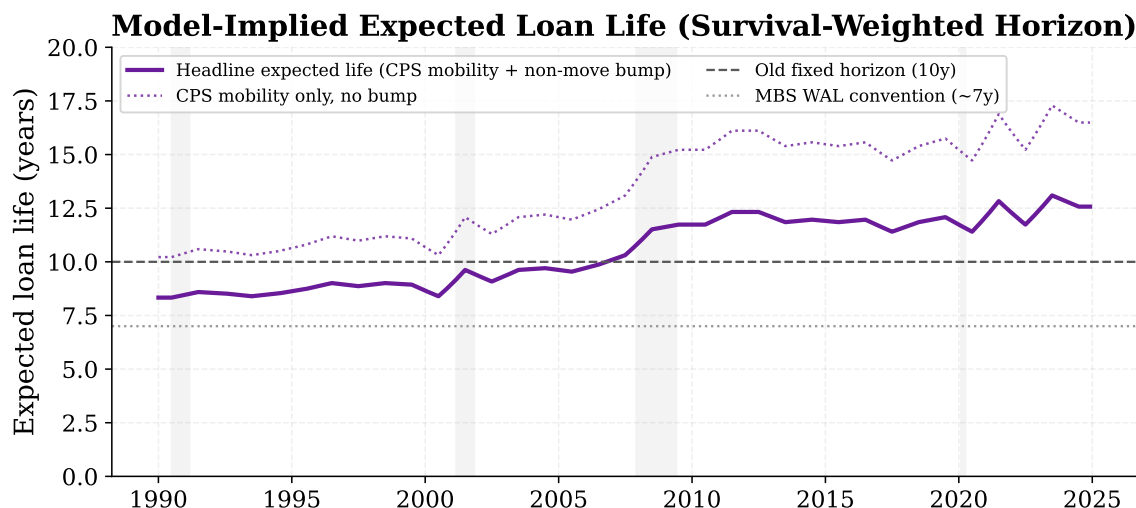
Specification	Full	1990s	2000s	2010s	2020s
Fixed baseline	25	38	19	5	47
+ Time-varying balance	22	26	16	5	50
+ Time-varying points	22	21	19	7	52
+ Time-varying mobility	23	22	20	8	53
+ Time-varying tax rate (= full)	24	23	21	8	55
+ Real-Treasury discount, survival horizon (= headline)	29	24	26	12	69

Notes: PV-equivalent option value in basis points (decade means). Each row adds one time-varying parameter while holding the others at their previous values. The fixed baseline uses \$250K balance, $\tau = 0.28$, $\mu = 0.10$, and 1% origination points. The sequential ordering is balance, origination points, mobility, tax rate. All decomposition rows hold the discount rate and horizon at the fixed-baseline settings ($\rho = 5\%$, fixed 10-year horizon) and use 5,000 Monte Carlo paths per month across 438 historical months; the final row additionally applies the observable real-Treasury discount and the survival-weighted horizon (the *headline* configuration, 10,000 paths), landing on the headline mean.

Appendix Figure 15 plots the survival-weighted expected loan life implied by the mobility series, the input behind the horizon channel in the decomposition.

P.2 Additional Views of the Option-Value Distribution

Appendix Table 21 summarizes the key inputs and outputs by rate regime; Appendix Figures 16 and 17 show the option-value distribution and its drivers.



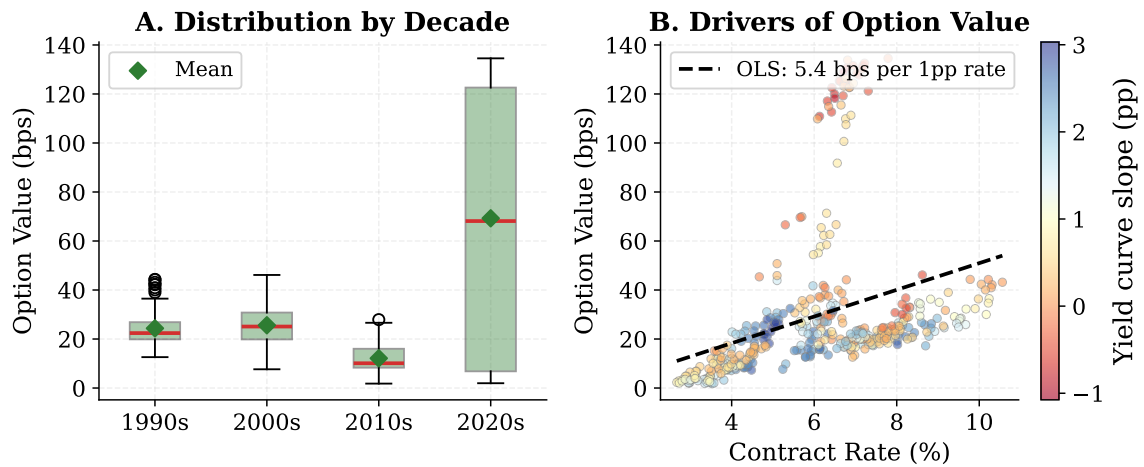
Appendix Figure 15: Model-implied expected loan life over time. The survival-weighted horizon uses the exogenous (non-refinancing) termination hazard $\mu_{\text{tot}} = \mu_t + \delta$, where μ_t is the CPS homeowner-mobility rate and $\delta = 0.025$ captures non-relocation terminations (in-place payoff, default, death). Expected life rises from about 8.5 years in the early 1990s to 12.4 years in the 2020s as mobility falls, sitting between the seven-year MBS weighted-average-life convention and the sixteen-year tenure implied by mobility alone; the figure plots both as reference lines, along with the mobility-only tenure. Gray bands mark NBER recessions.

Appendix Table 21: Rate Environment and Option Value by Regime

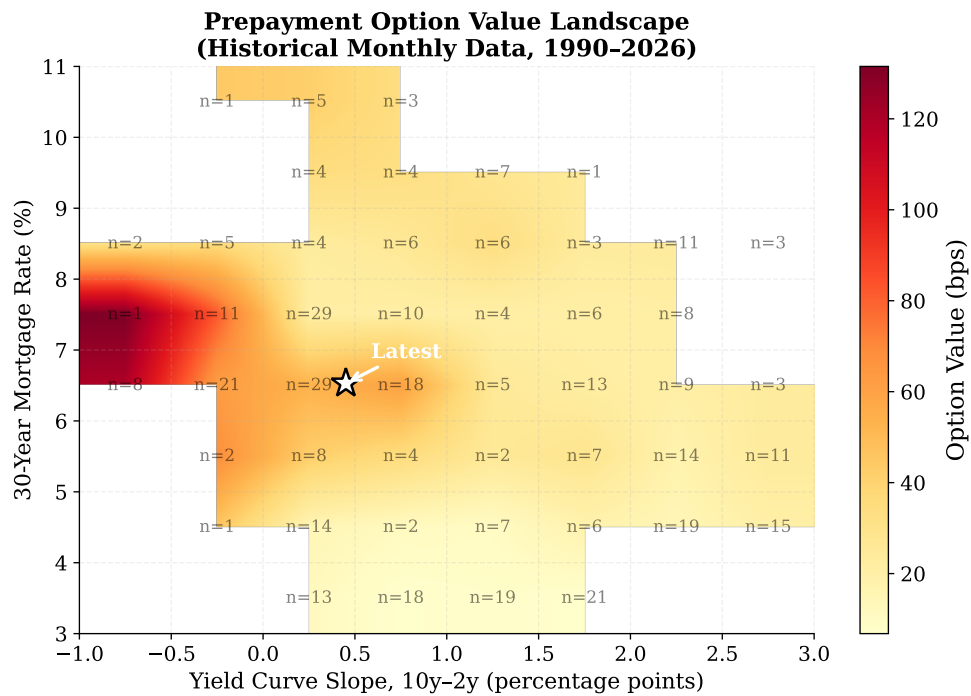
	Low (<4%)	Medium (4–6%)	High (>6%)	2022–24
<i>N</i> (months)	83	114	241	36
Mean rate (%)	3.5	4.9	7.4	6.2
Mean slope (pp)	1.06	1.72	0.73	–0.26
Mean volatility (%)	0.55	0.78	0.85	1.07
Mean spread (bps)	182	174	180	260
Mean threshold (bps)	68	88	128	92
Mean refinancings	0.32	0.71	1.24	2.44
Mean OV (bps)	7	21	41	103
Median OV (bps)	7	18	25	122

Notes: Columns group months by the 30-year fixed mortgage contract rate at origination; the 2022–24 column is a separate grouping by date. Mean rate = average contract rate in the bin. Slope = 10-year minus 2-year Treasury yield (pp). Volatility = trailing 36-month annualized standard deviation of monthly mortgage rate changes. Spread = PMMS 30-year rate minus 10-year Treasury yield. Threshold = Agarwal et al. (2013) optimal refinancing threshold. Mean refinancings = average number of refinancing events per path over the loan’s expected life. OV = option value ($m(0) - c^*$, basis points).

Summary Statistics: Option Value Distribution and Drivers



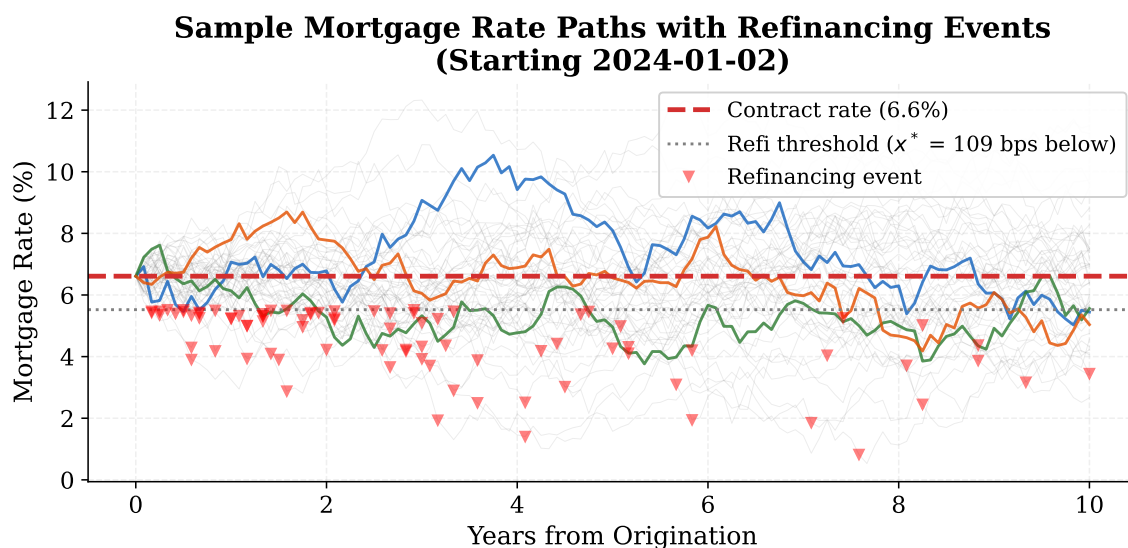
Appendix Figure 16: Panel A: box plots of the option value by decade (median line, mean diamond, IQR, 5th–95th whiskers). Panel B: option value versus contract rate, colored by yield curve slope. Based on 438 monthly observations, 10,000 paths per month, survival-weighted horizon. Option value computed using the PV-equivalent measure (option-adjusted cost rate). Any fitted association between the option value and the rate environment summarizes the model’s own comparative statics—the rate level, slope, and volatility are simulation inputs—and is not independent evidence on what drives the option value.



Appendix Figure 17: Cell color shows the mean option value (bps) for months falling in that (mortgage rate, yield curve slope) bin; the printed number in each cell is the count of months. The star marks the most recent observation. Based on 438 monthly observations, January 1990 through June 2026. Option value computed using the PV-equivalent measure (option-adjusted cost rate).

P.3 Simulated Rate Paths

Appendix Figure 18 shows sample Monte Carlo paths for a borrower originating a mortgage in January 2024. Under the January 2024 calibration-only configuration (\$345K balance, $\tau = 0.065$, $\mu = 0.043$, 0.7% points), the optimal threshold is 109 bps (100 bps under the headline configuration)—well above the naive NPV break-even of roughly 30 bps. The optimal borrower waits for roughly three times the naive break-even, because exercising the option today forecloses the possibility of exercising at an even better rate tomorrow. The paths illustrate the heterogeneity of outcomes: some generate multiple refinancing events as rates follow a downward trajectory, while others never trigger a refinance because rates rise or remain near the initial level.



Appendix Figure 18: Each thin line is one simulated mortgage rate path over 10 years from the January 2024 yield curve (contract rate 6.61%); three sample paths are highlighted in color. The dashed red line is the initial contract rate. The dotted gray line marks the optimal refinancing threshold (contract rate minus x^*) under the January 2024 *calibration_only* configuration (Appendix Table 12). The line is drawn for the initial contract; after each refinance the reference rate resets, so later exercises can occur below it. Red triangles mark refinancing events.

Appendix Table 22: Refinancing Event Distribution by Scenario (10,000 paths, survival-weighted horizon)

	Jan 2021 (2.67%)	Jan 2024 (6.61%)	Oct 2023 (7.31%)
P(0 refis)	87%	3%	1%
P(1 refi)	11%	16%	11%
P(2 refis)	2%	29%	27%
P(3+ refis)	0%	52%	61%
Mean refis	0.15	2.63	2.89
Median refis	0	3	3

Notes: Each column is a representative origination month; the contract rate and yield curve are taken from that date. $P(k \text{ refis})$ = fraction of 10,000 simulated paths in which the borrower refinances exactly k times (or ≥ 3 for the last row) over the loan's expected life, applying the Agarwal et al. (2013) optimal threshold at each monthly step. Borrower parameters are time-varying (Section 2.5); remaining parameters as in Appendix J. Simulated independently of Appendix Table 18; small differences in mean refinancings across the two tables reflect Monte Carlo sampling error.

P.4 Refinancing-Event Distribution

Q Replication Materials

The ex-ante simulation is written in Python using NumPy, SciPy, and Pandas; interest-rate and mortgage data are downloaded from the FRED API at a fixed July 2026 vintage (sample truncated at June 2026). Expected inflation is the Cleveland Fed model-based series, and the survey rate expectation used in the surprise decomposition is the Philadelphia Fed Survey of Professional Forecasters long-range 10-year Treasury forecast. Two discontinued or spliced series are handled explicitly: the points series MORTPTS30US (discontinued November 2022) is held at its 2022 average thereafter, and the average GSE balance splices the FHFA MIRS (through 2019) with the NMDB (2019 onward). The realized-gains pipeline additionally uses the HMDA Loan/Application Register files, the Freddie Mac single-family loan-level dataset and Quarterly Refinance Report, and the Survey of Consumer Finances; every headline table is regenerated from these public inputs by a single orchestrator script. The replication package is available at <https://www.adamlooney.com/data/mortgage-option-value>.